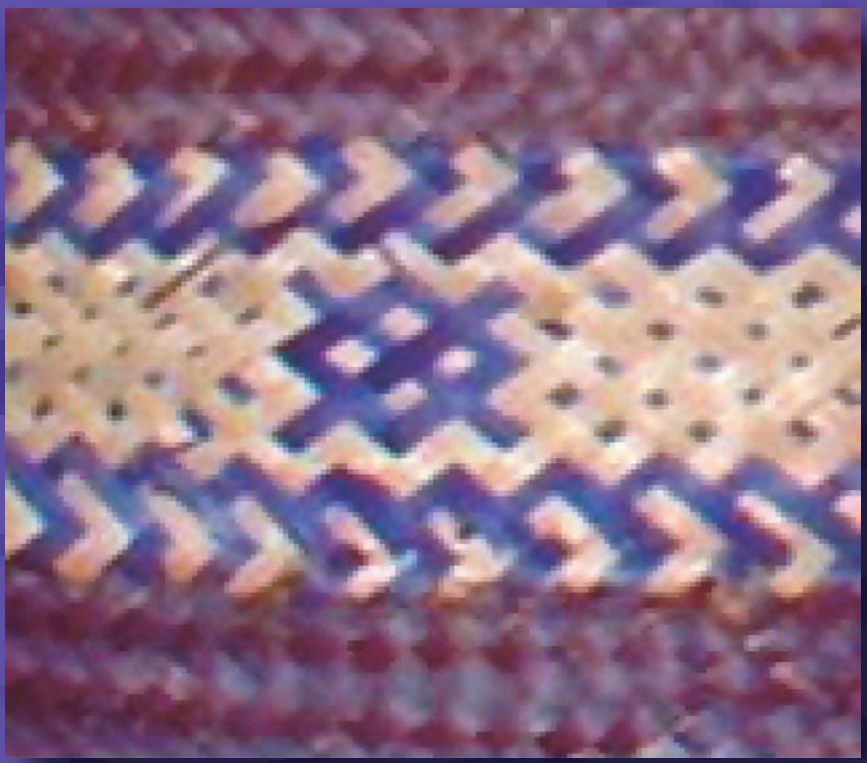
Adventures in the World of Matrices



Paulus Gerdes

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ADVENTURES IN THE WORLD OF MATRICES

PAULUS GERDES

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The photograph on the cover presents a detail of a mat woven by Mwani women in the coastal area of the Cabo Delgado Province in Northeast Mozambique (Africa). The decorative motive corresponds to the structure of a positive cycle matrix of dimensions 4 by 4 (See chapters 4, 5 and 6). The decorative motive on the backside of the mat corresponds to the structure of a negative cycle matrix of the same dimensions (cf. Gerdes, 2007).

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PREFACE

A mathematical matrix can be defined as a rectangular table consisting of abstract quantities that can be added and multiplied. While the term matrix is relatively recent in the literature (it was introduced in 1850 by James Joseph Sylvester), the use of matrices goes back to ancient China, with the study of systems of simultaneous linear equations. Matrices are universally present in mathematics as well as in various disciplines of science. The theory of matrices is one of the richest, most abstract and useful branches of mathematics. There are various classes of matrices, among them is the collection of cycle matrices, or matrices whose entries appear in somehow alternating or periodic manner. These cycle matrices were discovered by the author Paulus Gerdes while analyzing mathematical properties of African traditional drawings from Angola. Cycle matrices present interesting and beautiful visual characteristics.

This book by Paulus Gerdes, a prolific and well-known mathematics educator, is an exciting step-by-step introduction and fabulous journey into the magic world of cycle matrices. Beyond the abstract contribution, it also contains numerous carefully selected applications of matrices, including those arising in biology. It is accessible, attractive and easy to read. It presupposes very little background beyond elementary Arithmetic. The reader learns by actually “playing” and working with matrices. Thus this book is quite suitable for a self-study or a workshop with a diverse audience.

Baltimore, January 2007.
Gaston M. N’Guerekata

PRESENTATION

Matrices constitute a mathematical instrument ever more used in diverse fields of science and technology. Today it is difficult to enroll in a higher education program where the student does not meet or apply, in one form or another, the concept of matrices. From economy to agricultural science, engineering to veterinary and medicine, sociology to linguistics, psychology to computer science, and physics to biology, matrices appear.

Matrices also constitute an attractive field for mathematical exploration.

The present book deals with a special type of matrices. It analyzes what will be called *cycle matrices*. The book pretends to divulge some of my research results already published in mathematical journals. I gave a lecture on cycle matrices to the participants in the 13th Pan-African Mathematics Olympiad held in April 2003 in Maputo (Mozambique), and the pupils seemed delighted with the visual properties of these matrices. At the fifth Pan-African Congress of Mathematicians hold in September 2004 in Tunis (Tunisia), I presented the theme of cycle matrices and their variations. The interest displayed by the colleagues who were present and their questions and suggestions led me to write this introductory book about cycle matrices for a larger public.

I hope that the present book about cycle matrices may provoke the interest, the curiosity and the attention of a larger public of high school pupils, students in higher education, mathematics teachers and lecturers, professionals who use in their daily life mathematical instruments, and other readers. I hope that the surprising properties of cycle matrices may contribute to a major appreciation of the beauty of mathematical constructions, a better understanding of mathematics as a science of shapes and structures, and more pleasure in discovering un-imagined regularities.

The book is organized in short chapters. Each chapter includes some activities that contain exercises and/or questions for reflection by the reader. The proposed questions and problems are usually answered or solved in the same or a later chapter.

For those readers who have not yet studied matrices, the concept of matrix will be introduced in the first chapter and the operations of addition and multiplication of matrices will be presented. As cycle matrices are composed of cycles of certain properties, the notion of cycle is introduced in the second chapter.

In the third chapter we will begin the adventures in the world of matrices by analyzing some matrices composed of cycles. In chapters 4 through 12 alternating cycle matrices of even dimensions will be studied, culminating with the formulation of general hypotheses concerning those matrices.

In chapters 13 through 17 alternating cycle matrices of odd dimensions will be studied, after which we will return to alternating cycle matrices of even dimensions in chapter 18. In chapters 19 through 22 proofs for the theorems that govern the world of alternating cycle matrices will be constructed step by step. In the case that the reader finds chapters 19 through 22 too difficult, one can continue without any problems with the next chapters and return later to the earlier chapters. In any regards chapters 19 through 22 show how a more complex problem can be dissected into easier problems and how one can advance gradually with a proof.

In chapters 23 through 25 we will advance with the presentation of concrete examples of other types of cycle matrices. Chapter 26 presents a general summary of the theory of cycle matrices as analyzed in the book.

In chapter 27 I will describe the context that led me to enter, unexpectedly, into the world of cycle matrices. A possible application of cycle matrices in biology will be presented in chapter 28. In chapter 29, a note follows for those readers who would like to use a computer program to explore cycle matrices.

The book concludes with several appendices about additional properties of the special class of alternating cycle matrices. Chapters 31 and 32 deal with inverse matrices and determinants of alternating cycle matrices and are written for readers with access to a computer. Chapters 33 and 34, written for all readers, contain surprising and beautiful results concerning the transformation of these alternating cycle matrices, in particular, a geometric interpretation of the matrix transformations.

The readers are invited to accept the challenge to experiment further with cycle matrices and to become convinced of the veracity of the formulated hypotheses. I would like to wish the readers a lot of pleasure on their voyage of adventures through the world of cycle matrices.

I hope to continue this book with other adventures in the world of matrices to divulge results concerning helix and cylinder matrices that also display attractive geometric-algebraic properties that are elegantly preserved when adding or multiplying them.

Maputo, Mozambique
May 2006

Paulus Gerdes

Chapter 1

THE CONCEPT OF A MATRIX

ABSTRACT

In chapter 1 the concept of a matrix, the operations of addition and multiplication of matrices and the notation of the elements of a matrix will be introduced. The chapter closes with a brief historical note

A matrix is a rectangular array of numbers.

1	0	3	-4	2
5	4	2	3	6
-2	6	5	-3	4

Figure 1.1.

Figure 1.1 presents an example of a matrix. It is composed of 15 numbers, called the **elements** of the matrix. The elements are ordered in 3 (horizontal) **rows** and 5 (vertical) **columns**. One says that this matrix has **dimensions** 3 by 5.

The order of the rows and of the columns corresponds to the normal reading in English, going from the upper line downwards. The first row is the top row, etc. The last row is the bottom row of the matrix. Reading a line from the left to the right, we meet the first column, the second column, etc. (figure 1.2).

	1st column	2nd column	3rd column	4th column	5th column
1st row	1	0	3	-4	2
2nd row	5	4	2	3	6
3rd row	-2	6	5	-3	4

Figure 1.2.

MULTIPLICATION OF A MATRIX BY A SCALAR

A matrix may be multiplied by a number, called a **scalar**. The multiplication of a matrix by a scalar consists in the multiplication of all elements of the matrix by this number. The multiplication of a matrix by a scalar results in a new matrix. For instance, figure 1.3 presents the multiplication of the matrix of figure 1.1 by the scalar 3.

$$3 \times \begin{bmatrix} 1 & 0 & 3 & -4 & 2 \\ 5 & 4 & 2 & 3 & 6 \\ -2 & 6 & 5 & -3 & 4 \end{bmatrix} = \begin{bmatrix} 3 & 0 & 9 & -12 & 6 \\ 15 & 12 & 6 & 9 & 18 \\ -6 & 18 & 15 & -9 & 12 \end{bmatrix}$$

Figure 1.3.

ADDITION OF MATRICES

Two matrices of the same dimensions can be added. The addition of matrices consists of adding the **corresponding elements** of the two matrices, producing a new matrix. It is said that an element of the first matrix corresponds to an element of the second matrix if both are in the same row and in the same column of the respective matrices.

Figure 1.4 illustrates an example of the addition of two matrices of dimensions 3 by 5.

$$\begin{bmatrix} 1 & 0 & 3 & -4 & 2 \\ 5 & 4 & 2 & 3 & 6 \\ -2 & 6 & 5 & -3 & 4 \end{bmatrix} + \begin{bmatrix} 4 & -2 & -3 & 5 & 3 \\ -1 & 5 & -5 & 2 & 0 \\ 6 & 1 & -4 & 7 & -6 \end{bmatrix} = \begin{bmatrix} 1+4 & 0-2 & 3-3 & -4+5 & 2+3 \\ 5-1 & 4+5 & 2-5 & 3+2 & 6+0 \\ -2+6 & 6+1 & 5-4 & -3+7 & 4-6 \end{bmatrix} =$$

$$= \begin{bmatrix} 5 & -2 & 0 & 1 & 5 \\ 4 & 9 & -3 & 5 & 6 \\ 4 & 7 & 1 & 4 & -2 \end{bmatrix}$$

Figure 1.4.

In the present book we shall analyze only **square matrices**. Square matrices are matrices that have the same number of rows and columns. Figure 1.5 presents a square matrix of dimensions 4 by 4.

$$\begin{bmatrix} 3 & 6 & -4 & -7 \\ 4 & 9 & -3 & 5 \\ 5 & -2 & 0 & 1 \\ 4 & 7 & 1 & 4 \end{bmatrix}$$

Figure 1.5.

MULTIPLICATION OF SQUARE MATRICES

Square matrices of the same dimensions can be multiplied. The result of the multiplication is a new matrix of the same dimensions, which is called the **product** of the two matrices.

A concrete example of the multiplication of two matrices of dimensions 4 by 4 (figure 1.6) illustrates how two square matrices of the same dimensions are multiplied.

$$\begin{bmatrix} 3 & 6 & -4 & -7 \\ 4 & 9 & -3 & 5 \\ 5 & -2 & 0 & 1 \\ 4 & 7 & 1 & 4 \end{bmatrix} \times \begin{bmatrix} 2 & 0 & -4 & 5 \\ -1 & 5 & -6 & -2 \\ 4 & -2 & 1 & 3 \\ 6 & -3 & 2 & 7 \end{bmatrix} = \begin{bmatrix} & & & \\ & & ? & \\ & & & \\ & & & \end{bmatrix}$$

Figure 1.6.

The question mark is placed in the element that lies on the intersection of the second row and of the third column. This element results from the 'multiplication' of the second row of the first matrix (matrix **A**) by the third column of the second matrix (matrix **B**) (figure 1.7).

$$\begin{bmatrix} 3 & 6 & -4 & -7 \\ \boxed{4 \quad 9 \quad -3 \quad 5} \\ 5 & -2 & 0 & 1 \\ 4 & 7 & 1 & 4 \end{bmatrix} \times \begin{bmatrix} 2 & 0 & \boxed{-4} & 5 \\ -1 & 5 & \boxed{-6} & -2 \\ 4 & -2 & \boxed{1} & 3 \\ 6 & -3 & \boxed{2} & 7 \end{bmatrix} = \begin{bmatrix} & & \\ & ? & \\ & & \\ & & \end{bmatrix}$$

Figure 1.7.

One has to multiply the 1st element of the row under consideration by the 1st element of the column under consideration, that is, 4 x (-4); one has to multiply the 2nd element of the row by the 2nd element of the column, that is, 9 x (-6); one has to multiply the 3rd element of the row by the 3rd element of the column, that is, (-3) x 1; one has to multiply the 4th element of the row by the 4th element of the column, that is, 5 x 2; at the end one has to add the four partial products already obtained: 4 x (-4) + 9 x (-6) + (-3) x 1 + 5 x 2 = -16 –54 – 3 + 10 = - 63. The other elements of the product matrix are calculated in the same way.

ACTIVITIES

- Calculate the other elements of the new matrix, **C = AB**.
- Calculate in the same way a matrix **D = BA**. Compare the matrices **C** and **D**.

Figure 1.8 presents the result of the multiplication of the two matrices **A** and **B**.

$$\begin{bmatrix} 3 & 6 & -4 & -7 \\ 4 & 9 & -3 & 5 \\ 5 & -2 & 0 & 1 \\ 4 & 7 & 1 & 4 \end{bmatrix} \times \begin{bmatrix} 2 & 0 & -4 & 5 \\ -1 & 5 & -6 & -2 \\ 4 & -2 & 1 & 3 \\ 6 & -3 & 2 & 7 \end{bmatrix} = \begin{bmatrix} -58 & 59 & -66 & -58 \\ 17 & 36 & -63 & 28 \\ 18 & -13 & -6 & 36 \\ 29 & 21 & -49 & 37 \end{bmatrix}$$

Figure 1.8.

Figure 1.9 presents the result of the multiplication of the two matrices **A** and **B** in the inverse order: **BA**. The reader sees that **AB** and **BA** are very different: one says that the multiplication of matrices is not commutative.

$$\begin{bmatrix} 2 & 0 & -4 & 5 \\ -1 & 5 & -6 & -2 \\ 4 & -2 & 1 & 3 \\ 6 & -3 & 2 & 7 \end{bmatrix} \times \begin{bmatrix} 3 & 6 & -4 & -7 \\ 4 & 9 & -3 & 5 \\ 5 & -2 & 0 & 1 \\ 4 & 7 & 1 & 4 \end{bmatrix} = \begin{bmatrix} 6 & 55 & -3 & 2 \\ -21 & 37 & -13 & 18 \\ 21 & 25 & -7 & -25 \\ 44 & 54 & -8 & -27 \end{bmatrix}$$

Figure 1.9.

Only in very special cases is **AB = BA** valid for matrices. Some of these special cases will be encountered in this book.

IDENTITY MATRIX

A square matrix that has on its principal diagonal only 1's and has 0's in all other positions is called an **identity matrix** and is abbreviated by **I**. Figure 1.10 presents the identity matrix when the dimensions are 4x4.

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Identity matrix of dimensions 4x4.

Figure 1.10.

When one multiplies a square matrix by the identity matrix, the result is equal to the initial matrix. Figure 1.11 presents an example of dimensions 3x3. Verify!

$$\begin{bmatrix} 2 & 4 & 7 \\ -1 & 3 & 5 \\ 5 & 8 & 9 \end{bmatrix} \times \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 4 & 7 \\ -1 & 3 & 5 \\ 5 & 8 & 9 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} 2 & 4 & 7 \\ -1 & 3 & 5 \\ 5 & 8 & 9 \end{bmatrix}$$

Multiplication by the identity matrix.

Figure 1.11.

HISTORICAL NOTE

For more than 2000 years rectangular arrays of numbers, matrices, have been used in various cultures and continents – one of the oldest uses of matrices was in Chinese texts around 200 BC. The idea of the multiplication of matrices is relatively recent. It emerged for the first time in 1857 in the work of the English mathematician Arthur Cayley (1821-1895),

when he introduced, in his words, a ‘convenient way to express systems of linear equations’. An example illustrates the initial idea of Cayley. The system of two linear equations

$$3x + 4y = 2$$
$$5x - 2y = 12$$

can be written as

$$\boxed{\begin{matrix} 3 & 4 \\ 5 & -2 \end{matrix}} \boxed{\begin{matrix} x \\ y \end{matrix}} = \boxed{\begin{matrix} 2 \\ 12 \end{matrix}}$$

In this historic example the reader sees that matrices do not need to be square in order to be able to multiply them. It is sufficient that the number of elements of the rows of the first matrix is equal to the number of elements of the columns of the second matrix.

NOTATIONS

In the present book we shall represent – as we already did in this chapter – matrices always ‘embraced’ by **circumscribed rectangles**. In other literature different notations may be seen, as to place the whole array of the elements of a matrix between brackets.

As we proceed in the second half of the book with some proofs, it will be useful to have a notation to indicate the specific place where a certain element of a matrix **A** is positioned. **Indices** will be used. The element that is positioned in the second row and fourth column of matrix **A** will be represented by $\mathbf{A}_{(2,4)}$. The element that lies in the fifth row and third column will be represented by $\mathbf{A}_{(5,3)}$. And more in general, the element that is placed in the *i*-th row and *j*-th column will be indicated by $\mathbf{A}_{(i,j)}$, where *i* and *j* are called the indices of the considered element of matrix **A**.

If for all possible values of *i* and *j*, we have $\mathbf{A}_{(i,j)} = \mathbf{A}_{(j,i)}$, matrix **A** is called **symmetrical**, being the principal diagonal the symmetry axis. Figure 1.12 presents an example.

$$\boxed{\begin{matrix} 3 & 5 & 4 & 1 & 12 \\ 5 & 6 & 19 & 8 & -4 \\ 4 & 19 & -1 & 2 & 7 \\ 1 & 8 & 2 & 0 & 3 \\ 12 & -4 & 7 & 3 & 15 \end{matrix}}$$

Example of a symmetrical matrix.

Figure 1.12.

We have introduced more than enough concepts to initiate our adventurous voyage through the world of cycle matrices. We still have to clarify, in chapter 2, the notion of a cycle.

Chapter 2

CYCLES OF NUMBERS

ABSTRACT

In chapter 2 the concept of a cycle of numbers will be introduced. Some examples of cycles of numbers, like those of alternating cycles and of periodic cycles, will be presented.

We may picture a cycle of numbers as a sequence of numbers such that after the last number one returns to the first number.

For example, on a watch the natural numbers 1 to 12 belong to the cycle composed of those twelve numbers. On many watches the numbers are placed around a circle; on other watches the numbers are placed around a rectangle. In this way, both the circle and and the rectangle are two possible forms of representation of a cycle of numbers.

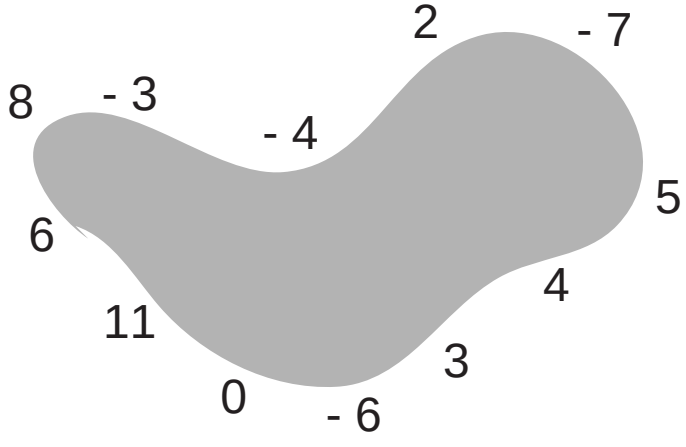


Figure 2.1.

Figure 2.1 presents another example of a cycle of numbers. Once more, all the numbers of the cycle are different. It may happen, however, that some numbers are equal, as illustrates the example of the cycle in figure 2.2.

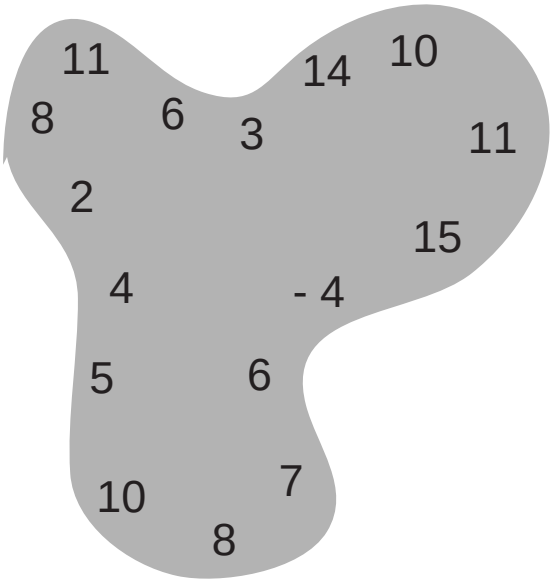


Figure 2.2.

If in a cycle only two numbers appear and these alternate, we may say that we have a **cycle of alternating numbers** or an **alternating cycle**. Figure 2.3 gives an example of an alternating cycle composed by 2's and 5's.

2	5	2	5	2	5
5					2
2					5
5					2
	2			5	
		5	2		

Figure 2.3.

An alternating cycle constitutes an example of a **periodic** cycle: it has period 2. The cycle in figure 2.4 has period 5: the five numbers always appear in the same order.

1	2	5	7	6	1
6					2
7					5
5					7
2					6
1	6	7	5	2	1

Figure 2.4.

In this book matrices will be analyzed that are composed of cycles. The analysis will present both surprising and attractive properties.

Chapter 3

SOME MATRICES OF ALTERNATING CYCLES

ABSTRACT

In chapter 3 some types of matrices composed of alternating cycles will be analyzed.

ACTIVITY

* Construct some matrices whose elements belong to alternating cycles. Do the matrices you constructed display more particularities?

3	4	3	4
4	1	5	3
3	5	1	4
4	3	4	3

Figure 3.1.

Let us observe the matrix in figure 3.1. Around the interior of the border we see a cycle of twelve elements, where the numbers 3 and 4 alternate (figure 3.2a). Around the center of the same matrix, one finds a cycle of four elements where the numbers 1 and 5 alternate (figure 3.2.b).

3	4	3	4
4			3
3			4
4	3	4	3

External cycle
a

1	5
5	1

Internal cycle
b

Figure 3.2.

Note also that the matrix displays a rotational symmetry: the matrix is invariant under a rotation of 180° about its center.

ACTIVITIES

- Construct more matrices of dimensions 4 by 4 that have alternating cycles of the same type as the matrix in figure 3.1.
- Add pairs of the constructed matrices and analyze the resulting matrices.
- Multiply pairs of matrices of the considered type and study the structure of the matrices that result from the multiplication.

Figure 3.3 shows us a second matrix with the same structure of alternating cycles as the first matrix in figure 3.1. When we add the two matrices we obtain the matrix on the right side, that has the same cyclic structure as the two added matrices: an external cycle of 5's and 3's and an internal cycle of (-1)'s and 8's. The matrix that results from the addition also displays rotational symmetry.

$$\begin{bmatrix} 3 & 4 & 3 & 4 \\ 4 & 1 & 5 & 3 \\ 3 & 5 & 1 & 4 \\ 4 & 3 & 4 & 3 \end{bmatrix} + \begin{bmatrix} 2 & -1 & 2 & -1 \\ -1 & -2 & 3 & 2 \\ 2 & 3 & -2 & -1 \\ -1 & 2 & -1 & 2 \end{bmatrix} = \begin{bmatrix} 5 & 3 & 5 & 3 \\ 3 & -1 & 8 & 5 \\ 5 & 8 & -1 & 3 \\ 3 & 5 & 3 & 5 \end{bmatrix}$$

Figure 3.3.

Figure 3.4 presents the multiplication of the same two matrices. This time the resulting matrix does not have an alternating external cycle. Nevertheless it displays rotational symmetry.

$$\begin{bmatrix} 3 & 4 & 3 & 4 \\ 4 & 1 & 5 & 3 \\ 3 & 5 & 1 & 4 \\ 4 & 3 & 4 & 3 \end{bmatrix} \text{x} \begin{bmatrix} 2 & -1 & 2 & -1 \\ -1 & -2 & 3 & 2 \\ 2 & 3 & -2 & -1 \\ -1 & 2 & -1 & 2 \end{bmatrix} = \begin{bmatrix} 4 & 6 & 8 & 10 \\ 14 & 15 & -2 & -1 \\ -1 & -2 & 15 & 14 \\ 10 & 8 & 6 & 4 \end{bmatrix}$$

Figure 3.4.

In general, we may conclude that the matrix resulting from the multiplication of two matrices with the considered cyclic structure does not have an alternating cycle. When multiplying the two matrices the structure of two alternating cycles is lost.

ACTIVITIES

- Consider the matrix in figure 3.5. Does it have a cyclic structure? Does it have alternating cycles?

2	1	3	-2
1	2	-2	3
2	1	3	-2
1	2	-2	3

Figure 3.5.

- Construct a matrix that has the same cyclic structure as the one in figure 3.5.
- Add the two matrices and analyze the resulting matrix.
- Multiply the two matrices and contemplate the structure of the matrix that results from the multiplication.

2	1	
1	2	
2	1	
1	2	

First cycle
a

	3	-2
	-2	3
	3	-2
	-2	3

Second cycle
b

Figure 3.6.

Figure 3.6 presents the two alternating cycles that, together, constitute the matrix of figure 3.5. The matrix of figure 3.7 has the same cyclic structure. We can add the two matrices and arrive at a matrix (figure 3.8) that has the same cyclic structure.

0	-3	1	5
-3	0	5	1
0	-3	1	5
-3	0	5	1

Figure 3.7.

2	1	3	-2		0	-3	1	5		2	-2	4	3
1	2	-2	3		-3	0	5	1		-2	2	3	4
2	1	3	-2	+	0	-3	1	5	=	2	-2	4	3
1	2	-2	3		-3	0	5	1		-2	2	3	4

Figure 3.8.

Let us multiply the two matrices, both the first times the second and the second times the first (figure 3.9). The resulting matrices are both distinct but present alternating cycles.

2	1	3	-2		0	-3	1	5		3	-15	1	24
1	2	-2	3		-3	0	5	1		-15	3	24	1
2	1	3	-2	x	0	-3	1	5	=	3	-15	1	24
1	2	-2	3		-3	0	5	1		-15	3	24	1

Figure 3.9 Continued on next page.

$$\begin{bmatrix} 0 & -3 & 1 & 5 \\ -3 & 0 & 5 & 1 \\ 0 & -3 & 1 & 5 \\ -3 & 0 & 5 & 1 \end{bmatrix} \times \begin{bmatrix} 2 & 1 & 3 & -2 \\ 1 & 2 & -2 & 3 \\ 2 & 1 & 3 & -2 \\ 1 & 2 & -2 & 3 \end{bmatrix} = \begin{bmatrix} 4 & 5 & -1 & 4 \\ 5 & 4 & 4 & -1 \\ 4 & 5 & -1 & 4 \\ 5 & 4 & 4 & -1 \end{bmatrix}$$

Figure 3.9.

Will it be that, in general, the matrices that result from multiplication have alternating cycles?

Let us consider the general case (see figure 3.10), where *a*, *b*, *c*, *d*, *e*, *f*, *g*, and *h* are arbitrary numbers.

$$\begin{bmatrix} a & b & c & d \\ b & a & d & c \\ a & b & c & d \\ b & a & d & c \end{bmatrix} \times \begin{bmatrix} e & f & g & h \\ f & e & h & g \\ e & f & g & h \\ f & e & h & g \end{bmatrix} = \quad ?$$

Figure 3.10.

Let us calculate, one by one, the elements of the resulting matrix. The first and the third row of the first matrix are equal and, therefore, the first and the third row of the resulting matrix are equal, etc. The element in the first row and the first column is equal to *ae+bf+ce+df*, etc. The matrix that results from the multiplication appears in figure 3.11.

ae+bf+ce+df	af+be+cf+de	ag+bh+cg+dh	ah+bg+ch+dg
be+af+de+cf	bf+ae+df+ce	bg+ah+dg+ch	bh+ag+dh+cg
ae+bf+ce+df	af+be+cf+de	ag+bh+cg+dh	ah+bg+ch+dg
be+af+de+cf	bf+ae+df+ce	bg+ah+dg+ch	bh+ag+dh+cg

Figure 3.11.

As we have

ae+bf+ce+df = bf+ae+df+ce, etc.

we may conclude that the matrix resulting from the multiplication has two alternating cycles, as we wanted to prove.

ACTIVITIES

- Invent other cyclic structures for matrices of dimensions 4 by 4.
- Which are the alternating cycles in the matrix of figure 3.12?

4	2	-3	1
2	1	4	-3
-3	4	1	2
1	-3	2	4

Figure 3.12.

- Construct more matrices that display alternating cycles.
- Add the matrices and analyze the resulting matrices.
- Multiply pairs of these matrices and observe the structure of the matrices resulting from the multiplication. Do alternating cycles appear?

Chapter 4

ALTERNATING CYCLE MATRICES OF DIMENSIONS 4 BY 4. FIRST PART

ABSTRACT

In chapter 4 a special type of matrices composed of alternating cycles will be analyzed. The matrices have dimensions 4 by 4 and are composed of two alternating cycles.

ACTIVITIES

- Observe the three matrices in figure 4.1. Do you observe some alternating cycles?

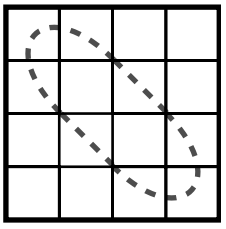
$$\begin{bmatrix} 3 & 4 & 0 & 0 \\ 4 & 0 & 3 & 0 \\ 0 & 3 & 0 & 4 \\ 0 & 0 & 4 & 3 \end{bmatrix} + \begin{bmatrix} 0 & 0 & 1 & 2 \\ 0 & 2 & 0 & 1 \\ 1 & 0 & 2 & 0 \\ 2 & 1 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 3 & 4 & 1 & 2 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 2 & 1 & 4 & 3 \end{bmatrix}$$

Figure 4.1.

- Construct some matrices that have the same cyclic structure as the third matrix in figure 4.1.
- Add pairs of the constructed matrices and analyze the resulting matrices.
- Multiply pairs of matrices of the considered type and study the structure of the matrices resulting from the multiplication.

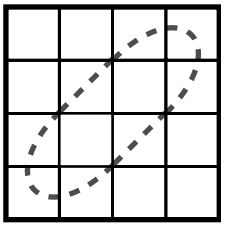
The first matrix of figure 4.1 presents the cycle composed by the numbers 3 and 4. The second matrix has an alternating cycle composed of 1's and 2's (see the scheme in figure 4.2).

3	4		
4		3	
	3		4
		4	3



a

		1	2
	2		1
1		2	
2	1		



b

Figure 4.2.

The third matrix is constituted by two ‘crossing’ alternating cycles (see the scheme in figure 4.3).

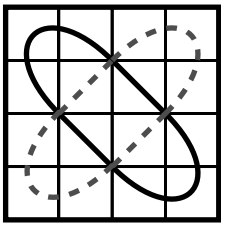


Figure 4.3.

The matrix in figure 4.4 has the same cyclic structure.

-1	5	-2	3
5	3	-1	-2
-2	-1	3	5
3	-2	5	-1

Figure 4.4.

When we add these two matrices, we obtain a matrix with the same cyclic structure, as can be seen in figure 4.5.

3	4	1	2
4	2	3	1
1	3	2	4
2	1	4	3

+

-1	5	-2	3
5	3	-1	-2
-2	-1	3	5
3	-2	5	-1

=

2	9	-1	5
9	5	2	-1
-1	2	5	9
5	-1	9	2

Figure 4.5.

Let us verify now what will happen if we multiply the two matrices, the first times the second and the second times the first (see figure 4.6).

$$\begin{bmatrix} 3 & 4 & 1 & 2 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 2 & 1 & 4 & 3 \end{bmatrix} \times \begin{bmatrix} -1 & 5 & -2 & 3 \\ 5 & 3 & -1 & -2 \\ -2 & -1 & 3 & 5 \\ 3 & -2 & 5 & -1 \end{bmatrix} = \begin{bmatrix} 21 & 22 & 3 & 4 \\ 3 & 21 & 4 & 22 \\ 22 & 4 & 21 & 3 \\ 4 & 3 & 22 & 21 \end{bmatrix}$$

$$\begin{bmatrix} -1 & 5 & -2 & 3 \\ 5 & 3 & -1 & -2 \\ -2 & -1 & 3 & 5 \\ 3 & -2 & 5 & -1 \end{bmatrix} \times \begin{bmatrix} 3 & 4 & 1 & 2 \\ 4 & 2 & 3 & 1 \\ 1 & 3 & 2 & 4 \\ 2 & 1 & 4 & 3 \end{bmatrix} = \begin{bmatrix} 21 & 3 & 22 & 4 \\ 22 & 21 & 4 & 3 \\ 3 & 4 & 21 & 22 \\ 4 & 22 & 3 & 21 \end{bmatrix}$$

Figure 4.6.

ACTIVITIES

- How can one characterize the matrices resulting from the multiplication? Do they have a cyclic structure? Are the two matrices very different?
- Construct other matrices with the same cyclic structure as the one schematically represented in figure 4.3, and multiply pairs of such matrices. What can be noted?

Observing the two matrices resulting from the multiplications in figure 4.6, we conclude that both their principal and secondary diagonals are equal and composed of the repetition of the number 21 in the case of the principal diagonal and of the repetition of the number 4 in the case of the secondary diagonal. Moreover, the two matrices resulting from the multiplications have an alternating cycle composed of 3's and 22's. The two cycles are different in their phases, where a 3 appears in the first cycle a 22 appears in the second and vice versa.

Is this phenomenon the consequence of the cyclic structure of the matrices under consideration that have to be multiplied or is it caused by the particular choice of the concrete numbers in the two matrices? Let us observe one more example (figure 4.7).

$$\begin{bmatrix} -2 & 3 & 4 & 1 \\ 3 & 1 & -2 & 4 \\ 4 & -2 & 1 & 3 \\ 1 & 4 & 3 & -2 \end{bmatrix} \times \begin{bmatrix} 4 & -3 & 2 & -1 \\ -3 & -1 & 4 & 2 \\ 2 & 4 & -1 & -3 \\ -1 & 2 & -3 & 4 \end{bmatrix} = \begin{bmatrix} -10 & 21 & 1 & 0 \\ 1 & -10 & 0 & 21 \\ 21 & 0 & -10 & 1 \\ 0 & 1 & 21 & -10 \end{bmatrix}$$

$$\begin{bmatrix} 4 & -3 & 2 & -1 \\ -3 & -1 & 4 & 2 \\ 2 & 4 & -1 & -3 \\ -1 & 2 & -3 & 4 \end{bmatrix} \times \begin{bmatrix} -2 & 3 & 4 & 1 \\ 3 & 1 & -2 & 4 \\ 4 & -2 & 1 & 3 \\ 1 & 4 & 3 & -2 \end{bmatrix} = \begin{bmatrix} -10 & 1 & 21 & 0 \\ 21 & -10 & 0 & 1 \\ 1 & 0 & -10 & 21 \\ 0 & 21 & 1 & -10 \end{bmatrix}$$

Figure 4.7.

Once again, the diagonals of the matrices that result from the multiplication are equal and constant, whereas two alternating cycles appear, one in each matrix, which differ one unit in phase. Figure 4.8 presents schematically the structure of the matrices resulting from the multiplication.

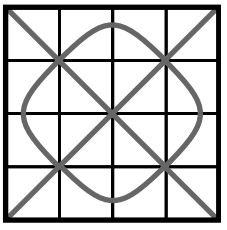


Figure 4.8.

ACTIVITIES

- Construct some more matrices with the same cyclic structure as the one schematically represented in figure 4.3, and multiply pairs of those matrices. Do the matrices resulting from the multiplication present the same structure as the one represented in figure 4.8?
- How can one prove that the same always happens when two matrices of the considered type are multiplied?

Let us consider the general case (see figure 4.9), where *a*, *b*, *c*, *d*, *e*, *f*, *g*, and *h* are any numbers.

$$\begin{bmatrix} a & b & c & d \\ b & d & a & c \\ c & a & d & b \\ d & c & b & a \end{bmatrix} \times \begin{bmatrix} e & f & g & h \\ f & h & e & g \\ g & e & h & f \\ h & g & f & e \end{bmatrix} = \quad ?$$

Figure 4.9.

ACTIVITIES

- Calculate, one by one, the elements of the resulting matrix. Compare the elements and analyze the structure of the matrix resulting from the multiplication.

The matrix resulting from the multiplication is presented in figure 4.10.

ae+bf+cg+dh	af+bh+ce+dg	ag+be+ch+df	ah+bg+cf+de
be+df+ag+ch	bf+dh+ae+cg	bg+de+ah+cf	bh+dg+af+ce
ce+af+dg+bh	cf+ah+de+bg	cg+ae+dh+bf	ch+ag+df+be
de+cf+bg+ah	df+ch+be+ag	dg+ce+bh+af	dh+cg+bf+ae

Figure 4.10.

The elements on the principal diagonal are equal as one can change the order of the parts to be added:

$$ae+bf+cg+dh = bf+dh+ae+cg = cg+ae+dh+bf = dh+cg+bf+ae.$$

Let $p = ae+bf+cg+dh$. Therefore we have on the principal diagonal of the matrix resulting from the multiplication of the two matrices in figure 4.9 four times the number p.

The elements on the secondary diagonal are equal as one may change the order of the parts to be added:

$$ah+bg+cf+de = bg+de+ah+cf = cf+ah+de+bg = de+cf+bg+ah$$

Let $q = ah+bg+cf+de$. By consequence, we have on the secondary diagonal of the matrix resulting from the multiplication of the matrices in figure 4.9 four times the number q.

We have still to analyze the cycle formed by the other elements (see figure 4.11).

p	**af+bh+ce+dg**	ag+be+ch+df	q
be+df+ag+ch	p	q	**bh+dg+af+ce**
ce+af+dg+bh	q	p	ch+ag+df+be
q	df+ch+be+ag	**dg+ce+bh+af**	p

Figure 4.11.

Let r be the element in the first row and the second column, that is, $r = af+bh+ce+dg$. As one may change the order of the four parts without changing the value of their sum, we have:

$$r = af+bh+ce+dg = bh+dg+af+ce = ce+af+dg+bh = dg+ce+bh+af.$$

Let s be the element in the first row and the third column, that is, $s = ag+be+ch+df$, we have in the same way:

$$s = ag+be+ch+df = be+df+ag+ch = ch+ag+df+be = df+ch+be+ag,$$

and, by consequence, the matrix resulting from the multiplication of the two matrices composed of two alternating cycles (figure 4.9) has **constant** diagonals, whose elements are equal to p and to q, respectively, and a cycle wherein the numbers r and s alternate (see figure 4.12).

p	r	s	q
s	p	q	r
r	q	p	s
q	s	r	p

Figure 4.12.

In other words, we have proved the following theorem:

Theorem 4.1.

The multiplication of two matrices of dimensions 4 by 4, composed of two alternating cycles, where *a*, *b*, *c*, *d*, *e*, *f*, *g* and *h* are any numbers, results in a matrix whose diagonals are constant and that has an alternating cycle (figure 4.13) where $p = ae+bf+cg+dh$, $q = ah+bg+cf+de$, $r = af+bh+ce+dg$ and $s = ag+be+ch+df$.

a	b	c	d
b	d	a	c
c	a	d	b
d	c	b	a

x

e	f	g	h
f	h	e	g
g	e	h	f
h	g	f	e

=

p	r	s	q
s	p	q	r
r	q	p	s
q	s	r	p

Figure 4.13.

ACTIVITY

- Show that, if one inverts the order of the matrices to be multiplied, the resulting matrix has the same form as in figure 4.12, with the unique difference that the order of the numbers *r* and *s* is inverted (see figure 4.14).

e	f	g	h
f	h	e	g
g	e	h	f
h	g	f	e

x

a	b	c	d
b	d	a	c
c	a	d	b
d	c	b	a

=

p	s	r	q
r	p	q	s
s	q	p	r
q	r	s	p

Figure 4.14.

Let us now observe another particularity.

When we introduced the number *p* by $p = ae+bf+cg+dh$, we had for the elements on the principal diagonal: $p = ae+bf+cg+dh = bf+dh+ae+cg = cg+ae+dh+bf = dh+cg+bf+ae$. In the first element, that is $ae+bf+cg+dh$, the part *ae* appears in the first place (1); in the second element, that is $bf+dh+ae+cg$, the part *ae* appears in the third place (3); in the third element, the part *ae* appears in the second place (2) and finally, in the fourth element, the part *ae* appears in the fourth place (4). Noting in the same way the successive positions of the parts *bf* (2, 1, 4, 3), *cg* and *dh*, we obtain the matrix shown on the right side in figure 4.15. Surprisingly, this matrix has the same structure as the one in figure 4.12

ae	1	3	2	4
bf	2	1	4	3
cg	3	4	1	2
dh	4	2	3	1

Figure 4.15.

ACTIVITIES

- Observe in the same way the order of the parts in the expressions for *q*, *r* and *s*, and construct the corresponding matrices. What particularity may be noted?
- Choose a matrix composed of two alternating cycles. What particularity may one observe when multiplying the matrix by itself? That is, what particularity does the square of the matrix display?

Figure 4.16 shows an example of the multiplication of a cycle matrix by itself. One notes that in the cycle the number -7 is repeated.

$$\begin{bmatrix} -2 & 3 & 4 & 1 \\ 3 & 1 & -2 & 4 \\ 4 & -2 & 1 & 3 \\ 1 & 4 & 3 & -2 \end{bmatrix} \times \begin{bmatrix} -2 & 3 & 4 & 1 \\ 3 & 1 & -2 & 4 \\ 4 & -2 & 1 & 3 \\ 1 & 4 & 3 & -2 \end{bmatrix} = \begin{bmatrix} 30 & -7 & -7 & 20 \\ -7 & 30 & 20 & -7 \\ -7 & 20 & 30 & -7 \\ 20 & -7 & -7 & 30 \end{bmatrix}$$

Figure 4.16.

Let us analyze this question in general. Observing figures 4.13 and 4.14, and taking into account the equality in figure 4.17 resulting from the calculation of the square of the matrix, one arrives at the equality in figure 4.18.

$$\begin{bmatrix} a & b & c & d \\ b & d & a & c \\ c & a & d & b \\ d & c & b & a \end{bmatrix} = \begin{bmatrix} e & f & g & h \\ f & h & e & g \\ g & e & h & f \\ h & g & f & e \end{bmatrix}$$

Figure 4.17.

$$\begin{bmatrix} p & r & S & q \\ s & p & q & r \\ r & q & P & s \\ q & s & r & p \end{bmatrix} = \begin{bmatrix} p & s & r & q \\ r & p & q & s \\ s & q & p & r \\ q & r & s & p \end{bmatrix}$$

Figure 4.18.

By consequence, we have $r = s$ and the general structure of the square of a matrix of two alternating cycles of dimensions 4 by 4 is the one in figure 4.19.

p	r	r	q
r	p	q	r
r	q	p	r
q	r	r	p

Figure 4.19.

ACTIVITY

- Choose various matrices that have the matrix structure displayed in figure 4.12 and multiply pairs of these matrices. Will it be possible to note some interesting particularity of the matrices resulting from the multiplication? If yes, experiment with some pairs and try to prove a general result.

The question formulated in the last activity will be the theme to be analyzed in the following chapter.

Chapter 5

ALTERNATING CYCLE MATRICES OF DIMENSIONS 4 BY 4. SECOND PART

ABSTRACT

In chapter 5, properties of the special type of matrices of dimensions 4 by 4 (figure 5.1), resulting from the multiplication of two matrices composed of two alternating cycles, will be analyzed.

p	r	s	q
s	p	q	r
r	q	p	s
q	s	r	p

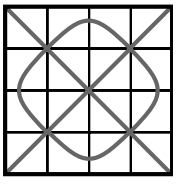


Figure 5.1.

ACTIVITIES

- Construct some matrices that have the structure presented in figure 5.1.
- Add pairs of the constructed matrices and analyze the resulting matrices.
- Multiply pairs of matrices of the considered type and study the structure of the matrices resulting from the multiplication.

Figure 5.2 displays two matrices with the cyclic structure presented in figure 5.1. When we add them, we obtain a matrix with the same structure.

$$\begin{bmatrix} 3 & -1 & 5 & 2 \\ 5 & 3 & 2 & -1 \\ -1 & 2 & 3 & 5 \\ 2 & 5 & -1 & 3 \end{bmatrix} + \begin{bmatrix} -2 & 1 & -3 & 4 \\ -3 & -2 & 4 & 1 \\ 1 & 4 & -2 & -3 \\ 4 & -3 & 1 & -2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 2 & 6 \\ 2 & 1 & 6 & 0 \\ 0 & 6 & 1 & 2 \\ 6 & 2 & 0 & 1 \end{bmatrix}$$

Figure 5.2.

Let us prove that this is true in general.

Let us consider two arbitrary matrices that present the cyclic structure in figure 5.1 and let us add them (figure 5.3). Immediately one sees that the matrix resulting from the addition has the same cyclic structure.

$$\begin{bmatrix} p & r & s & q \\ s & p & q & r \\ r & q & p & s \\ q & s & r & p \end{bmatrix} + \begin{bmatrix} t & v & w & u \\ w & t & u & v \\ v & u & t & w \\ u & w & v & t \end{bmatrix} = \begin{bmatrix} p+t & r+v & s+w & q+u \\ s+w & p+t & q+u & r+v \\ r+v & q+u & p+t & s+w \\ q+u & s+w & r+v & p+t \end{bmatrix}$$

Figure 5.3.

Let us verify now what will happen if we multiply the two matrices in figure 5.2, both the first times the second and the second times the first (see figure 5.3).

$$\begin{bmatrix} 3 & -1 & 5 & 2 \\ 5 & 3 & 2 & -1 \\ -1 & 2 & 3 & 5 \\ 2 & 5 & -1 & 3 \end{bmatrix} \text{x} \begin{bmatrix} -2 & 1 & -3 & 4 \\ -3 & -2 & 4 & 1 \\ 1 & 4 & -2 & -3 \\ 4 & -3 & 1 & -2 \end{bmatrix} = \begin{bmatrix} 10 & 19 & -21 & -8 \\ -21 & 10 & -8 & 19 \\ 19 & -8 & 10 & -21 \\ -8 & -21 & 19 & 10 \end{bmatrix}$$

$$\begin{bmatrix} -2 & 1 & -3 & 4 \\ -3 & -2 & 4 & 1 \\ 1 & 4 & -2 & -3 \\ 4 & -3 & 1 & -2 \end{bmatrix} \text{x} \begin{bmatrix} 3 & -1 & 5 & 2 \\ 5 & 3 & 2 & -1 \\ -1 & 2 & 3 & 5 \\ 2 & 5 & -1 & 3 \end{bmatrix} = \begin{bmatrix} 10 & 19 & -21 & -8 \\ -21 & 10 & -8 & 19 \\ 19 & -8 & 10 & -21 \\ -8 & -21 & 19 & 10 \end{bmatrix}$$

Figure 5.3.

Surprisingly, the matrices resulting from the two multiplications are equal. They also present the same cyclic structure as the two initial matrices. Is this an exceptional case? Let us experiment with two more matrices (figure 5.4).

$$\begin{bmatrix} 4 & 2 & 5 & 1 \\ 5 & 4 & 1 & 2 \\ 2 & 1 & 4 & 5 \\ 1 & 5 & 2 & 4 \end{bmatrix} \text{x} \begin{bmatrix} 3 & -2 & -3 & 5 \\ -3 & 3 & 5 & -2 \\ -2 & 5 & 3 & -3 \\ 5 & -3 & -2 & 3 \end{bmatrix} = \begin{bmatrix} 1 & 20 & 11 & 4 \\ 11 & 1 & 4 & 20 \\ 20 & 4 & 1 & 11 \\ 4 & 11 & 20 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 3 & -2 & -3 & 5 \\ -3 & 3 & 5 & -2 \\ -2 & 5 & 3 & -3 \\ 5 & -3 & -2 & 3 \end{bmatrix} \text{x} \begin{bmatrix} 4 & 2 & 5 & 1 \\ 5 & 4 & 1 & 2 \\ 2 & 1 & 4 & 5 \\ 1 & 5 & 2 & 4 \end{bmatrix} = \begin{bmatrix} 1 & 20 & 11 & 4 \\ 11 & 1 & 4 & 20 \\ 20 & 4 & 1 & 11 \\ 4 & 11 & 20 & 1 \end{bmatrix}$$

Figure 5.4.

Once more, the matrices resulting from the two multiplications are equal and display the same structure as the two initial matrices. It appears that the moment to analyze the general case has come.

Let us consider the general case (see figure 5.5), where *p*, *q*, *r*, *s*, *t*, *u*, *v*, and *w* are any numbers.

$$\begin{bmatrix} p & r & s & q \\ s & p & q & r \\ r & q & p & s \\ q & s & r & p \end{bmatrix} \times \begin{bmatrix} t & v & w & u \\ w & t & u & v \\ v & u & t & w \\ u & w & v & t \end{bmatrix} = ?$$

Figure 5.5.

ACTIVITIES

- Calculate, one by one, the elements of the resulting matrix. Compare its elements and analyze the structure of the matrix resulting from the multiplication.

The matrix resulting from the multiplication is presented in figure 5.6.

pt+rw+sv+qu	pv+rt+su+qw	pw+ru+st+qv	pu+rv+sw+qt
st+pw+qv+ru	sv+pt+qu+rw	sw+pu+qt+rv	su+pv+qw+rt
rt+qw+pv+su	rv+qt+pu+sw	rw+qu+pt+sv	ru+qv+pw+st
qt+sw+rv+pu	qv+st+ru+pw	qw+su+rt+pv	qu+sv+rw+pt

Figure 5.6.

The elements in the principal diagonal are equal:

$$pt+rw+sv+qu = sv+pt+qu+rw = rw+qu+pt+sv = qu+sv+rw+pt.$$

The elements in the secondary diagonal are equal:

$$pu+rv+sw+qt = sw+pu+qt+rv = rv+qt+pu+sw = qt+sw+rv+pu.$$

The other elements of the matrix constitute an alternating cycle, as we have:

$$pv+rt+su+qw = su+pv+qw+rt = rt+qw+pv+su = qw+su+rt+pv$$

and

$$pw+ru+st+qv = st+pw+qv+ru = ru+qv+pw+st = qv+st+ru+pw.$$

In other words, we proved the following theorem:

Theorem 5.1.

The multiplication of two matrices of dimensions 4 by 4, composed of two constant diagonals and an alternating cycle, where *p*, *q*, *r*, *s*, *t*, *u v* and *w* are any numbers, results in a matrix that has the same structure (figure 5.7) where $a = pt+rw+sv+qu$, $b = pu+rv+sw+qt$, $c = pv+rt+su+qw$ and $d = pw+ru+st+qv$.

$$\begin{bmatrix} p & r & s & q \\ s & p & q & r \\ r & q & p & s \\ q & s & r & p \end{bmatrix} \times \begin{bmatrix} t & v & w & u \\ w & t & u & v \\ v & u & t & w \\ u & w & v & t \end{bmatrix} = \begin{bmatrix} a & c & d & b \\ d & a & b & c \\ c & b & a & d \\ b & d & c & a \end{bmatrix}$$

Figure 5.7.

Let us now invert the order of the two matrices to be multiplied (figure 5.8).

$$\begin{bmatrix} t & v & w & u \\ w & t & u & v \\ v & u & t & w \\ u & w & v & t \end{bmatrix} \times \begin{bmatrix} p & r & s & q \\ s & p & q & r \\ r & q & p & s \\ q & s & r & p \end{bmatrix} = \quad ?$$

Figure 5.8.

In agreement with the theorem that we just proved, we know already that the structure of the matrix resulting from the multiplication is the same as that of the matrices to be multiplied. Therefore we need to calculate only the four elements of the first row in order to know all elements of the matrix resulting from the multiplication.

The first element of the first row is equal to $tp+vs+wr+uq$. Let us compare it with the first element of the first row of the other matrix (figure 5.6): $pt+rw+sv+qu$. The two elements are equal as the order of the addition of the parts is free and the multiplication of the numbers is commutative ($pt = tp$, $vs = sv$, $wr = rw$, $uq = qu$):

$$\boldsymbol{tp+vs+wr+uq} = tp+wr+vs+uq = \boldsymbol{pt+rw+sv+qu.}$$

In the same way one shows that the other elements of the first row of the matrix resulting from the second multiplication (figure 5.8) are equal to the corresponding elements of the matrix resulting from the first multiplication (figure 5.7).

Let us observe now another particularity.

When we introduce the number *a* through $a = pt+rw+sv+qu$, we had for the elements in the principal diagonal $a = pt+rw+sv+qu = sv+pt+qu+rw = rw+qu+pt+sv = qu+sv+rw+pt$. In the first element, $pt+rw+sv+qu$, the part *pt* appears in the first place (1); in the second element, that is $sv+pt+qu+rw$, the part *pt* appears in the second place (2); in the third element, the part *pt* appears in the third place (3) and finally, in the fourth element, the part *pt* appears in the fourth place (4). Observing in the same way the successive positions of the parts *rw* (2, 4, 1, 3), *sv* and *qu*, we obtain the matrix shown at the right side in figure 5.9. Surprisingly, this matrix has the structure of a matrix of two alternating cycles, analyzed in chapter 4 (see the scheme in figure 5.10 [= figure 4.3]).

pt	1	2	3	4
rw	2	4	1	3
sv	3	1	4	2
qu	4	3	2	1

Figure 5.9.

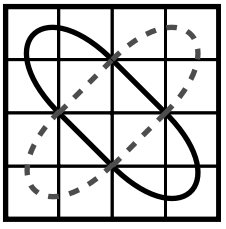


Figure 5.10.

ACTIVITY

- Analyze in the same way the order of the parts in the elements equal to *b*, *c*, and *d*, respectively. Will the same surprising particularity be observed?

We may conclude that the matrices of the two types, characterized by the structures of figures 5.1 ( ) and 5.10 ( ) are very much related. If we do something with a matrix of one type we arrive easily at a matrix of the other type and vice-versa.

ACTIVITIES

- Add matrices of the two distinct types. Does something special happen?
- Multiply matrices of the two distinct types. Does something special happen? Will it be possible to discover some multiplication rule?

Chapter 6

ALTERNATING CYCLE MATRICES OF DIMENSIONS 4 BY 4. THIRD PART: MULTIPLICATION TABLE

ABSTRACT

In chapter 6 we shall analyze some more properties of alternating cycle matrices of dimensions 4 by 4 and, in particular, construct the multiplication table of the two types of alternating cycle matrices (and).

In the last activity of the previous chapter we posed the question of the addition and of the multiplication of matrices belonging to the two types of alternating cycle matrices.

Figure 6.1 presents an example of the addition of two matrices of different cycle structures (and).

$$\begin{bmatrix} 3 & -2 & -3 & 7 \\ -3 & 3 & 7 & -2 \\ -2 & 7 & 3 & -3 \\ 7 & -3 & -2 & 3 \end{bmatrix} + \begin{bmatrix} 2 & 3 & 6 & -1 \\ 3 & -1 & 2 & 6 \\ 6 & 2 & -1 & 3 \\ -1 & 6 & 3 & 2 \end{bmatrix} = \begin{bmatrix} 5 & 1 & 3 & 6 \\ 0 & 2 & 9 & 4 \\ 4 & 9 & 2 & 0 \\ 6 & 3 & 1 & 5 \end{bmatrix}$$

Figure 6.1.

The matrix resulting from the addition does not have an alternating cyclic structure. Nevertheless it presents a particularity. It displays rotational symmetry: it is invariant under a rotation of 180° about the center of the matrix. Figure 6.2 illustrates some pairs of positions corresponding under this rotation. The numbers in these corresponding positions are equal.

$$\begin{matrix} 5 & 1 & \boxed{3} & 6 \\ 0 & 2 & 9 & 4 \\ 4 & 9 & 2 & 0 \\ 6 & \boxed{3} & 1 & 5 \end{matrix} \qquad \begin{matrix} 5 & 1 & 3 & 6 \\ \boxed{0} & 2 & 9 & 4 \\ 4 & 9 & 2 & \boxed{0} \\ 6 & 3 & 1 & 5 \end{matrix} \qquad \begin{matrix} 5 & 1 & 3 & \boxed{6} \\ 0 & 2 & 9 & 4 \\ 4 & 9 & 2 & 0 \\ \boxed{6} & 3 & 1 & 5 \end{matrix}$$

Figure 6.2.

Are we dealing with a particular case or will rotational symmetry always occur? Let us analyze the situation. Figure 6.3 shows the addition of two matrices of distinct cycle structures, and respectively, where *p*, *q*, *r*, *s*, *a*, *b*, *c* and *d* represent any numbers.

p	r	s	q
s	p	q	r
r	q	p	s
q	s	r	p

+

a	b	c	d
b	d	a	c
c	a	d	b
d	c	b	a

=

p+a	r+b	s+c	q+d
s+b	p+d	q+a	r+c
r+c	q+a	p+d	s+b
q+d	s+c	r+b	p+a

Figure 6.3.

The matrix resulting from the addition always displays rotational symmetry of 180°, which is independent from the value of the numbers.

ACTIVITIES

- Under which conditions does the matrix resulting from the addition have a structure of two alternating cycles ?
- Under which conditions does the matrix resulting from the addition have a structure of an alternating cycle and two constant diagonals ?

ANSWERS

- If *r*=*s* and *p*=*q* (figure 6.4a), the matrix resulting from the addition has the structure of two alternating cycles
- If *a*=*d* and *b*=*c* (figure 6.4b), the matrix resulting from the addition has the structure of an alternating cycle and of two constant diagonals .

p	r	r	p
r	p	p	r
r	p	p	r
p	r	r	p

A

a	b	b	a
b	a	a	b
b	a	a	b
a	b	b	a

b

Figure 6.4.

The two matrices in figures 6.4a and 6.4b have the same structure. Matrices of this type present both the structure of two alternating cycles as the structure of two constant diagonals and an alternating cycle where – what is very particular – in the last cycle the same number is repeated, that is, it alternates with itself.

ACTIVITY

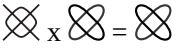
- Multiply some pairs of alternating cycle matrices, where one has structure and the other structure . Do the matrices resulting from the multiplication present some particularity?

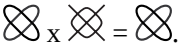
$$\begin{bmatrix} 3 & -2 & -3 & 7 \\ -3 & 3 & 7 & -2 \\ -2 & 7 & 3 & -3 \\ 7 & -3 & -2 & 3 \end{bmatrix} \times \begin{bmatrix} 2 & 3 & 6 & -1 \\ 3 & -1 & 2 & 6 \\ 6 & 2 & -1 & 3 \\ -1 & 6 & 3 & 2 \end{bmatrix} = \begin{bmatrix} -25 & 47 & 38 & -10 \\ 47 & -10 & -25 & 38 \\ 38 & -25 & -10 & 47 \\ -10 & 38 & 47 & -25 \end{bmatrix}$$

$$\begin{bmatrix} 2 & 3 & 6 & -1 \\ 3 & -1 & 2 & 6 \\ 6 & 2 & -1 & 3 \\ -1 & 6 & 3 & 2 \end{bmatrix} \times \begin{bmatrix} 3 & -2 & -3 & 7 \\ -3 & 3 & 7 & -2 \\ -2 & 7 & 3 & -3 \\ 7 & -3 & -2 & 3 \end{bmatrix} = \begin{bmatrix} -22 & 50 & 35 & -13 \\ 50 & -13 & -22 & 35 \\ 35 & -22 & -13 & 50 \\ -13 & 35 & 50 & -22 \end{bmatrix}$$

Figure 6.5.

Figure 6.5 shows the multiplication of the two matrices considered earlier in the example of addition (figure 6.1). Both products are shown, the product of the first times the second, and the product of the second times the first. The two matrices resulting from the multiplication differ. However, observing the distribution of numbers in the product matrices, we can see that both matrices have the structure of two alternating cycles . We may say that we have in this example as far as the structures of the involved matrices are concerned, the following:





Is this a generalization or will it only result from particular numbers that appear as elements of those two matrices?

ACTIVITY

- Analyze, in general, the multiplication of alternating cycle matrices where one presents the structure and the other the structure .

Let us analyze in the first place the situation where the first matrix has the structure and the second matrix the structure (see figure 6.6), where *p*, *q*, *r*, *s*, *a*, *b*, *c* and *d* denote any numbers.

p	r	s	q
s	p	q	r
r	q	p	s
q	s	r	p

x

a	b	c	d
b	d	a	c
c	a	d	b
d	c	b	a

= ?

Figure 6.6.

ACTIVITIES

- Calculate, one by one, the elements of the resulting matrix. Compare the elements and analyze the structure of the matrix resulting from the multiplication.

The matrix resulting from the multiplication is presented in figure 6.7.

pa+rb+sc+qd	pb+rd+sa+qc	pc+ra+sd+qb	pd+rc+sb+qa
sa+pb+qc+rd	sb+pd+qa+rc	sc+pa+qd+rb	sd+pc+qb+ra
ra+qb+pc+sd	rb+qd+pa+sc	rc+qa+pd+sb	rd+qc+pb+sa
qd+sc+rb+pa	qb+sd+ra+pc	qc+sa+rd+pb	qd+sc+rb+pa

Figure 6.7.

Let us observe the elements of the first cycle (see figure 6.8).

pa+rb+sc+qd	pb+rd+sa+qc		
sa+pb+qc+rd		**sc+pa+qd+rb**	
	rb+qd+pa+sc		rd+qc+pb+sa
		qc+sa+rd+pb	**qd+sc+rb+pa**

Figure 6.8.

The elements represented in bold are all equal:

$pa+rb+sc+qd = sc+pa+qd+rb = rb+qd+pa+sc = qd+sc+rb+pa$ (= f)

as they are the sum of the same parts *pa*, *rb*, *sc* and *qd* added in different orders.

For the same reason the other elements of the same cycle are also equal:

$pb+rd+sa+qc = sa+pb+qc+rd = rd+qc+pb+sa = qc+sa+rd+pb$ (= g).

We are dealing with an alternating cycle: *f, g, f, g, …*

Let us observe now the elements of the second cycle (figure 6.9).

		pc+ra+sd+qb	pd+rc+sb+qa
	sb+pd+qa+rc		**sd+pc+qb+ra**
ra+qb+pc+sd		rc+qa+pd+sb	
qd+sc+rb+pa	**qb+sd+ra+pc**		

Figure 6.9.

On the basis of the same arguments, we see that the bold elements are equal:

$$pc+ra+sd+qb = sd+pc+qb+ra = ra+qb+pc+sd = qb+sd+ra+pc \ (= h),$$

just as the other elements are equal between them:

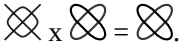
$$pd+rc+sb+qa = sb+pd+qa+rc = rc+qa+pd+sb = qd+sc+rb+pa \ (= i)$$

By consequence, the second cycle is equally an alternating cycle *h*, *i*, *h*, *i*, … and the structure of the matrix resulting from the multiplication of the two matrices has itself the structure of two alternating cycles (figure 6.10).

f	g	h	i
g	i	f	h
h	f	i	g
i	h	g	f

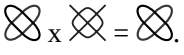
Figure 6.10.

We have proved that the following is valid under multiplication for matrices consisting of the given structures:

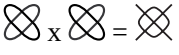


ACTIVITY

- Show that for alternating cycle matrices with the given structures the following holds under multiplication:



When we summarize the main conclusions of chapters 4 and 5, the following is valid for the structures of alternating cycle matrices when they are multiplied:



and

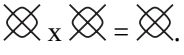


Figure 6.11 joins the four results in the multiplication table for alternating cycle matrices of dimensions 4 by 4.

x		

Figure 6.11.

ACTIVITY

- Compare the multiplication table for alternating cycle matrices of dimensions 4 by 4 with other multiplication tables that the reader already knows. Do you observe some similarity with one of those tables?

Figure 6.12 shows us the table (of signs) of multiplication of numbers: ‘negative’ times ‘negative’ gives ‘positive’, etc.

x	—	+
—	+	—
+	—	+

Figure 6.12.

The two tables are very similar. This led me to introduce names for the two types of alternating cycle matrices that allows us to remember the respective multiplication table. We may call the alternating cycle matrices with the structure negative and the alternating cycle matrices with the structure positive.

In other words,

1. **negative** alternating cycle matrices of dimensions 4 by 4 have the structure of two alternating cycles;
2. **positive** alternating cycle matrices of dimensions 4 by 4 have the structure of two constant diagonals and one alternating cycle.

ACTIVITIES

- Will it be possible to picture alternating cycle matrices of dimensions 6 by 6? When possible, analyze if it is possible to distinguish them into two types, negative and positive.
- Try to construct negative and positive alternating cycle matrices of dimensions 8 by 8. Experiment with multiplying pairs of those matrices and analyze the structure of the products obtained in this way. Does there exist some parallelism with the situation we have already studied of negative and positive alternating cycle matrices of dimensions 4 by 4?

Chapter 7

MORE PROPERTIES OF ALTERNATING CYCLE MATRICES OF DIMENSIONS 4 BY 4

ABSTRACT

In chapter 7 some more properties of alternating cycle matrices of dimensions 4 by 4 will be analyzed, in particular, symmetries and other invariants.

ACTIVITIES

- Construct some negative alternating cycle matrices of dimensions 4 by 4 ( ), and calculate the sums of the elements in their rows and columns. What do you observe? Will this always happen?
- Construct some positive alternating cycle matrices of dimensions 4 by 4 ( ), and calculate the sums of the elements in their rows and columns. What do you observe? Will this always happen?

Let us consider the negative alternating cycle matrix in figure 7.1.

2	3	6	-1
3	-1	2	6
6	2	-1	3
-1	6	3	2

Figure 7.1.

The sum of the elements of the first row is equal to 10, like the sum of the elements of the second row, of the third and of the fourth. The sum of the elements of the first column is equal to 10, etc.

a	b	c	d
b	d	a	c
c	a	d	b
d	c	b	a

Figure 7.2.

Let *a*, *b*, *c* and *d* denote any numbers. We observe that the sum of the elements of any of the rows or of any of the columns of the negative alternating cycle matrix in figure 7.2 is always equal to $a+b+c+d$. Indeed, each of the numbers *a*, *b*, *c*, and *d* appears exactly once in each row and in each column.

In other words, negative alternating cycle matrices of dimensions 4 by 4 are **Latin squares**.

Latin squares

A Latin square is a numerical square in which each number appears once in each column and once in each row.

In the same way it may be shown that positive alternating cycle matrices are also Latin squares. Letting *p*, *q*, *r* and *s* represent any numbers, the sum of the elements of any row or of any column of the positive alternating cycle matrix in figure 7.3, is equal to $p+q+r+s$.

p	r	s	q
s	p	q	r
r	q	p	s
q	s	r	p

Figure 7.3.

ACTIVITY

- Construct some Latin squares of dimensions 4 by 4 that are not alternating cycle matrices.

Let us now analyze the alternating cycle matrices, of dimensions 4 by 4, from the point of view of the symmetries they present. Can these matrices display rotational symmetry? Can they have axes of symmetry?

ACTIVITIES

- Analyze which symmetries negative alternating cycle matrices of dimensions 4 by 4 have ( )?

- Find the symmetries that positive alternating cycle matrices of dimensions 4 by 4 display ( )?
- Do alternating cycle matrices exist that are the same time negative and positive? If they exist, which are the symmetries that those matrices present?

Consider any negative alternating cycle matrix ( ), of dimensions 4 by 4 (figure 7.4).

a	b	c	d
b	d	a	c
c	a	d	b
d	c	b	a

Figure 7.4.

We may see that the first row (*a*, *b*, *c*, *d*) is equal to the first column; the second row (*b*, *d*, *a*, *c*) is equal to second column; the third row (*c*, *a*, *d*, *b*) is equal to the third column; and the fourth row (*d*, *c*, *b*, *a*) is equal to the fourth column. In other words, the principal diagonal of the matrix constitutes a symmetry axis of the matrix. The numbers in positions symmetrical to the principal diagonal are equal, as in figure 7.5.

a	*b*	*c*	*d*
b	*d*	*a*	$\boxed{c}$
c	*a*	*d*	*b*
d	$\boxed{c}$	*b*	*a*

a	*b*	*c*	*d*
b	*d*	$\boxed{a}$	*c*
c	$\boxed{a}$	*d*	*b*
d	*c*	*b*	*a*

a	*b*	*c*	$\boxed{d}$
b	*d*	*a*	*c*
c	*a*	*d*	*b*
$\boxed{d}$	*c*	*b*	*a*

Figure 7.5.

The first row read from right to left (*d*, *c*, *b*, *a*) is equal to the fourth column read from the top to the bottom; the second row read from right to left (*c*, *a*, *d*, *b*) is equal to the third column read from the top to the bottom; the third row read from right to the left (*b*, *d*, *a*, *c*) is equal to the second column read from the top to the bottom; and the fourth row read from right to left (*a*, *b*, *c*, *d*) is equal to the first column read from the top to the bottom. By consequence, the secondary diagonal of the matrix constitutes a symmetry axis too: the numbers in places symmetrical to the secondary diagonal are equal, as in figure 7.6.

a	b	$\boxed{c}$	d
b	d	a	$\boxed{c}$
c	a	d	b
d	c	b	a

a	$\boxed{b}$	c	d
b	d	a	c
c	a	d	$\boxed{b}$
d	c	b	a

$\boxed{a}$	b	c	d
b	d	a	c
c	a	d	b
d	c	b	$\boxed{a}$

Figure 7.6.

In figure 7.7 the two symmetry axes of negative alternating cycle matrices of dimensions 4 by 4 are presented.

a	b	c	d
b	d	a	c
c	a	d	b
d	c	b	a

Figure 7.7.

If we read the fourth row from right to left, we have (*a*, *b*, *c*, *d*), which is the same as reading the first row from left to right. Reading the third row from right to left, we have (*b*, *d*, *a*, *c*), which is the same as reading the second row from left to right. In the same way, reading the first column from the top to the bottom we have (*a*, *b*, *c*, *d*), which is the same as reading the last column from the bottom to the top, etc. This means that the matrix has a rotational symmetry of 180°: the numbers in positions symmetrical to the center of the matrix are equal. Figure 7.8 illustrates some examples.

a	b	c	d
$\boxed{b}$	d	a	c
c	a	d	$\boxed{b}$
d	c	b	a

a	b	$\boxed{c}$	d
b	d	a	c
c	a	d	b
d	$\boxed{c}$	b	a

a	b	c	d
b	d	$\boxed{a}$	c
c	$\boxed{a}$	d	b
d	c	b	a

Figure 7.8.

After this analysis of the symmetries of negative alternating cycle matrices of dimensions 4 by 4, the reader is invited to produce a similar analysis of positive alternating cycle matrices of the same dimensions.

ACTIVITIES

- Find the symmetries of positive alternating cycle matrices of dimensions 4 by 4 (⊗).
- Do alternating cycle matrices exist that are simultaneously negative and positive? If yes, what symmetries do those matrices have?

Now consider any positive alternating cycle matrix (⊗) of dimensions 4 by 4 (figure 7.9).

p	r	s	q
s	p	q	r
r	q	p	s
q	s	r	p

Figure 7.9.

In general, this matrix does not have symmetry axes. Only in the case that $r = s$ do the diagonals of the matrix constitute symmetry axes. This positive alternating cycle matrix always has a rotational symmetry of 180°: the numbers in positions symmetrical to the center of the matrix are equal.

q	s	r	p
r	q	p	s
s	p	q	r
p	r	s	q

a

q	s	r	p
r	q	p	s
s	p	q	r
p	r	s	q

b

Figure 7.10.

If we reflect a positive alternating cycle matrix (figure 7.9) across the horizontal line in the middle (figure 7.10a) or the vertical middle line (figure 7.10b), we obtain the same matrix that differs from the original matrix in the inversion of the positions. On one side, the numbers p and q are inverted and on the other side r and s. When this occurs, one says that the horizontal and vertical middle lines of the matrix are **anti-symmetry axes** of the matrix.

ACTIVITY

- Determine if negative alternating cycle matrices of dimensions 4 by 4 have anti-symmetry axes (figure 7.11).

a	b	c	d
b	d	a	c
c	a	d	b
d	c	b	a

d	c	b	a
c	a	d	b
b	d	a	c
a	b	c	d

d	c	b	a
c	a	d	b
b	d	a	c
a	b	c	d

Figure 7.11.

In the previous chapter we saw that there exist alternating cycle matrices of dimensions 4 by 4 that are simultaneously positive and negative. The general form of those matrices is presented in figure 7.12.

p	r	r	p
r	p	p	r
r	p	p	r
p	r	r	p

Figure 7.12.

The horizontal and vertical middle lines constitute, together with the diagonals, symmetry axes. In addition the matrices of this form display rotational symmetry of 90°. Rotating the matrix 90° about the center of the matrix places each number where an identical value was. Figure 7.13 illustrates some examples of corresponding places.

p	r	r	p
r	p	p	r
r	p	p	r
p	r	r	p

p	r	r	p
r	p	p	r
r	p	p	r
p	r	r	p

p	r	r	p
r	p	p	r
r	p	p	r
p	r	r	p

Figure 7.13.

ACTIVITY

- Analyze the structure and the properties of the matrices of dimensions 6 by 6, presented in figure 7.14.

2	3	-1	5	4	6
3	5	2	6	-1	4
-1	2	4	3	6	5
5	6	3	4	2	-1
4	-1	6	2	5	3
6	4	5	-1	3	2

3	4	1	-5	0	2
1	3	0	4	2	-5
4	-5	3	2	1	0
0	1	2	3	-5	4
-5	2	4	0	3	1
2	0	-5	1	4	3

Figure 7.14.

Chapter 8

NEGATIVE ALTERNATING CYCLE MATRICES OF DIMENSIONS 6 BY 6

ABSTRACT

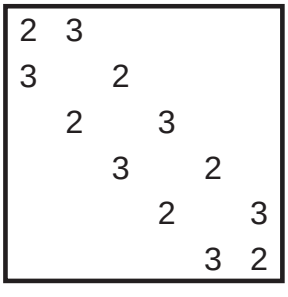
In chapter 8 negative alternating cycle matrices of dimensions 6 by 6 will be introduced and some of their properties will be analyzed.

At the end of the previous chapter two matrices were presented and the question was posed if their structure had some particularity. Figure 8.1 reproduces the first matrix.

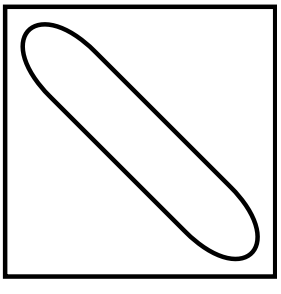
$$\mathbf{A} = \begin{bmatrix} 2 & 3 & -1 & 5 & 4 & 6 \\ 3 & 5 & 2 & 6 & -1 & 4 \\ -1 & 2 & 4 & 3 & 6 & 5 \\ 5 & 6 & 3 & 4 & 2 & -1 \\ 4 & -1 & 6 & 2 & 5 & 3 \\ 6 & 4 & 5 & -1 & 3 & 2 \end{bmatrix}$$

Figure 8.1.

When we observe the distribution of the numbers in matrix **A**, we can verify the existence of three alternating cycles (figure 8.2).



a



b

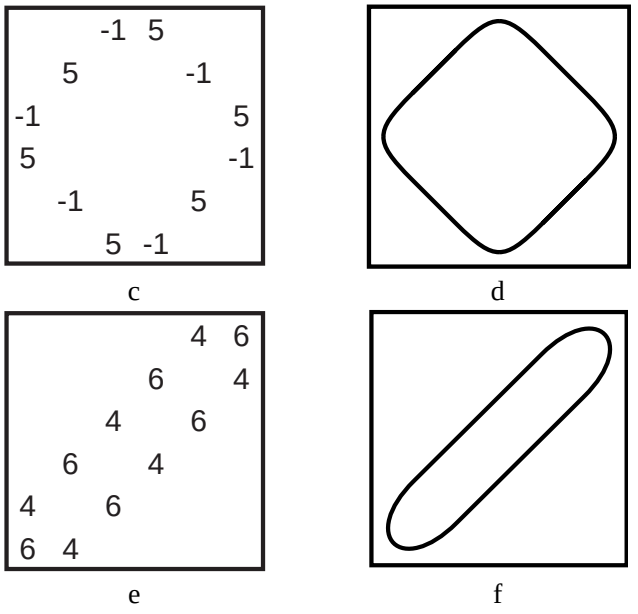


Figure 8.2.

Figure 8.3 illustrates the matrix and the three cycles at the same time.

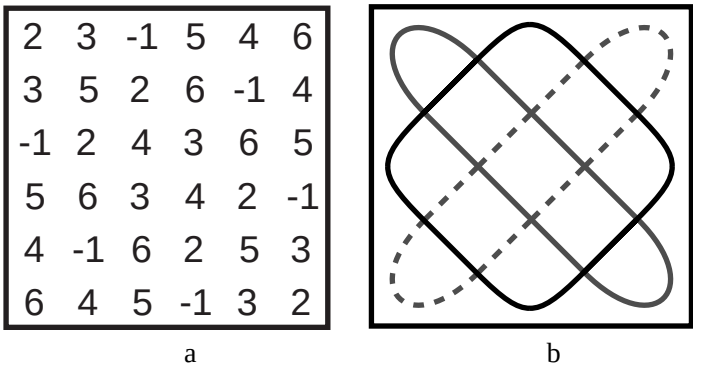


Figure 8.3.

Let us observe the first cycle (figure 8.2 a). In the first row a 2 and a 3 appear once. In the second row from left to right, a 3 and then a 2 appears once. In the third row a 2 and then a 3 each appear once. The pattern continues. A similar pattern can be noted with the columns. In the first column a 2 and then a 3 appears once. In the second column from top to bottom a 3 and then a 2 each appear once. In the third column first a 2 and then a 3 each appear once. This pattern also continues. In other words, the two numbers that alternate in the first cycle, that is the 2 and 3, appear once in each row and once in each column. From one row to the next, the order of the two numbers is inverted; in the same way, from one column to the next, the order of the two numbers is inverted.

The same pattern occurs with the numbers -1 and 5 of the second cycle (see figure 8.2c) and with the numbers 4 and 6 of the third cycle (figure 8.2e).

By consequence, we can conclude that, like we had seen in the case of cycle matrices of dimensions 4 by 4, each of the six numbers appears exactly once in each row and once in each column. The considered matrix is a Latin square and the sum of the numbers in each row and in each column is constant (= 19).

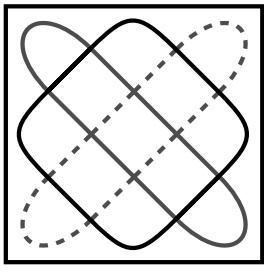
ACTIVITY

- Construct other matrices of dimensions 6 by 6 that have the same cyclic structure as that of matrix **A** (figure 8.3b). Will all matrices thus constructed be Latin squares?

Let *a*, *b*, *c*, *d*, *e*, and *f* be any six numbers. With them we can construct an alternating cycle matrix with the same structure as that of matrix **A**, as figure 8.4 illustrates.

a	b	c	d	e	f
b	d	a	f	c	e
c	a	e	b	f	d
d	f	b	e	a	c
e	c	f	a	d	b
f	e	d	c	b	a

a



b

Figure 8.4.

The sum of the numbers in each row and in each column is equal to $a+b+c+d+e+f$. In the case that the six numbers are different, the matrix is a Latin square.

Taking into consideration that the structure of the matrix is similar to the structure of a negative alternating cycle matrix of dimensions 4 by 4 ( ), we can call it a negative alternating cycle matrix of dimensions 6 by 6.

ACTIVITY

- What are the symmetries of negative alternating cycle matrices of dimensions 6 by 6?

Reading the numbers in the first row from left to right we have the sequence (*a*, *b*, *c*, *d*, *e*, *f*), just as when we read the numbers of the last row from right to left. Reading the numbers in the second row from left to right we have the sequence (*b*, *d*, *a*, *f*, *c*, *e*), the same as when we read the numbers of the fifth row from right to left. Reading the numbers in the third row from left to right we have the sequence (*c*, *a*, *e*, *b*, *f*, *d*), the same as when we read the numbers of the fourth row from right to left. In other words, negative alternating cycle

matrices of dimensions 6 by 6 present a rotational symmetry of 180°, just like negative alternating cycle matrices of dimensions 4 by 4.

In addition, as in the case of negative alternating cycle matrices of dimensions 4 by 4 (figure 7.7), the negative alternating cycle matrices of dimensions 6 by 6 have two diagonal axes of symmetry (figure 8.5).

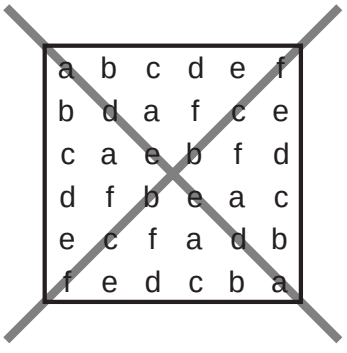


Figure 8.5.

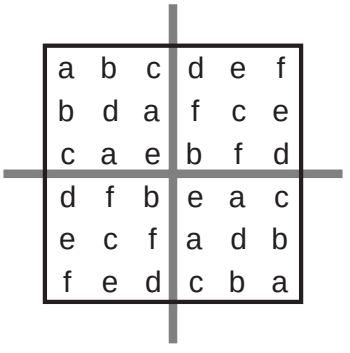


Figure 8.6.

The horizontal and vertical middle lines of a negative alternating cycle matrix of dimensions 6 by 6 constitute anti-symmetry axes (figure 8.6). When we reflect the matrix across one of these axes the respective positions of the numbers *a* and *f* are inverted as are the positions of the numbers *b* and *e* and the numbers *c* and *d*.

ACTIVITIES

- Construct some negative alternating cycle matrices of dimensions 6 by 6.
- When one multiplies a negative alternating cycle matrix of dimensions 6 by 6 by any number, what can be said about the resulting matrix?
- When any two negative alternating cycle matrices of dimensions 6 by 6 are added, what can be said about the matrix resulting from the addition?

$$3 \times \begin{bmatrix} 2 & 3 & -1 & 5 & 4 & 6 \\ 3 & 5 & 2 & 6 & -1 & 4 \\ -1 & 2 & 4 & 3 & 6 & 5 \\ 5 & 6 & 3 & 4 & 2 & -1 \\ 4 & -1 & 6 & 2 & 5 & 3 \\ 6 & 4 & 5 & -1 & 3 & 2 \end{bmatrix} = \begin{bmatrix} 6 & 9 & -3 & 15 & 12 & 18 \\ 9 & 15 & 6 & 18 & -3 & 12 \\ -3 & 6 & 12 & 9 & 18 & 15 \\ 15 & 18 & 9 & 12 & 6 & -3 \\ 12 & -3 & 18 & 6 & 15 & 9 \\ 18 & 12 & 15 & -3 & 9 & 6 \end{bmatrix}$$

Figure 8.7.

As an example, figure 8.7 presents the multiplication of matrix **A** by the number 3. The elements of the resulting matrix (3**A**) are 3 times the corresponding elements of the matrix **A** (figure 8.7). The first cycle of the matrix A generated the first cycle of the matrix 3**A**, where the numbers 6 and 9 alternate; in the second cycle of the matrix 3**A** the numbers -3 and 15

alternate, and in the third the numbers 12 and 18 alternate. We can say that the matrix 3**A** is a negative alternating cycle matrix of dimensions 6 by 6.

The reader is invited to prove in general that any multiple of a negative alternating cycle matrix of dimensions 6 by 6 constitutes a negative alternating cycle matrix of dimensions 6 by 6 too.

$$\begin{bmatrix} 2 & 3 & -1 & 5 & 4 & 6 \\ 3 & 5 & 2 & 6 & -1 & 4 \\ -1 & 2 & 4 & 3 & 6 & 5 \\ 5 & 6 & 3 & 4 & 2 & -1 \\ 4 & -1 & 6 & 2 & 5 & 3 \\ 6 & 4 & 5 & -1 & 3 & 2 \end{bmatrix} + \begin{bmatrix} 1 & -4 & 2 & 3 & -6 & -2 \\ -4 & 3 & 1 & -2 & 2 & -6 \\ 2 & 1 & -6 & -4 & -2 & 3 \\ 3 & -2 & -4 & -6 & 1 & 2 \\ -6 & 2 & -2 & 1 & 3 & -4 \\ -2 & -6 & 3 & 2 & -4 & 1 \end{bmatrix} = \begin{bmatrix} 3 & -1 & 1 & 8 & -2 & 4 \\ -1 & 8 & 3 & 4 & 1 & -2 \\ 1 & 3 & -2 & -1 & 4 & 8 \\ 8 & 4 & -1 & -2 & 3 & 1 \\ -2 & 1 & 4 & 3 & 8 & -1 \\ 4 & -2 & 8 & 1 & -1 & 3 \end{bmatrix}$$

Figure 8.8.

Figure 8.8 presents the addition of the matrices **A** and **B** that are both negative alternating cycle matrices of dimensions 6 by 6. In the resulting matrix **A**+**B** three cycles may be observed: in the first the numbers 3 and -1 alternate; in the second the numbers 1 and 8 and in the third the numbers -2 and 4. By consequence, the matrix **A**+**B** is a negative alternating cycle matrix of dimensions 6 by 6.

The reader is invited to prove in general that the sum of any two negative alternating cycle matrices of dimensions 6 by 6 constitutes a negative alternating cycle matrix of dimensions 6 by 6.

ACTIVITY

- Choose a pair of negative alternating cycle matrices of dimensions 6 by 6. Multiply the two matrices, both the first times the second and the second times the first. What can be said about the structure of the resulting matrices? Compare the two matrices resulting from the multiplications. Repeat the experience with other pairs of negative alternating cycle matrices of dimensions 6 by 6. Does the same result occur?

Chapter 9

MULTIPLICATION OF NEGATIVE ALTERNATING CYCLE MATRICES OF DIMENSIONS 6 BY 6

ABSTRACT

In chapter 9 the properties of the multiplication of negative alternating cycle matrices of dimensions 6 by 6 will be analyzed. The concept of positive alternating cycle matrices of dimensions 6 by 6 will be introduced.

Let us consider the two negative alternating cycle matrices of dimensions 6 by 6 presented in figure 9.1.

2	3	-1	5	4	6
3	5	2	6	-1	4
-1	2	4	3	6	5
5	6	3	4	2	-1
4	-1	6	2	5	3
6	4	5	-1	3	2

A

-1	2	-2	1	3	4
2	1	-1	4	-2	3
-2	-1	3	2	4	1
1	4	2	3	-1	-2
3	-2	4	-1	1	2
4	3	1	-2	2	-1

B

Figure 9.1.

Calculating the products **AB** and **BA**, we obtain the matrices presented in figure 9.2.

47	38	22	11	7	8
22	47	7	38	8	11
38	11	47	8	22	7
7	22	8	47	11	38
11	8	38	7	47	22
8	7	11	22	38	47

AB

47	22	38	7	11	8
38	47	11	22	8	7
22	7	47	8	38	11
11	38	8	47	7	22
7	8	22	11	47	38
8	11	7	38	22	47

BA

Figure 9.2.

ACTIVITIES

- Analyze the structure of the matrices in figure 9.2. Do they have alternating cycles?
- What is the relationship between the two matrices **AB** and **BA**?
- What can be said about the matrix **AB**+**BA** ?

The principal diagonals of the two matrices are equal and on both diagonals the number 47 is repeated. In the same way, the secondary diagonals are equal and their elements are all equal to 8.

In the first matrix there exist two alternating cycles: on the first the numbers 38 and 22 alternate; on the second the numbers 11 and 7 alternate (figure 9.3). In the second matrix, there are also two cycles in which the same numbers appear as in the first matrix, but the cycles are different in phase. In other words, the numbers in the alternating cycles of the second matrix are inverted relative to the numbers in the alternating cycles of the first matrix. When one reflects the matrix **AB** across its secondary diagonal one obtains the matrix **BA** (verify!).

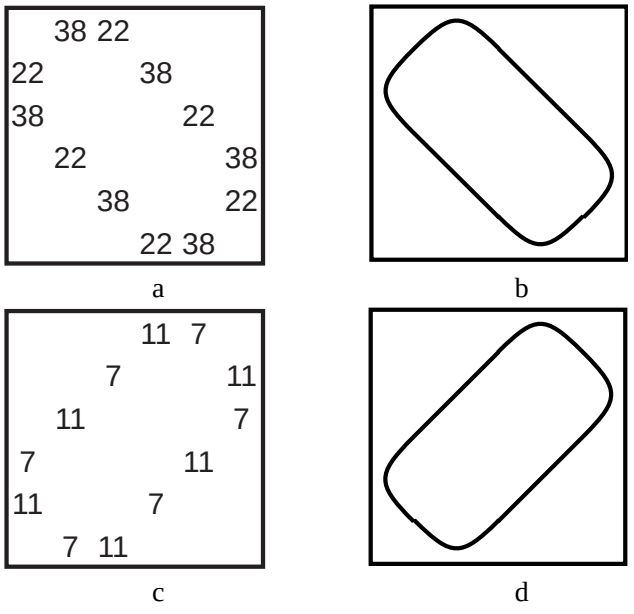


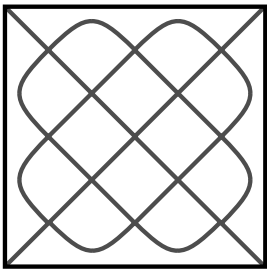
Figure 9.3.

As the structure is similar to that of positive alternating cycle matrices of dimensions 4 by 4 (⊗), we can say that the products **AB** and **BA** constitute **positive alternating cycle matrices of dimensions 6 by 6**. Besides the constant diagonals, two alternating cycles appear instead of only one, as was the case of positive alternating cycle matrices of dimensions 4 by 4.

Figure 9.4 presents the general form and the cyclic structure of positive alternating cycle matrices of dimensions 6 by 6.

p	r	s	t	u	q
s	p	u	r	q	t
r	t	p	q	s	u
u	s	q	p	t	r
t	q	r	u	p	s
q	u	t	s	r	p

General form
a



Cyclic structure
b

Positive alternating cycle matrices of dimensions 6 by 6

Figure 9.4.

ACTIVITIES

- Construct additional pairs of negative alternating cycle matrices and calculate their products, multiplying the matrices in both possible ways. Compare the results.
- Calculate the squares of some negative alternating cycle matrices. Do you note some particularity?

Figure 9.5 presents the squares $\mathbf{A}^2$ and $\mathbf{B}^2$ of the negative alternating cycle matrices **A** and **B**. The squares $\mathbf{A}^2$ and $\mathbf{B}^2$ are positive alternating cycle matrices. On the cycles of each of these squares a certain number is repeated. The diagonals of the squares $\mathbf{A}^2$ and $\mathbf{B}^2$ are symmetry axes.

91	69	69	47	47	38
69	91	47	69	38	47
69	47	91	38	69	47
47	69	38	91	47	69
47	38	69	47	91	69
38	47	47	69	69	91

$\mathbf{A}^2$

35	12	12	-5	-5	0
12	35	-5	12	0	-5
12	-5	35	0	12	-5
-5	12	0	35	-5	12
-5	0	12	-5	35	12
0	-5	-5	12	12	35

$\mathbf{B}^2$

Figure 9.5.

Will it be that the results observed in this chapter are consequences of the very particular numbers that appear in the negative alternating cycle matrices under consideration, or do these beautiful results reflect the general character of negative alternating cycle matrices of dimensions 6 by 6?

ACTIVITIES

- Formulate general hypotheses about the multiplication of negative alternating cycle matrices of dimensions 6 by 6.
- How can one verify if the general hypotheses about the multiplication of negative alternating cycle matrices of dimensions 6 by 6 are true or not?
- Construct some positive alternating cycle matrices of dimensions 6 by 6. Experiment with the addition and the multiplication of positive alternating cycle matrices of dimensions 6 by 6.

Chapter 10

MULTIPLICATION OF POSITIVE AND NEGATIVE ALTERNATING CYCLE MATRICES OF DIMENSIONS 6 BY 6

ABSTRACT

In chapter 10 the properties of addition and multiplication of positive and negative alternating cycle matrices of dimensions 6 by 6 will be analyzed.

Let us consider the two positive alternating cycle matrices of dimensions 6 by 6 presented in figure 10.1.

-1	7	5	3	-2	4
5	-1	-2	7	4	3
7	3	-1	4	5	-2
-2	5	4	-1	3	7
3	4	7	-2	-1	5
4	-2	3	5	7	-1

P

4	-1	2	-3	3	5
2	4	3	-1	5	-3
-1	-3	4	5	2	3
3	2	5	4	-3	-1
-3	5	-1	3	4	2
5	3	-3	2	-1	4

Q

Figure 10.1.

When we add the two matrices, we obtain the matrix **P**+**Q**, illustrated in figure 10.2, which is also a positive alternating cycle matrix.

3	6	7	0	1	9
7	3	1	6	9	0
6	0	3	9	7	1
1	7	9	3	0	6
0	9	6	1	3	7
9	1	0	7	6	3

P+Q

Figure 10.2.

Calculating the products **PQ** and **QP**, we arrive at the surprising result that **PQ** = **QP**, that is, the matrices **P** and **Q** commute. The matrix **PQ** also constitutes a positive alternating cycle matrix.

40	22	44	35	21	-2
44	40	21	22	-2	35
22	35	40	-2	44	21
21	44	-2	40	35	22
35	-2	22	21	40	44
-2	21	35	44	22	40

PQ

Figure 10.3.

ACTIVITIES

- Construct some more pairs of positive alternating cycle matrices and calculate the respective products. Do the matrices always commute? Are the results of the multiplication of two positive alternating cycle matrices also always positive alternating cycle matrices? In other words, will it be that ‘positive’ times ‘positive’ equals ‘positive’ is valid in the case of alternating cycle matrices of dimensions 6 by 6?
- Multiply a negative alternating cycle matrix by a positive one, and vice versa. What is the structure of the matrices resulting from these multiplications? Will it be that we have ‘negative’ times ‘positive’ equals ‘negative’ in the case of alternating cycle matrices of dimensions 6 by 6?

Matrix **S** in figure 10.4 is a negative alternating cycle matrix of dimensions 6 by 6 and matrix **T** is positive.

-1	2	-2	1	3	4
2	1	-1	4	-2	3
-2	-1	3	2	4	1
1	4	2	3	-1	-2
3	-2	4	-1	1	2
4	3	1	-2	2	-1

S

2	7	5	3	-2	4
5	2	-2	7	4	3
7	3	2	4	5	-2
-2	5	4	2	3	7
3	4	7	-2	2	5
4	-2	3	5	7	2

T

Figure 10.4.

When we add the two matrices, we obtain the matrix **S**+**T** (figure 10.5), that is not an alternating cycle matrix. It presents only a rotational symmetry of 180°.

1	9	3	4	1	8
7	3	-3	11	2	6
5	2	5	6	9	-1
-1	9	6	5	2	5
6	2	11	-3	3	7
8	1	4	3	9	1

S+T

Figure 10.5.

Calculating the products **ST** and **TS**, we see that both are negative alternating cycle matrices. By consequence, we have ‘negative’ times ‘positive’ equals ‘negative’ and also ‘positive’ times ‘negative’ equals ‘negative’. The reader is invited to experiment with other pairs of positive and negative alternating cycle matrices to see if a similar result occurs.

17	0	24	19	37	36
0	19	17	36	24	37
24	17	37	0	36	19
19	36	0	37	17	24
37	24	36	17	19	0
36	37	19	24	0	17

ST

15	34	6	43	15	20
34	43	15	20	6	15
6	15	15	34	20	43
43	20	34	15	15	6
15	6	20	15	43	34
20	15	43	6	34	15

TS

Figure 10.6.

In chapter 9 we considered already an example of the multiplication of two negative alternating cycle matrices of dimensions 6 by 6.

In the examples given so far we observed that the properties of the multiplication of alternating cycle matrices of dimensions 6 by 6 are very similar to the properties of the multiplication of alternating cycle matrices of dimensions 4 by 4.

ACTIVITIES

- Construct some more positive and negative alternating cycle matrices of dimensions 6 by 6. Multiply pairs of them and verify if they satisfy the general properties that the reader was already thinking of. Formulate the ideas in terms of hypotheses.
- Try to prove the hypotheses you formulated.
- Try to picture alternating cycle matrices of other dimensions and analyze their properties.

We shall return to these questions in the following chapters.

Chapter 11

ALTERNATING CYCLE MATRICES OF EVEN DIMENSIONS

ABSTRACT

In chapter 11 examples of positive and negative alternating cycle matrices of even dimensions will be analyzed.

Until now we have met alternating cycle matrices of dimensions 4 by 4 and dimensions 6 by 6. We saw that in both cases two classes exist that we can call positive and negative. As we observed, the properties of the addition and multiplication of positive and negative matrices are rather similar in each case. This similarity offers us the opportunity to determine whether or not it is possible to construct cycle matrices of other dimensions. We shall begin this chapter with analyzing other even dimension matrices. Later on we shall determine the possibility of constructing cycle matrices of odd dimensions.

ACTIVITIES

- The smallest even number is 2. Will it be possible to picture alternating cycle matrices of dimensions 2 by 2? If yes, analyze positive and negative alternating cycle matrices of dimensions 2 by 2.
- Construct some alternating cycle matrices of dimensions 8 by 8 and perform some matrix operations with them. Can you discover some interesting properties?
- Construct some alternating cycle matrices of dimensions 10 by 10 and perform some matrix operations with them. Can you discover some interesting properties?

A matrix of dimensions 2 by 2 has no elements outside its diagonals. If the diagonals are constant we can say that we are dealing with a positive matrix (see figure 11.1).

a	b
b	a

Figure 11.1.

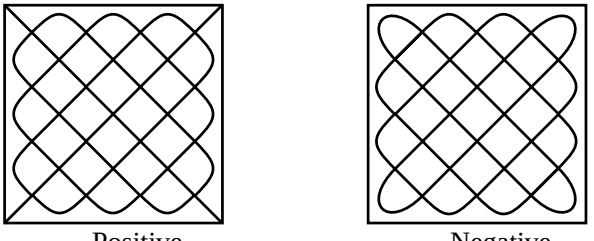
At the same time we can say that this matrix is also a negative alternating cycle matrix, as it is composed of a unique cycle of four elements (*a*, *b*, *a*, *b*) around the center of the matrix. In other words, in the special case of dimensions 2 by 2 the classes of positive and negative alternating cycle matrices coincide.

$$\begin{bmatrix} a & b \\ b & a \end{bmatrix} \times \begin{bmatrix} c & d \\ d & c \end{bmatrix} = \begin{bmatrix} ac+bd & ad+bc \\ bc+ad & bd+ac \end{bmatrix}$$

Figure 11.2.

Consider the product of two alternating cycle matrices of dimensions 2 by 2 (*a*, *b*, *c* and *d* are any numbers). We see that the resulting matrix is also a matrix of constant diagonals, as $ac+bd = bd+ac$ and $ad+bc = bc+ad$, as a consequence of the commutativity of the addition of numbers (figure 11.2).

Similar to the experience of the alternating cycle matrices of the dimensions 4 by 4 and 6 by 6, we can choose two structures, a positive and a negative, to picture cycle matrices of dimensions 8 by 8. Figure 11.3 illustrates the two structures.



Positive
a

Negative
b

Figure 11.3.

In the first case we have two constant diagonals and three cycles. In the second case we have four cycles. Figure 11.4 presents the general form of the two types of alternating cycle matrices of dimensions 8 by 8.

a	b	c	d	e	f	g	h
c	a	e	b	g	d	h	f
b	d	a	f	c	h	e	g
e	c	g	a	h	b	f	d
d	f	b	h	a	g	c	e
g	e	h	c	f	a	d	b
f	h	d	g	b	e	a	c
h	g	f	e	d	c	b	a

Positive
a

a	b	c	d	e	f	g	h
b	d	a	f	c	h	e	g
c	a	e	b	g	d	h	f
d	f	b	h	a	g	c	e
e	c	g	a	h	b	f	d
f	h	d	g	b	e	a	c
g	e	h	c	f	a	d	b
h	g	f	e	d	c	b	a

Negative
b

Figure 11.4.

ACTIVITY

- Under what conditions will an alternating cycle matrix of dimensions 8 by 8 be simultaneously positive and negative?

An alternating cycle matrix of dimensions 8 by 8 will be simultaneously positive and negative if it presents at the same time the structures of a positive matrix and of a negative matrix (figure 11.4). Therefore all corresponding elements of the matrices in figure 11.4 have to be equal. Let us compare the elements of the seconds rows: the first elements are equal, $c=b$; the second elements are equal, $a=d$; the third elements are equal, $e=a$; the fourth elements are equal, $b=f$; the fifth elements are equal, $g=c$; the sixth elements are equal, $d=h$. The other equalities of corresponding elements do not give us any new information. As a consequence, $b=c=f=g$ and $a=d=e=h$ both have to be valid. Figure 11.5 presents the general form of an alternating cycle matrix of dimensions 8 by 8 which is simultaneously positive and negative.

a	b	b	a	a	b	b	a
b	a	a	b	b	a	a	b
b	a	a	b	b	a	a	b
a	b	b	a	a	b	b	a
a	b	b	a	a	b	b	a
b	a	a	b	b	a	a	b
b	a	a	b	b	a	a	b
a	b	b	a	a	b	b	a

Positive and negative

Figure 11.5.

Let us now consider the two positive alternating cycle matrices of dimensions 8 by 8 presented in figure 11.6.

3	1	4	-2	5	0	-1	2
4	3	5	1	-1	-2	2	0
1	-2	3	0	4	2	5	-1
5	4	-1	3	2	1	0	-2
-2	0	1	2	3	-1	4	5
-1	5	2	4	0	3	-2	1
0	2	-2	-1	1	5	3	4
2	-1	0	5	-2	4	1	3

A

2	3	-1	4	-2	5	1	-3
-1	2	-2	3	1	4	-3	5
3	4	2	5	-1	-3	-2	1
-2	-1	1	2	-3	3	5	4
4	5	3	-3	2	1	-1	-2
1	-2	-3	-1	5	2	4	3
5	-3	4	1	3	-2	2	-1
-3	1	5	-2	4	-1	3	2

B

Figure 11.6.

Calculating the products **AB** and **BA**, we arrive at **AB** = **BA**, that is, the positive matrices **A** and **B** commute. The matrix AB also constitutes a positive alternating cycle matrix of dimensions 8 by 8 (figure 11.7).

30	59	22	11	12	6	-19	-13
22	30	12	59	-19	11	-13	6
59	11	30	6	22	-13	12	-19
12	22	-19	30	-13	59	6	11
11	6	59	-13	30	-19	22	12
-19	12	-13	22	6	30	11	59
6	-13	11	-19	59	12	30	22
-13	-19	6	12	11	22	59	30

AB = **BA**

Figure 11.7.

The matrices **C** and **D** in figure 11.8 are examples of negative alternating cycle matrices of dimensions 8 by 8.

-2	3	1	-3	4	-5	-4	2
3	-3	-2	-5	1	2	4	-4
1	-2	4	3	-4	-3	2	-5
-3	-5	3	2	-2	-4	1	4
4	1	-4	-2	2	3	-5	-3
-5	2	-3	-4	3	4	-2	1
-4	4	2	1	-5	-2	-3	3
2	-4	-5	4	-3	1	3	-2

C

4	0	2	3	-1	-2	1	5
0	3	4	-2	2	5	-1	1
2	4	-1	0	1	3	5	-2
3	-2	0	5	4	1	2	-1
-1	2	1	4	5	0	-2	3
-2	5	3	1	0	-1	4	2
1	-1	5	2	-2	4	3	0
5	1	-2	-1	3	2	0	4

D

Figure 11.8.

Multiplying the positive matrix **A** and the negative matrix **C**, we obtain the products **AC** and **CA** presented in figure11.9. Both matrices **AC** and **CA** are negative and are not equal as in the previous case of the products of two positive matrices.

35	1	-21	-9	10	2	-18	-48
1	-9	35	2	-21	-48	10	-18
-21	35	10	1	-18	-9	-48	2
-9	2	1	-48	35	-18	-21	10
10	-21	-18	35	-48	1	2	-9
2	-48	-9	-18	1	10	35	-21
-18	10	-48	-21	2	35	-9	1
-48	-18	2	10	-9	-21	1	35

AC

-7	-42	15	0	-11	-38	29	6
-42	0	-7	-38	15	6	-11	29
15	-7	-11	-42	29	0	6	-38
0	-38	-42	6	-7	29	15	-11
-11	15	29	-7	6	-42	-38	0
-38	6	0	29	-42	-11	-7	15
29	-11	6	15	-38	-7	0	-42
6	29	-38	-11	0	15	-42	-7

CA

Figure 11.9.

Let us consider next the products of the negative alternating cycle matrices **C** and **D**. On the base of our experience with negative alternating cycle matrices of dimensions 4 by 4 and 6 by 6 we can make the assumption that **CD** and **D**C are positive alternating cycle matrices and that the cycles of **CD** are different in phase. Will this indeed happen?

ACTIVITY

- Calculate the matrices **CD** and **DC**.

Figure 11.10 presents the matrices **CD** and **DC**. In fact, both matrices are positive alternating cycle matrices as expected: ‘negative’ times ‘negative’ equals ‘positive’. This result justifies once more the utilization of the expressions ‘positive’ and ‘negative’ to characterize the two types of alternating cycle matrices of dimensions 8 by 8. In addition, the two matrices are symmetrical in the sense that a reflection across the secondary diagonal transforms matrix **CD** in matrix **DC**. The corresponding cycles of the two matrices differ one place in phase. In the matrix **CD** the first cycle passes through the first row in the sequence 8 and then -28. In the matrix **DC** the first cycle passes through the first row in the inverse order -28 followed by 8, etc.

-3	8	-28	-26	31	12	-46	4
-28	-3	31	8	-46	-26	4	12
8	-26	-3	12	-28	4	31	-46
31	-28	-46	-3	4	8	12	-26
-26	12	8	4	-3	-46	-28	31
-46	31	4	-28	12	-3	-26	8
12	4	-26	-46	8	31	-3	-28
4	-46	12	31	-26	-28	8	-3

CD

-3	-28	8	31	-26	-46	12	4
8	-3	-26	-28	12	31	4	-46
-28	31	-3	-46	8	4	-26	12
-26	8	12	-3	4	-28	-46	31
31	-46	-28	4	-3	12	8	-26
12	-26	4	8	-46	-3	31	-28
-46	4	31	12	-28	-26	-3	8
4	12	-46	-26	31	8	-28	-3

DC

Figure 11.10.

ACTIVITIES

- The reader is invited to formulate hypotheses about the multiplication of alternating cycle matrices of dimensions 8 by 8.
- The reader is invited to formulate hypotheses about the multiplication of alternating cycle matrices of dimensions 10 by 10 (see the examples in figure 11.11).
- The reader is invited to formulate more general hypotheses about the multiplication of alternating cycle matrices of the same even dimensions.

3	1	4	-2	5	0	-1	2	6	-4
4	3	5	1	-1	-2	6	0	-4	2
1	-2	3	0	4	2	5	-4	-1	6
5	4	-1	3	6	1	-4	-2	2	0
-2	0	1	2	3	-4	4	6	5	-1
-1	5	6	4	-4	3	2	1	0	-2
0	2	-2	-4	1	6	3	-1	4	5
6	-1	-4	5	2	4	0	3	-2	1
2	-4	0	6	-2	-1	1	5	3	4
-4	6	2	-1	0	5	-2	4	1	3

P

2	0	1	4	-2	-4	-3	-1	5	3
1	2	-2	0	-3	4	5	-4	3	-1
0	4	2	-4	1	-1	-2	3	-3	5
-2	1	-3	2	5	0	3	4	-1	-4
4	-4	0	-1	2	3	1	5	-2	-3
-3	-2	5	1	3	2	-1	0	-4	4
-4	-1	4	3	0	5	2	-3	1	-2
5	-3	3	-2	-1	1	-4	2	4	0
-1	3	-4	5	4	-3	0	-2	2	1
3	5	-1	-3	-4	-2	4	1	0	2

Q

Two positive alternating cycle matrices of dimensions 10x10

a

2	0	3	4	1	-2	5	-1	-3	6
0	4	2	-2	3	-1	1	6	5	-3
3	2	1	0	5	4	-3	-2	6	-1
4	-2	0	-1	2	6	3	-3	1	5
1	3	5	2	-3	0	6	4	-1	-2
-2	-1	4	6	0	-3	2	5	3	1
5	1	-3	3	6	2	-1	0	-2	4
-1	6	-2	-3	4	5	0	1	2	3
-3	5	6	1	-1	3	-2	2	4	0
6	-3	-1	5	-2	1	4	3	0	2

R

-2	3	1	-3	4	-5	-4	2	0	5
3	-3	-2	-5	1	2	4	5	-4	0
1	-2	4	3	-4	-3	0	-5	5	2
-3	-5	3	2	-2	5	1	0	4	-4
4	1	-4	-2	0	3	5	-3	2	-5
-5	2	-3	5	3	0	-2	-4	1	4
-4	4	0	1	5	-2	2	3	-5	-3
2	5	-5	0	-3	-4	3	4	-2	1
0	-4	5	4	2	1	-5	-2	-3	3
5	0	2	-4	-5	4	-3	1	3	-2

S

Two negative alternating cycle matrices of dimensions 10x10

b

27	-11	-3	22	33	-10	-39	13	17	21
-3	27	33	-11	-39	22	17	-10	21	13
-11	22	27	-10	-3	13	33	21	-39	17
33	-3	-39	27	17	-11	21	22	13	-10
22	-10	-11	13	27	21	-3	17	33	-39
-39	33	17	-3	21	27	13	-11	-10	22
-10	13	22	21	-11	17	27	-39	-3	33
17	-39	21	33	13	-3	-10	27	22	-11
13	21	-10	17	22	-39	-11	33	27	-3
21	17	13	-39	-10	33	22	-3	-11	27

PQ = **QP**

c

9	4	30	-32	-26	19	9	17	35	-50
30	9	-26	4	9	-32	35	19	-50	17
4	-32	9	19	30	17	-26	-50	9	35
-26	30	9	9	35	4	-50	-32	17	19
-32	19	4	17	9	-50	30	35	-26	9
9	-26	35	30	-50	9	17	4	19	-32
19	17	-32	-50	4	35	9	9	30	-26
35	9	-50	-26	17	30	19	9	-32	4
17	-50	19	35	-32	9	4	-26	9	30
-50	35	17	9	19	-26	-32	30	4	9

RS

d

Figure 11.11 continued on next page.

9	30	4	-26	-32	9	19	35	17	-50
4	9	-32	30	19	-26	17	9	-50	35
30	-26	9	9	4	35	-32	-50	19	17
-32	4	19	9	17	30	-50	-26	35	9
-26	9	30	35	9	-50	4	17	-32	19
19	-32	17	4	-50	9	35	30	9	-26
9	35	-26	-50	30	17	9	19	4	-32
17	19	-50	-32	35	4	9	9	-26	30
35	-50	9	17	-26	19	30	-32	9	4
-50	17	35	19	9	-32	-26	4	30	9

SR

e

-34	84	79	-1	11	19	1	23	43	-15
84	-1	-34	19	79	23	11	-15	1	43
79	-34	11	84	1	-1	43	19	-15	23
-1	19	84	23	-34	-15	79	43	11	1
11	79	1	-34	43	84	-15	-1	23	19
19	23	-1	-15	84	43	-34	1	79	11
1	11	43	79	-15	-34	23	84	19	-1
23	-15	19	43	-1	1	84	11	-34	79
43	1	-15	11	23	79	19	-34	-1	84
-15	43	23	1	19	11	-1	79	84	-34

PR

f

-7	61	8	-47	66	59	-3	-11	41	43
61	-47	-7	59	8	-11	66	43	-3	41
8	-7	66	61	-3	-47	41	59	43	-11
-47	59	61	-11	-7	43	8	41	66	-3
66	8	-3	-7	41	61	43	-47	-11	59
59	-11	-47	43	61	41	-7	-3	8	66
-3	66	41	8	43	-7	-11	61	59	-47
-11	43	59	41	-47	-3	61	66	-7	8
41	-3	43	66	-11	8	59	-7	-47	61
43	41	-11	-3	59	66	-47	8	61	-7

RP

g

Figure 11.11.

ACTIVITY

- Try to picture the structure of cycle matrices of dimensions 5 by 5 and of other odd dimensions.

Chapter 12

GENERAL HYPOTHESES ABOUT ALTERNATING CYCLE MATRICES OF EVEN DIMENSIONS

ABSTRACT

In chapter 12 some general hypotheses about the properties of positive and negative alternating cycle matrices of even dimensions will be presented. The hypotheses will summarize and generalize the experiences obtained in the previous chapters.

On the basis of experimenting with alternating cycle matrices of dimensions 4 by 4, 6 by 6, 8 by 8, and 10 by 10 and also taking into account the proof realized in the case of dimensions 4 by 4, we can try to formulate some general hypotheses concerning the properties of cycle matrices of even dimensions.

ACTIVITIES

- Count the number of cycles in the positive alternating cycle matrices of dimensions 4 by 4, 6 by 6, 8 by 8, and 10 x 10. How many cycles will positive alternating cycle matrices of even dimensions have in general?
- Count the number of cycles in the negative alternating cycle matrices of dimensions 4 by 4, 6 by 6, 8 by 8, and 10 x 10. How many cycles will negative alternating cycle matrices of even dimensions have in general?

Any even number can be written as $2m$, where m represents a natural number. In this way let us consider square matrices of dimensions ($2m$) by ($2m$), another notation is written ($2m$) x ($2m$).

We can distinguish two classes of alternating cycle matrices of dimensions ($2m$) x ($2m$). The matrices of the first class we shall call **positive**. They have constant diagonals and (m – 1) alternating cycles. The matrices of the second class we shall call **negative**. They have m alternating cycles.

The first general properties of alternating cycle matrices of dimensions ($2m$) x ($2m$) that we can verify refer to the distribution of the numbers throughout the matrix and to the sum of the numbers in each row and in each column.

As the two numbers that alternate in a cycle appear in each row and in each column of a negative alternating cycle matrix of dimensions ($2m$) x ($2m$), the sums of the elements of the rows and of the columns are all equal. In the case of a positive alternating cycle matrix the same happens, but it is necessary to take into account that in each row and in each column one finds one element of the principal diagonal and one element of the secondary diagonal. If the numbers that appear in the first row of a positive or negative alternating cycle matrix are different, then these numbers are repeated in all rows and in all columns; in other words, the matrix is a **Latin square**.

ACTIVITY

- Taking into account the experiences of the previous chapters, formulate general hypotheses concerning the multiplication by any number and the addition of positive and negative alternating cycle matrices of dimensions ($2m$) x ($2m$).

We can formulate the following hypotheses concerning the multiplication of an alternating cycle matrix by any number:

Hypothesis 1a

If a positive alternating cycle matrix of dimensions ($2m$) x ($2m$) is multiplied by an arbitrary number, the matrix resulting from this operation is also a positive alternating cycle matrix of the same dimensions.

Hypothesis 1b

If a negative alternating cycle matrix of dimensions ($2m$) x ($2m$) is multiplied by an arbitrary number, the matrix resulting from this operation is also a negative alternating cycle matrix of the same dimensions.

Concerning the addition of two alternating cycle matrices we can formulate the following two similar general hypotheses:

Hypothesis 2a

If **A** and **B** are two positive alternating cycle matrices of dimensions ($2m$) x ($2m$), then **A**+**B** is also a positive alternating cycle matrix of the same dimensions.

Hypothesis 2b

If **A** and **B** are two negative alternating cycle matrices of dimensions (2*m*) x (2*m*), then **A**+**B** is also a negative alternating cycle matrix of the same dimensions.

ACTIVITIES

- Attempt to verify the hypotheses 1a and 1b and find proofs.
- Attempt to verify the hypotheses 2a and 2b and find proofs.

Let us consider hypothesis 1a. Let **P** be a positive alternating cycle matrix of dimensions (2*m*) x (2*m*) and let *s* be any number. All elements of the principal diagonal of **P** are equal, let us say equal to the number *c*. Consequently, all elements of the principal diagonal of the matrix *s***P** are equal to *sc*, and thus the principal diagonal of *s***P** is constant. In the same way it is shown that the secondary diagonal of *s***P** is constant. Let us consider now the first cycle of the matrix **P** where two numbers, let us say *f* and *g*, alternate. The corresponding elements of matrix *s***P** are equal to *sf* and *sg*, respectively. We can conclude that matrix *s***P** has a first cycle and that the numbers *sf* and *sg* alternate on the cycle. The same reasoning may be repeated for the other ($m - 2$) cycles of matrix **P**. In this manner we arrive at the final conclusion that *s***P** has constant diagonals and has ($m - 2$) alternating cycles, that is, *s***P** is a positive alternating cycle matrix of dimensions (2*m*) x (2*m*), as we wanted to prove.

In a similar way one may find a proof in the case of hypothesis 1b and of hypotheses 2a and b. Once proven the hypotheses become true affirmations, and we may call them **theorems**.

Even more interesting and truly spectacular is the situation of the multiplication of positive and negative alternating cycle matrices of dimensions (2*m*) x (2*m*).

ACTIVITY

- Taking into account the experiences of the previous chapters, formulate general hypotheses concerning the multiplication of positive and negative alternating cycle matrices of dimensions (2*m*) x (2*m*).

Let us remember that our choice of the nomenclature ‘positive’ and ‘negative’ for the two types of alternating cycle matrices of dimensions 4 x 4, 6 x 6, 8 x 8, etc. reflected some properties of the multiplication of those matrices. Thus we can formulate the following hypotheses about the multiplication table of alternating cycle matrices of dimensions (2*m*) x (2*m*):

Hypothesis 3a

If **A** and **B** are two positive alternating cycle matrices of dimensions (2*m*) x (2*m*), then **AB** is also a positive alternating cycle matrix of the same dimensions.

Hypothesis 3b

If **A** is a positive alternating cycle matrix of dimensions (2*m*) x (2*m*) and **B** is a negative alternating cycle matrix of dimensions (2*m*) x (2*m*), then **AB** and **BA** are negative alternating cycle matrices of the same dimensions.

Hypothesis 3c

If **A** and **B** are two negative alternating cycle matrices of dimensions (2*m*) x (2*m*), then **AB** is a positive alternating cycle matrix of the same dimensions.

Besides the rules of 'signs', we have observed two more interesting particularities that lead us to formulate two additional general hypotheses:

Hypothesis 4

If **A** and **B** are two positive alternating cycle matrices of dimensions (2*m*) x (2*m*), then **A** and **B** commute, that is, **AB** = **BA**.

Hypothesis 5

If **A** and **B** are two negative alternating cycle matrices of dimensions (2*m*) x (2*m*), then **AB** and **BA** are symmetrical in the sense that a reflection across the secondary diagonal of **AB** transforms matrix **AB** into matrix **BA**, and the cycles of **AB** and **BA** differ in phase.

So far we shall postpone the verification and the subsequent proof of hypotheses 3, 4 and 5, and the most surprising conjectures concerning positive and negative alternating cycle matrices of dimensions (2*m*) x (2*m*) to a later chapter. In the following chapter we shall proceed with analyzing the question if it is possible to picture and construct alternating cycle matrices of odd dimensions.

ACTIVITY

- Construct positive and negative alternating cycle matrices of dimensions 5 by 5 and study their properties.

Chapter 13

ALTERNATING CYCLE MATRICES OF DIMENSIONS 5 BY 5

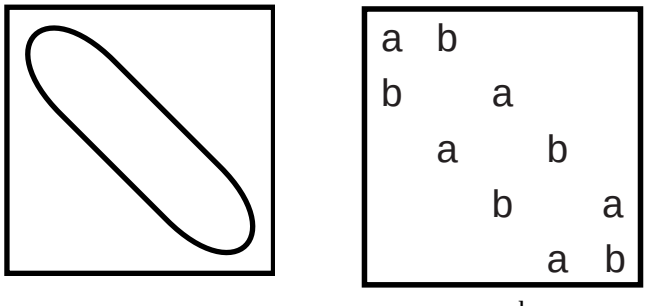
ABSTRACT

In chapter 13 negative and positive alternating cycle matrices of dimensions 5 by 5 will be constructed and analyzed.

ACTIVITY

- Attempt to construct positive and negative alternating cycle matrices of dimensions 5 by 5 and study their properties.

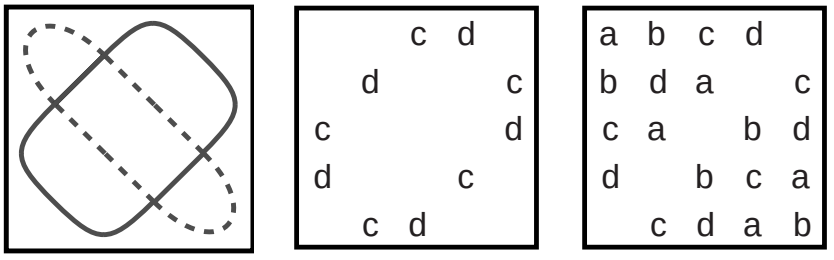
Let us consider a 5 by 5 square. Picture a first cycle in it (figure 13.1a) and construct it as an alternating cycle. Figure 13.1b shows this.



a b

Figure 13.1.

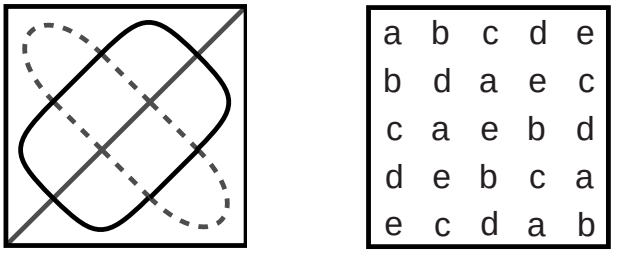
Next, we can try to construct a second cycle. Figure 13.2 shows that this is indeed possible.



a b c

Figure 13.2.

Our choice for the elements of the secondary diagonal remain. Let us opt for a constant diagonal and in this way we arrive at the general cyclic structure and form presented in figure 3.3. Matrices of this form have only one constant diagonal instead of two, as is the case of positive alternating cycle matrices of even dimensions.



Cyclic structure General form
a b

Figure 13.3.

ACTIVITY

- Determine the symmetries of the matrices that consist of the form in figure 13.3b.

The principal diagonal constitutes a symmetry axis, whereas the secondary diagonal is an anti-symmetry axis that inverts the numbers of the alternating cycles.

We are dealing with a form (figure 13.3.b) of alternating cycle matrices. The question emerges if we can call this type of matrices ‘positive’ or ‘negative’.

ACTIVITIES

- Construct some pairs of matrices that present the cyclic structure and form in figure 13.3.
- Add the matrices of the constructed pairs and analyze the results.
- Multiply the matrices of the constructed pairs and analyze the results.

Figure 13.4 presents two matrices, **A** and **B**, with the form and the cyclic structure in figure 13.3.

2	3	0	-1	4
3	-1	2	4	0
0	2	4	3	-1
-1	4	3	0	2
4	0	-1	2	3

A

3	-2	1	5	-3
-2	5	3	-3	1
1	3	-3	-2	5
5	-3	-2	1	3
-3	1	5	3	-2

B

Figure 13.4.

Calculating the products **AB** and **BA**, we obtain the matrices presented in figure 13.5.

-17	18	33	12	-14
33	-17	-14	18	12
18	12	-17	-14	33
-14	33	12	-17	18
12	-14	18	33	-17

AB

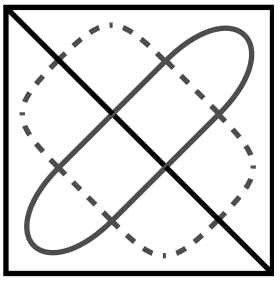
-17	33	18	-14	12
18	-17	12	33	-14
33	-14	-17	12	18
12	18	-14	-17	33
-14	12	33	18	-17

BA

Figure 13.5.

The structure of the matrices **AB** and **BA** is different from the cyclic structure of the matrices **A** and **B** themselves, but, nevertheless, may be considered cyclic. This time, however, the principal diagonal is constant. We can observe also that the matrices **AB** and **BA** are symmetrical: a reflection of matrix **AB** in the principal diagonal transforms it into matrix **BA**. Moreover, the alternating cycles of matrices **AB** and **BA** differ in phase, that is, the numbers of the corresponding cycles are inverted.

The properties of the matrices **AB** and **BA** are rather similar to those we found already for the products of negative alternating cycle matrices of even dimensions. Taking into account this similarity, we call the structure and form of the matrices of **A** and **B negative** (figure 13.3) and we call the alternating cyclic structure and form of the matrices **AB** and **BA positive**, presented in figure 13.6.



a

a	b	c	d	e
c	a	e	b	d
b	d	a	e	c
e	c	d	a	b
d	e	b	c	a

b

Structure and form of positive alternating cycle matrices of dimensions 5 by 5

Figure 13.6.

ACTIVITIES

- Construct some pairs of positive and negative alternating cycle matrices of dimensions 5 by 5.
- Add the matrices of the constructed pairs and analyze the results.
- Multiply the matrices of the constructed pairs and analyze the results.

Figure 13.7 presents two positive alternating cycle matrices of dimensions 5 by 5, **C** and **D**.

-2	3	1	4	-5
1	-2	-5	3	4
3	4	-2	-5	1
-5	1	4	-2	3
4	-5	3	1	-2

C

4	-1	2	3	6
2	4	6	-1	3
-1	3	4	6	2
6	2	3	4	-1
3	6	-1	2	4

D

Figure 13.7.

Calculating the products **CD** and **DC**, we note that **CD** = **DC** (figure 13.8), and that the matrix resulting from the multiplication is also a positive alternating cycle matrix. In other words, in the case of the considered example we have ‘positive’ times ‘positive’ equals ‘positive’.

6	-5	35	3	-25
35	6	-25	-5	3
-5	3	6	-25	35
-25	35	3	6	-5
3	-25	-5	35	6

CD = **DC**

Figure 13.8.

Multiplying the negative matrix **A** and the positive matrix **C**, we obtain the matrices **AC** and **CA** (figure 13.9) which are negative alternating cycle matrices.

20	-21	-5	23	-9
-21	23	20	-9	-5
-5	20	-9	-21	23
23	-9	-21	-5	20
-9	-5	23	20	-21

AC

-19	9	27	7	-16
9	7	-19	-16	27
27	-19	-16	9	7
7	-16	9	27	-19
-16	27	7	-19	9

CA

Figure 13.9.

Multiplying matrices **B** and **D** (figure 13.10), we see that it is verified in these examples, on the one side, ‘negative’ times ‘positive’ equals ‘negative’ and, on the other side, ‘positive’ times ‘negative’ equals ‘negative’.

28	-16	16	31	-3
-16	31	28	-3	16
16	28	-3	-16	31
31	-3	-16	16	28
-3	16	31	28	-16

BD

13	-10	19	40	-6
-10	40	13	-6	19
19	13	-6	-10	40
40	-6	-10	19	13
-6	19	40	13	-10

DB

Figure 13.10.

ACTIVITY

- Can a matrix of dimensions 5 by 5 be, at the same time, a negative and a positive alternating cycle matrix? If the answer is yes, determine the general form of these matrices.

a	b	c	d	e
b	d	a	e	c
c	a	e	b	d
d	e	b	c	a
e	c	d	a	b

Negative
a

a	b	c	d	e
c	a	e	b	d
b	d	a	e	c
e	c	d	a	b
d	e	b	c	a

Positive
b

Figure 13.11.

Figure 13.11 presents the negative and positive forms of the alternating cycle matrices of dimensions 5 by 5. Let us determine if a matrix can have both forms simultaneously. Superimposing the two forms in figure 13.11, the corresponding elements have to be equal in order for a matrix with the first row (a, b, c, d, e) to be simultaneously negative and positive. Let us compare the second rows. Their respective first elements have to be equal, $b = c$; the second elements have to be equal, $d = a$; the third elements have to be equal, $a = e$; the fourth elements have to be the equal, $e = b$; the fifth elements have to be equal, $c = d$. To satisfy all these conditions, it follows that $a = b = c = d = e$. In other words, the only alternating cycle matrices of dimensions 5 by 5 that are positive and negative at the same time are constant matrices, where all elements are equal. In their cycles the constant number alternates with itself.

ACTIVITY

- In the case of alternating cycle matrices of even dimensions we saw that the sums of the elements of the rows and of the columns are constant. Will the same occur in the case of dimensions 5 by 5?

The sums of the elements of the rows and of the columns of a positive or negative alternating cycle matrix, of which the first row is (a, b, c, d, e), is always equal to $a+b+c+d+e$. In the case that the five numbers differ, a positive or negative alternating cycle matrix constitutes a Latin square.

ACTIVITIES

- Formulate the apparent rules, observed in the previous examples, as hypotheses.
- Try to prove the formulated hypotheses.
- Try to picture alternating cycle matrices of other odd dimensions and analyze their properties.

In the next chapters we shall return to the questions posed.

Chapter 14

MULTIPLICATION OF POSITIVE AND NEGATIVE ALTERNATING CYCLE MATRICES OF DIMENSIONS 3 BY 3

ABSTRACT

In chapter 14 the concepts of positive and negative alternating cycle matrices of dimensions 3 by 3 will be introduced and the properties of the multiplication of these matrices will be analyzed. The multiplication tables of basic positive and negative alternating cycle matrices of dimensions 3 by 3 will be constructed and analyzed.

ACTIVITIES

- Picture the notions of positive and negative alternating cycle matrices of dimensions 3 by 3.
- Construct some pairs of positive and negative alternating cycle matrices of dimensions 3 by 3 and calculate their respective products.
- Do the matrices commute?
- Will the products of two positive alternating cycle matrices of dimensions 3 by 3 always be positive alternating cycle matrices too? In other words, will ‘positive’ times ‘positive’ equals ‘positive’ be valid with alternating cycle matrices of dimensions 3 by 3?
- Multiply a negative alternating cycle matrix by a positive one, and vice versa. What is the structure of the product? Will ‘negative’ times ‘positive’ equals ‘negative’ be valid in the case of alternating cycle matrices of dimensions 3 by 3?

Figure 14.1 presents the general form and the cyclic structure of positive and negative alternating cycle matrices of dimensions 3 by 3.

a	b	c
c	a	b
b	c	a

positive
a

a	b	c
b	c	a
c	a	b

negative
b

General form and cyclic structure of alternating cycle matrices of dimensions 3 by 3

Figure 14.1.

Let us consider the two positive alternating cycle matrices of dimensions 3 by 3, **P** and **Q**, presented in figure 14.2 and multiply the first times the second and the second times the first.

$$\begin{bmatrix} 3 & 2 & 1 \\ 1 & 3 & 2 \\ 2 & 1 & 3 \end{bmatrix} \times \begin{bmatrix} 2 & 4 & -1 \\ -1 & 2 & 4 \\ 4 & -1 & 2 \end{bmatrix} = \begin{bmatrix} 8 & 15 & 7 \\ 7 & 8 & 15 \\ 15 & 7 & 8 \end{bmatrix}$$

P x **Q**

$$\begin{bmatrix} 2 & 4 & -1 \\ -1 & 2 & 4 \\ 4 & -1 & 2 \end{bmatrix} \times \begin{bmatrix} 3 & 2 & 1 \\ 1 & 3 & 2 \\ 2 & 1 & 3 \end{bmatrix} = \begin{bmatrix} 8 & 15 & 7 \\ 7 & 8 & 15 \\ 15 & 7 & 8 \end{bmatrix}$$

Q x **P**

Example of ‘positive’ times ‘positive’

Figure 14.2.

Calculating the products **PQ** and **QP**, we arrive at the surprising result that **PQ** = **QP**, that is, the matrices **P** and **Q** commute under multiplication. The matrix **PQ** also constitutes a positive alternating cycle matrix.

$$\begin{bmatrix} 3 & 2 & 1 \\ 1 & 3 & 2 \\ 2 & 1 & 3 \end{bmatrix} \times \begin{bmatrix} 4 & -1 & -2 \\ -1 & -2 & 4 \\ -2 & 4 & -1 \end{bmatrix} = \begin{bmatrix} 8 & -3 & 1 \\ -3 & 1 & 8 \\ 1 & 8 & -3 \end{bmatrix}$$

P x **N**

Example of ‘positive’ times ‘negative’

a

$$\begin{bmatrix} 4 & -1 & -2 \\ -1 & -2 & 4 \\ -2 & 4 & -1 \end{bmatrix} \times \begin{bmatrix} 3 & 2 & 1 \\ 1 & 3 & 2 \\ 2 & 3 & 1 \end{bmatrix} = \begin{bmatrix} 7 & 3 & -4 \\ 3 & -4 & 7 \\ -4 & 7 & 3 \end{bmatrix}$$

N x **P**

Example of ‘negative’ times ‘positive’

b

Figure 14.3.

Figure 14.3 presents examples of the multiplication of two alternating cycle matrices of dimensions 3 by 3, one being positive and the other negative. The two products are negative cycle matrices and are different: **PN** ≠ **NP**.

$$\begin{bmatrix} 4 & -1 & -2 \\ -1 & -2 & 4 \\ -2 & 4 & -1 \end{bmatrix} \times \begin{bmatrix} 3 & 2 & 5 \\ 2 & 5 & 3 \\ 5 & 3 & 2 \end{bmatrix} = \begin{bmatrix} 0 & -3 & 13 \\ 13 & 0 & -3 \\ -3 & 13 & 0 \end{bmatrix}$$

N x **M**

$$\begin{bmatrix} 3 & 2 & 5 \\ 2 & 5 & 3 \\ 5 & 3 & 2 \end{bmatrix} \times \begin{bmatrix} 4 & -1 & -2 \\ -1 & -2 & 4 \\ -2 & 4 & -1 \end{bmatrix} = \begin{bmatrix} 0 & 13 & -3 \\ -3 & 0 & 13 \\ 13 & -3 & 0 \end{bmatrix}$$

M x **N**

Example of ‘negative’ times ‘negative’

Figure 14.4.

In the example of multiplying two negative alternating cycle matrices of dimensions 3 by 3 we observe that the products **NM** and **MN** are both positive alternating cycle matrices (figure 14.4). The matrices **NM** and **MN** are symmetrical (a reflection of the matrix **NM** across the principal diagonal leads to matrix **MN**), and the alternating cycles of the matrices **NM** and **MN** differ in phase.

How can we prove that the properties observed in the examples are verified in general?

ACTIVITY

- Formulate general hypotheses concerning the multiplication of alternating cycle matrices of dimensions 3 by 3 and verify them in some concrete cases. When you conclude that the conjectures seem to be true in general, try to find proofs for them.

One way that may assist us in attaining proofs is the following.

Each positive alternating cycle matrix may be written as the sum of three matrices, as figure 14.5 illustrates.

$$\begin{bmatrix} a & b & c \\ c & a & b \\ b & c & a \end{bmatrix} = \begin{bmatrix} a & 0 & 0 \\ 0 & a & 0 \\ 0 & 0 & a \end{bmatrix} + \begin{bmatrix} 0 & b & 0 \\ 0 & 0 & b \\ b & 0 & 0 \end{bmatrix} + \begin{bmatrix} 0 & 0 & c \\ c & 0 & 0 \\ 0 & c & 0 \end{bmatrix}$$

Figure 14.5.

Each of these three matrices is a multiple of a **basic** positive alternating cycle matrix and thus the initial matrix can be written as the sum of multiples of basic positive alternating cycle matrices as figure 14.6 illustrates.

$$\begin{bmatrix} a & b & c \\ c & a & b \\ b & c & a \end{bmatrix} = a \times \begin{bmatrix} \mathbf{1} & 0 & 0 \\ 0 & \mathbf{1} & 0 \\ 0 & 0 & \mathbf{1} \end{bmatrix} + b \times \begin{bmatrix} 0 & \mathbf{1} & \mathbf{0} \\ \mathbf{0} & 0 & \mathbf{1} \\ \mathbf{1} & \mathbf{0} & 0 \end{bmatrix} + c \times \begin{bmatrix} 0 & \mathbf{0} & \mathbf{1} \\ \mathbf{1} & 0 & \mathbf{0} \\ \mathbf{0} & \mathbf{1} & 0 \end{bmatrix}$$

Figure 14.6.

Denoting these basic positive alternating cycle matrices by **P**(1), **P**(2) and **P**(3), where the index number 1, 2, and 3 indicates the place of the first row where in each matrix the number 1 appears (figure 14.7).

$$\mathbf{P}(1) = \begin{bmatrix} \mathbf{1} & 0 & 0 \\ 0 & \mathbf{1} & 0 \\ 0 & 0 & \mathbf{1} \end{bmatrix} \qquad \mathbf{P}(2) = \begin{bmatrix} 0 & \mathbf{1} & \mathbf{0} \\ \mathbf{0} & 0 & \mathbf{1} \\ \mathbf{1} & \mathbf{0} & 0 \end{bmatrix} \qquad \mathbf{P}(3) = \begin{bmatrix} 0 & \mathbf{0} & \mathbf{1} \\ \mathbf{1} & 0 & \mathbf{0} \\ \mathbf{0} & \mathbf{1} & 0 \end{bmatrix}$$

Figure 14.7.

In this manner, our positive alternating cycle matrix **P**(a,b,c) can be written uniquely as the sum of multiples of the three basic positive alternating cycle matrices :

$$\mathbf{P}(a,b,c) = a\,\mathbf{P}(1) + b\,\mathbf{P}(2) + c\,\mathbf{P}(3),$$

where **P**(a,b,c) represents the positive alternating cycle matrix of dimensions 3 by 3 , in which the numbers a, b, and c appear in its first row.

ACTIVITIES

- How can we write a negative alternating cycle matrices of dimensions 3 by 3 as the sum of some basic negative alternating cycle matrices?
- Construct the basic negative alternating cycle matrices.
- Calculate the multiplication tables of the basic positive and negative alternating cycle matrices of dimensions 3 by 3.
- Can the reader discover some particularity of the calculated multiplication tables?

By multiplying basic alternating cycle matrices, we can construct four multiplication tables. When multiplying two basic positive alternating cycle matrices, a basic positive alternating cycle matrix is always the result. Figure 14.8 presents the multiplication table of the basic positive alternating cycle matrices.

x	P(1)	P(2)	P(3)
P(1)	**P(1)**	**P(2)**	**P(3)**
P(2)	**P(2)**	**P(3)**	**P(1)**
P(3)	**P(3)**	**P(1)**	**P(2)**

Multiplication table of the basic positive alternating cycle matrices of dimensions 3 by 3

Figure 14.8.

By consequence, the product of two positive alternating cycle matrices is always positive, as may be proven in the following way.

Let $\mathbf{P}(a,b,c)$ and $\mathbf{P}(d,e,f)$ be any two positive alternating cycle matrices of dimensions 3 by 3. We have

$$\begin{aligned}
&\mathbf{P}(a,b,c) \text{ x } \mathbf{P}(d,e,f) = \\
&\{a\,\mathbf{P}(1) + b\,\mathbf{P}(2) + c\,\mathbf{P}(3)\} \text{ x } \{d\,\mathbf{P}(1) + e\,\mathbf{P}(2) + f\,\mathbf{P}(3)\} = \\
&= ad\,\{\mathbf{P}(1) \text{ x } \mathbf{P}(1)\} + ae\,\{\mathbf{P}(1) \text{ x } \mathbf{P}(2)\} + af\,\{\mathbf{P}(1) \text{ x } \mathbf{P}(3)\} + \\
&+ bd\,\{\mathbf{P}(2) \text{ x } \mathbf{P}(1)\} + be\,\{\mathbf{P}(2) \text{ x } \mathbf{P}(2)\} + bf\,\{\mathbf{P}(2) \text{ x } \mathbf{P}(3)\} + \\
&+ cd\,\{\mathbf{P}(3) \text{ x } \mathbf{P}(1)\} + ce\,\{\mathbf{P}(3) \text{ x } \mathbf{P}(2)\} + cf\,\{\mathbf{P}(3) \text{ x } \mathbf{P}(3)\} = \\
&= ad\,\mathbf{P}(1) + ae\,\mathbf{P}(2) + af\,\mathbf{P}(3) + bd\,\mathbf{P}(2) + be\,\mathbf{P}(3) + bf\,\mathbf{P}(1) + cd\,\mathbf{P}(3) + ce\,\mathbf{P}(1) + cf\,\mathbf{P}(2) = \\
&= (ad+bf+ce)\,\mathbf{P}(1) + (ae+bd+cf)\,\mathbf{P}(2) + (af+be+cd)\,\mathbf{P}(3) = \\
&= \mathbf{P}(ad+bf+ce,\ ae+bd+cf,\ af+be+cd).
\end{aligned}$$

Summarizing, we conclude:

$$\mathbf{P}(a,b,c) \text{ x } \mathbf{P}(d,e,f) = \mathbf{P}(ad+bf+ce,\ ae+bd+cf,\ af+be+cd).$$

In the same manner, calculating the inverse product $\mathbf{P}(d,e,f)$ x $\mathbf{P}(a,b,c)$, we obtain:

$$\mathbf{P}(d,e,f) \text{ x } \mathbf{P}(a,b,c) = \mathbf{P}(da+ec+fb,\ db+ea+fc,\ dc+eb+fa).$$

As $ad+bf+ce = da+ec+fb$, $ae+bd+cf = db+ea+fc$ and $af+be+cd = dc+eb+fa$, it follows that the multiplication of the positive alternating cycle matrices of dimensions 3 by 3 is commutative:

$$\mathbf{P}(a,b,c) \text{ x } \mathbf{P}(d,e,f) = \mathbf{P}(d,e,f) \text{ x } \mathbf{P}(a,b,c).$$

The reasoning used in the given proof will be useful later on when we analyze positive alternating cycle matrices of other dimensions.

In the way we introduced basic positive alternating cycle matrices, we can introduce basic negative alternating cycle matrices (see figure 14.9).

$$\mathbf{N}(1) = \begin{bmatrix} \mathbf{1} & \mathbf{0} & 0 \\ \mathbf{0} & 0 & \mathbf{1} \\ 0 & \mathbf{1} & \mathbf{0} \end{bmatrix} \qquad \mathbf{N}(2) = \begin{bmatrix} \mathbf{0} & \mathbf{1} & 0 \\ \mathbf{1} & 0 & \mathbf{0} \\ 0 & \mathbf{0} & \mathbf{1} \end{bmatrix} \qquad \mathbf{N}(3) = \begin{bmatrix} 0 & 0 & \mathbf{1} \\ 0 & \mathbf{1} & 0 \\ \mathbf{1} & 0 & 0 \end{bmatrix}$$

The basic negative alternating cycle matrices of dimensions 3 by 3

Figure 14.9.

If we denote the negative alternating cycle matrix of dimensions 3 by 3, that has in its first row the numbers *p*, *q* and *r*, by **N**(*p*,*q*,*r*), we can express **N**(*p*,*q*,*r*) as the sum of multiples of the three basic negative alternating cycle matrices as illustrates figure 14.10.

$$\begin{bmatrix} p & q & r \\ q & r & p \\ r & p & q \end{bmatrix} = p \times \begin{bmatrix} \mathbf{1} & \mathbf{0} & 0 \\ \mathbf{0} & 0 & \mathbf{1} \\ 0 & \mathbf{1} & \mathbf{0} \end{bmatrix} + q \times \begin{bmatrix} \mathbf{0} & \mathbf{1} & 0 \\ \mathbf{1} & 0 & \mathbf{0} \\ 0 & \mathbf{0} & \mathbf{1} \end{bmatrix} + r \times \begin{bmatrix} 0 & 0 & \mathbf{1} \\ 0 & \mathbf{1} & 0 \\ \mathbf{1} & 0 & 0 \end{bmatrix}$$

Figure 14.10.

In this manner, we have

$$\mathbf{N}(p,q,r) = p\,\mathbf{N}(1)+q\,\mathbf{N}(2)+r\,\mathbf{N}(3).$$

When multiplying a basic positive alternating cycle matrix and a basic negative alternating cycle matrix, we always obtain a basic negative alternating cycle matrix. Figures 14.11 and 14.12 present the multiplication tables of the basic positive and negative alternating cycle matrices.

x	N(1)	N(2)	N(3)
P(1)	**N(1)**	**N(2)**	**N(3)**
P(2)	**N(3)**	**N(1)**	**N(2)**
P(3)	**N(2)**	**N(3)**	**N(1)**

Multiplication table of the basic positive alternating cycle matrices times the basic negative alternating cycle matrices

Figure 14.11.

x	P(1)	P(2)	P(3)
N(1)	**N(1)**	**N(2)**	**N(3)**
N(2)	**N(2)**	**N(3)**	**N(1)**
N(3)	**N(3)**	**N(1)**	**N(2)**

Multiplication table of the basic negative alternating cycle matrices times the basic positive alternating cycle matrices

Figure 14.12.

When multiplying two basic negative alternating cycle matrices, a positive alternating cycle matrix basic is always result. Figure 14.13 presents the multiplication table of the basic negative alternating cycle matrices.

x	N(1)	N(2)	N(3)
N(1)	**P(1)**	**P(2)**	**P(3)**
N(2)	**P(3)**	**P(1)**	**P(2)**
N(3)	**P(2)**	**P(3)**	**P(1)**

Multiplication table of the basic negative alternating cycle matrices

Figure 14.13.

ACTIVITIES

- Using the multiplication tables of the basic positive and negative alternating cycle matrices of dimensions 3 by 3, prove that the product of two positive and negative alternating cycle matrices of these dimensions is always a negative alternating cycle matrix.
- Using the multiplication table of the basic negative alternating cycle matrices of dimensions 3 by 3, show that the product of two negative alternating cycle matrices of these dimensions is always a positive alternating cycle matrix.

14.1. A Surprising Property of the Multiplication Tables

Let us now note a surprising and beautiful aspect of the four multiplication tables we have constructed. We are dealing with multiplication tables of matrices, but starting with each table we can construct a new matrix as figure 14.14 illustrates. Considering the multiplication table of the basic positive alternating cycle matrices, we see that the products always are basic positive alternating cycle matrices.

x	P(1)	P(2)	P(3)
P(1)	**P(1)**	**P(2)**	**P(3)**
P(2)	**P(2)**	**P(3)**	**P(1)**
P(3)	**P(3)**	**P(1)**	**P(2)**

↓

P(1)	P(2)	P(3)
P(2)	P(3)	P(1)
P(3)	P(1)	P(2)

↓

1	2	3
2	3	1
3	1	2

Figure 14.14.

Isolating the numbers that appear between brackets, we obtain a matrix that is surprisingly and unexpectedly a negative alternating cycle matrix! Thus, there suddenly appears a negative matrix (figure 14.15a) in the exclusive context of positive matrices.

In the same way, the multiplication table of the basic positive alternating cycle matrices times the basic negative matrices (figure 14.11) leads us to the positive matrix in figure 14.15b, whereas the multiplication table of the basic negative alternating cycle matrices times the basic positive matrices (figure 14.12) conducts us once more to the negative matrix in figure 14.15a.

1	2	3
2	3	1
3	1	2

a

1	2	3
3	1	2
2	3	1

b

Figure 14.15.

Finally, the multiplication table of the basic negative alternating cycle matrices leads us once again to the basic positive alternating cycle matrix in figure 14.15b.

The positive and negative alternating cycle matrices are intrinsically related; they are neatly ‘interwoven’.

Chapter 15

ALTERNATING CYCLE MATRICES OF DIMENSIONS 7 BY 7

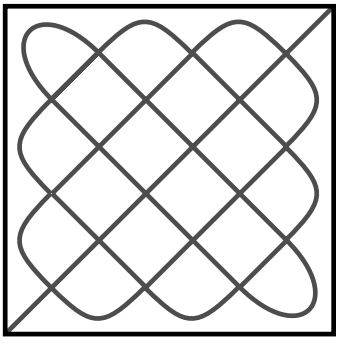
ABSTRACT

In chapter 15 the properties of negative and positive alternating cycle matrices of dimensions 7 by 7 will be analyzed.

ACTIVITIES

- Try to picture positive and negative alternating cycle matrices of dimensions 7 by 7 and study their properties.
- Determine the symmetries that they display.

Similar to the experience of conceiving alternating cycle matrices of dimensions 5 by 5 (chapter 13) and 3 by 3 (chapter 14) we can picture negative and positive alternating cycle matrices of dimensions 7 by 7 as having the general structure and form presented in figures 15.1 and 15.2, respectively.



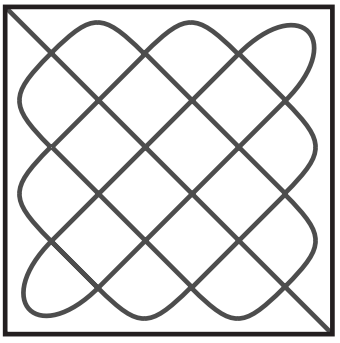
a

a	b	c	d	e	f	g
b	d	a	f	c	g	e
c	a	e	b	g	d	f
d	f	b	g	a	e	c
e	c	g	a	f	b	d
f	g	d	e	b	c	a
g	e	f	c	d	a	b

b

General structure and form of a negative alternating cycle matrix of dimensions 7 by 7

Figure 15.1.



a

a	b	c	d	e	f	g
c	a	e	b	g	d	f
b	d	a	f	c	g	e
e	c	g	a	f	b	d
d	f	b	g	a	e	c
g	e	f	c	d	a	b
f	g	d	e	b	c	a

b

General structure and form of a positive alternating cycle matrix of dimensions 7 by 7

Figure 15.2.

The negative alternating cycle matrices of dimensions 7 by 7 have their principal diagonal as a symmetry axis, whereas the positive alternating cycle matrices of dimensions 7 by 7 have their secondary diagonal as a symmetry axis.

The negative alternating cycle matrices of dimensions 7 by 7 have their secondary diagonal as an anti-symmetry axis (inverting the numbers of the same alternating cycle), whereas positive alternating cycle matrices of dimensions 7 by 7 have their principal diagonal as an anti-symmetry axis.

The sums of the elements of the rows and columns of a negative or positive alternating cycle matrix of dimensions 7 by 7, that has as first row (a, b, c, d, e, f, g), is always equal to $a+b+c+d+e+f+g$. In the case that the seven numbers are different, such a negative or positive alternating cycle matrix constitutes a Latin square.

As in the cases of the even and odd dimensions considered in the previous chapters it is easy to show that the sum of two negative or positive alternating cycle matrices of dimensions 7 by 7 is an alternating cycle matrix of the same type. Also it is easy to show that any multiple of a negative or positive alternating cycle matrix of dimensions 7 by 7 is equally an alternating cycle matrix of the same type. We shall analyze next the products of negative and positive alternating cycle matrices.

ACTIVITIES

- Construct some pairs of negative and positive alternating cycle matrices of dimensions 7 by 7.
- Multiply the matrices of the constructed pairs and analyze the results.

Figure 15.3 presents two negative alternating cycle matrices of dimensions 7 by 7, let us say **A** and **B**.

5	3	0	-1	4	2	-2
3	-1	5	2	0	-2	4
0	5	4	3	-2	-1	2
-1	2	3	-2	5	4	0
4	0	-2	5	2	3	-1
2	-2	-1	4	3	0	5
-2	4	2	0	-1	5	3

A

2	4	1	3	-2	0	-3
4	3	2	0	1	-3	-2
1	2	-2	4	-3	3	0
3	0	4	-3	2	-2	1
-2	1	-3	2	0	4	3
0	-3	3	-2	4	1	2
-3	-2	0	1	3	2	4

B

Figure 15.3.

Calculating the products **AB** and **BA**, we obtain the matrices presented in figure 15.4.

17	31	1	20	-7	7	-14
1	17	-7	31	-14	20	7
31	20	17	7	1	-14	-7
-7	1	-14	17	7	31	20
20	7	31	-14	17	-7	1
-14	-7	7	1	20	17	31
7	-14	20	-7	31	1	17

AB

17	1	31	-7	20	-14	7
31	17	20	1	7	-7	-14
1	-7	17	-14	31	7	20
20	31	7	17	-14	1	-7
-7	-14	1	7	17	20	31
7	20	-14	31	-7	17	1
-14	7	-7	20	1	31	17

BA

Figure 15.4.

As might be expected, the matrices **AB** and **BA** are positive alternating cycle matrices, and they are symmetrical (a reflection of matrix **AB** across its principal diagonal transforms it into matrix **BA**). The alternating cycles of the matrices **AB** and **BA** differ in phase, that is, the numbers of the corresponding cycles are inverted. Splendidly, the negative alternating cycle matrices of dimensions 7 by 7 have the same properties as those of dimensions 3 by 3 and 5 by 5.

Figure 15.5 presents two positive alternating cycle matrices of dimensions 7 by 7 , **C** and **D**, respectively.

4	3	1	-2	-5	6	0
1	4	-5	3	0	-2	6
3	-2	4	6	1	0	-5
-5	1	0	4	6	3	-2
-2	6	3	0	4	-5	1
0	-5	6	1	-2	4	3
6	0	-2	-5	3	1	4

C

3	-4	0	1	4	-2	-1
0	3	4	-4	-1	1	-2
-4	1	3	-2	0	-1	4
4	0	-1	3	-2	-4	1
1	-2	-4	-1	3	4	0
-1	4	-2	0	1	3	-4
-2	-1	1	4	-4	0	3

D

Figure 15.5.

Exactly, as in the case of the other dimensions already considered, the products **CD** and **DC** are equal (figure 15.6), and the matrix resulting from the multiplication is equally a positive alternating cycle matrix. In other words, in this example we have once more ‘positive’ times ‘positive’ equals ‘positive’.

-11	28	25	-11	8	0	-32
25	-11	8	28	-32	-11	0
28	-11	-11	0	25	-32	8
8	25	-32	-11	0	28	-11
-11	0	28	-32	-11	8	25
-32	8	0	25	-11	-11	28
0	-32	-11	8	28	25	-11

CD = **DC**

Figure 15.6.

When we multiply the negative and positive matrices **A** and **C**, we obtain the matrices **AC** and **CA** that are negative alternating cycle matrices (figure 15.7).

8	40	18	7	-25	7	22
40	7	8	7	18	22	-25
18	8	-25	40	22	7	7
7	7	40	22	8	-25	18
-25	18	22	8	7	40	7
7	22	7	-25	40	18	8
22	-25	7	18	7	8	40

AC

23	-2	17	8	12	-22	41
-2	8	23	-22	17	41	12
17	23	12	-2	41	8	-22
8	-22	-2	41	23	12	17
12	17	41	23	-22	-2	8
-22	41	8	12	-2	17	23
41	12	-22	17	8	23	-2

CA

Figure 15.7.

Will the properties observed in this chapter only be the properties of the examples we considered or will they be verified in general? How can we be certain? We are looking for a methodology to lead us to general proofs. In the previous chapter we were successful by giving a proof that used the multiplication tables of the **basic** alternating cycle matrices of dimensions 5 by 5.

Chapter 16

MULTIPLICATION TABLES OF BASIC POSITIVE AND NEGATIVE ALTERNATING CYCLE MATRICES OF DIMENSIONS 7 BY 7

ABSTRACT

In chapter 16 the multiplication tables of basic positive and negative alternating cycle matrices of dimensions 7 by 7 will be constructed and analyzed.

ACTIVITY

- Similar to what was discussed in chapter 14, picture **basic** positive and negative alternating cycle matrices of dimensions 7 by 7 and construct the respective multiplication tables of these basic matrices.

Just as we did in chapter 14 in the case of the dimensions 3 by 3, we can write any positive alternating cycle matrix of dimensions 7 by 7 as the sum of multiples of basic positive alternating cycle matrices.

For the reader accustomed with the language of **linear algebra**, we can say that each positive alternating cycle matrix, of dimensions 7 by 7, can be written as a **linear combination** of the basic positive alternating cycle matrices of dimensions 7 by 7. These basic positive matrices constitute a **basis** of the **vector space** of the positive alternating cycle matrices of dimensions 7 by 7. The considered basis is composed of 7 elements, that is, the **dimension** of this vector space is equal to 7. The dimension of the vector space of all matrices of dimensions 7 by 7 is equal to 49.

The basic positive alternating cycle matrices may be denoted by **P**(1), **P**(2), …, **P**(7), where the index number indicates the place of the first row of the corresponding matrix where the number 1 appears. Figure 16.1 presents the seven basic positive alternating cycle matrices.

$$\begin{bmatrix} \mathbf{1} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \mathbf{1} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \mathbf{1} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \mathbf{1} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \mathbf{1} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \mathbf{1} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \mathbf{1} \end{bmatrix}$$

P(1)

$$\begin{bmatrix} 0 & \mathbf{1} & \mathbf{0} & 0 & 0 & 0 & 0 \\ \mathbf{0} & 0 & 0 & \mathbf{1} & 0 & 0 & 0 \\ \mathbf{1} & 0 & 0 & 0 & \mathbf{0} & 0 & 0 \\ 0 & \mathbf{0} & 0 & 0 & 0 & \mathbf{1} & 0 \\ 0 & 0 & \mathbf{1} & 0 & 0 & 0 & \mathbf{0} \\ 0 & 0 & 0 & \mathbf{0} & 0 & 0 & \mathbf{1} \\ 0 & 0 & 0 & 0 & \mathbf{1} & \mathbf{0} & 0 \end{bmatrix}$$

P(2)

$$\begin{bmatrix} 0 & \mathbf{0} & \mathbf{1} & 0 & 0 & 0 & 0 \\ \mathbf{1} & 0 & 0 & \mathbf{0} & 0 & 0 & 0 \\ \mathbf{0} & 0 & 0 & 0 & \mathbf{1} & 0 & 0 \\ 0 & \mathbf{1} & 0 & 0 & 0 & \mathbf{0} & 0 \\ 0 & 0 & \mathbf{0} & 0 & 0 & 0 & \mathbf{1} \\ 0 & 0 & 0 & \mathbf{1} & 0 & 0 & \mathbf{0} \\ 0 & 0 & 0 & 0 & \mathbf{0} & \mathbf{1} & 0 \end{bmatrix}$$

P(3)

$$\begin{bmatrix} 0 & 0 & 0 & \mathbf{1} & \mathbf{0} & 0 & 0 \\ 0 & 0 & \mathbf{0} & 0 & 0 & \mathbf{1} & 0 \\ 0 & \mathbf{1} & 0 & 0 & 0 & 0 & \mathbf{0} \\ \mathbf{0} & 0 & 0 & 0 & 0 & 0 & \mathbf{1} \\ \mathbf{1} & 0 & 0 & 0 & 0 & \mathbf{0} & 0 \\ 0 & \mathbf{0} & 0 & 0 & \mathbf{1} & 0 & 0 \\ 0 & 0 & \mathbf{1} & \mathbf{0} & 0 & 0 & 0 \end{bmatrix}$$

P(4)

$$\begin{bmatrix} 0 & 0 & 0 & \mathbf{0} & \mathbf{1} & 0 & 0 \\ 0 & 0 & \mathbf{1} & 0 & 0 & \mathbf{0} & 0 \\ 0 & \mathbf{0} & 0 & 0 & 0 & 0 & \mathbf{1} \\ \mathbf{1} & 0 & 0 & 0 & 0 & 0 & \mathbf{0} \\ \mathbf{0} & 0 & 0 & 0 & 0 & \mathbf{1} & 0 \\ 0 & \mathbf{1} & 0 & 0 & \mathbf{0} & 0 & 0 \\ 0 & 0 & \mathbf{0} & \mathbf{1} & 0 & 0 & 0 \end{bmatrix}$$

P(5)

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & \mathbf{1} & \mathbf{0} \\ 0 & 0 & 0 & 0 & \mathbf{0} & 0 & \mathbf{1} \\ 0 & 0 & 0 & \mathbf{1} & 0 & \mathbf{0} & 0 \\ 0 & 0 & \mathbf{0} & 0 & \mathbf{1} & 0 & 0 \\ 0 & \mathbf{1} & 0 & \mathbf{0} & 0 & 0 & 0 \\ \mathbf{0} & 0 & \mathbf{1} & 0 & 0 & 0 & 0 \\ \mathbf{1} & \mathbf{0} & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

P(6)

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & \mathbf{0} & \mathbf{1} \\ 0 & 0 & 0 & 0 & \mathbf{1} & 0 & \mathbf{0} \\ 0 & 0 & 0 & \mathbf{0} & 0 & \mathbf{1} & 0 \\ 0 & 0 & \mathbf{1} & 0 & \mathbf{0} & 0 & 0 \\ 0 & \mathbf{0} & 0 & \mathbf{1} & 0 & 0 & 0 \\ \mathbf{1} & 0 & \mathbf{0} & 0 & 0 & 0 & 0 \\ \mathbf{0} & \mathbf{1} & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

P(7)

The seven basic positive alternating cycle matrices of dimensions 7x7

Figure 16.1.

Matrix **P**(1) is the identity matrix. Observe that **P**(2) and **P**(3) are symmetrical (a reflection across the principal diagonal of the matrix transforms **P**(2) in **P**(3) and vice versa). In matrix theory it is said that matrix **P**(3) is the **transposed** of matrix **P**(2) and one denotes

P(3) = **P**(2)^T. In the same way, **P**(5) = **P**(4)^T and **P**(7) = **P**(6)^T. The only non-zero cycles of **P**(2) and **P**(3) differ in phase. The same happens with the couple **P**(4) and **P**(5) and with the pair **P**(6) and **P**(7), respectively.

As an example, the positive alternating cycle matrix presented in figure 16.2 can be written as 4 **P**(1) + 3 **P**(2) + **P**(3) – 2 **P**(4) – 5 **P**(5) + 6 **P**(6) –3 **P**(7).

4	3	1	-2	-5	6	-3
1	4	-5	3	-3	-2	6
3	-2	4	6	1	-3	-5
-5	1	-3	4	6	3	-2
-2	6	3	-3	4	-5	1
-3	-5	6	1	-2	4	3
6	-3	-2	-5	3	1	4

Figure 16.2.

All negative alternating cycle matrices of dimensions 7 by 7 can be written as sums of multiples of basic negative alternating cycle matrices. Figure 16.3 presents the seven basic negative alternating cycle matrices of dimensions 7 by 7.

1	**0**	0	0	0	0	0
0	0	**1**	0	0	0	0
0	**1**	0	**0**	0	0	0
0	0	**0**	0	**1**	0	0
0	0	0	**1**	0	**0**	0
0	0	0	0	**0**	0	**1**
0	0	0	0	0	**1**	**0**

N(1)

0	**1**	0	0	0	0	0
1	0	**0**	0	0	0	0
0	**0**	0	**1**	0	0	0
0	0	**1**	0	**0**	0	0
0	0	0	**0**	0	**1**	0
0	0	0	0	**1**	0	**0**
0	0	0	0	0	**0**	**1**

N(2)

0	0	**1**	**0**	0	0	0
0	**0**	0	0	**1**	0	0
1	0	0	0	0	**0**	0
0	0	0	0	0	0	**1**
0	**1**	0	0	0	0	**0**
0	0	**0**	0	0	**1**	0
0	0	0	**1**	**0**	0	0

N(3)

0	0	**0**	**1**	0	0	0
0	**1**	0	0	**0**	0	0
0	0	0	0	0	**1**	0
1	0	0	0	0	0	**0**
0	**0**	0	0	0	0	**1**
0	0	**1**	0	0	**0**	0
0	0	0	**0**	**1**	0	0

N(4)

0	0	0	0	**1**	**0**	0
0	0	0	**0**	0	0	**1**
0	0	**1**	0	0	0	**0**
0	**0**	0	0	0	**1**	0
1	0	0	0	**0**	0	0
0	0	0	**1**	0	0	0
0	**1**	**0**	0	0	0	0

N(5)

0	0	0	0	**0**	**1**	0
0	0	0	**1**	0	0	**0**
0	0	**0**	0	0	0	**1**
0	**1**	0	0	0	**0**	0
0	0	0	0	**1**	0	0
1	0	0	**0**	0	0	0
0	**0**	**1**	0	0	0	0

N(6)

0	0	0	0	0	0	**1**
0	0	0	0	0	**1**	0
0	0	0	0	**1**	0	0
0	0	0	**1**	0	0	0
0	0	**1**	0	0	0	0
0	**1**	0	0	0	0	0
1	0	0	0	0	0	0

N(7)

The seven basic negative alternating cycle matrices

Figure 16.3.

The matrices **N**(1), **N**(2), …, **N**(7) are symmetrical. The only non-zero cycles of **N**(1) and **N**(2) (bold in figure 16.3) differ in phase. The same phenomenon may be verified in the case of the pair of matrices **N**(3) and **N**(4) and of the pair **N**(5) and **N**(6).

ACTIVITIES

- Construct a multiplication table of the basic positive alternating cycle matrices of dimensions 7 by 7.
- Construct a multiplication table of the basic negative alternating cycle matrices of dimensions 7 by 7.
- Construct two multiplication tables of the basic positive and negative alternating cycle matrices of dimensions 7 by 7.

When we multiply two arbitrary basic positive alternating cycle matrices of dimensions 7 by 7, we observe the surprising fact that there does not appear to be any specific matrix but we always obtain a basic positive alternating cycle matrix. The same phenomenon has already occurred in the case of dimensions 3 by 3 (See chapter 14).

Figure 16.4 presents the multiplication table of the basic positive alternating cycle matrices of dimensions 7 by 7.

x	**P(1)**	**P(2)**	**P(3)**	**P(4)**	**P(5)**	**P(6)**	**P(7)**
P(1)	P(1)	P(2)	P(3)	P(4)	P(5)	P(6)	P(7)
P(2)	P(2)	P(4)	P(1)	P(6)	P(3)	P(7)	P(5)
P(3)	P(3)	P(1)	P(5)	P(2)	P(7)	P(4)	P(6)
P(4)	P(4)	P(6)	P(2)	P(7)	P(1)	P(5)	P(3)
P(5)	P(5)	P(3)	P(7)	P(1)	P(6)	P(2)	P(4)
P(6)	P(6)	P(7)	P(4)	P(5)	P(2)	P(3)	P(1)
P(7)	P(7)	P(5)	P(6)	P(3)	P(4)	P(1)	P(2)

Multiplication table of the basic positive alternating cycle matrices of dimensions 7 by 7

Figure 16.4.

Let **A** and **B** be any two positive alternating cycle matrices of dimensions 7 by 7. Both **A** and **B** may be written as the sum of multiples of basic positive alternating cycle matrices of dimensions 7 by 7. Hence, the multiplication of **A** and **B** corresponds to the multiplication of two sums of multiples of basic positive alternating cycle matrices of dimensions 7 by 7, and, in consequence, the product **AB** itself corresponds to the sum of multiples of products of basic positive alternating cycle matrices of dimensions 7 by 7. Conforming with the multiplication table (figure 16.4), these last products are basic positive alternating cycle matrices themselves. In this way, **AB** is equal to the sum of certain multiples of basic positive alternating cycle matrices. Any sum of multiples of positive alternating cycle matrices is also a positive alternating cycle matrix.

Thus, we arrive at the conclusion that **AB** is a positive alternating cycle matrix of dimensions 7 by 7.

Let us now compare **AB** and **BA**. When analyzing **AB** we encounter parts of the type m**P**(r) x **P**(s), where m is any number and r and s any natural numbers from 1 to 7. When analyzing **BA**, however, we find a part m**P**(s) x **P**(r) instead of the part m**P**(r) x **P**(s). According to the symmetrical multiplication table (figure 16.4), we have always **P**(r) x **P**(s) = **P**(s) x **P**(r). Being that all corresponding parts are equal, we conclude that the sums are also equal, that is, we have **AB** = **BA**.

In this way we proved the following theorem:

If **A** and **B** are any two positive alternating cycle matrices of dimensions 7 by 7, then **AB** and **BA** are positive alternating cycle matrices of the same dimensions and **AB** = **BA**.

Let us consider the multiplication of matrices **A** and **C**, where **A** is any positive alternating cycle matrix of dimensions 7 by 7 and **C** is any negative alternating cycle matrix of the same dimensions. Matrix **A** may be written as the sum of certain multiples of basic positive alternating cycle matrices, whereas matrix **C** may be written as the sum of certain multiples of basic negative alternating cycle matrices. The calculation of the products **AC** and **CA** corresponds to the multiplication of these two sums of multiples of basic positive and basic negative alternating cycle matrices, that is, it corresponds to sums of parts of the types m**P**(r) x **N**(s) and m**N**(s) x **P**(r), respectively. We can construct the multiplication tables of the basic positive times basic negative alternating cycle matrices (figure 16.5) and of the basic negative times the basic positive alternating cycle matrices (figure 16.6).

x	**N(1)**	**N(2)**	**N(3)**	**N(4)**	**N(5)**	**N(6)**	**N(7)**
P(1)	N(1)	N(2)	N(3)	N(4)	N(5)	N(6)	N(7)
P(2)	N(3)	N(1)	N(5)	N(2)	N(7)	N(4)	N(6)
P(3)	N(2)	N(4)	N(1)	N(6)	N(3)	N(7)	N(5)
P(4)	N(5)	N(3)	N(7)	N(1)	N(6)	N(2)	N(4)
P(5)	N(4)	N(6)	N(2)	N(7)	N(1)	N(5)	N(3)
P(6)	N(7)	N(5)	N(6)	N(3)	N(4)	N(1)	N(2)
P(7)	N(6)	N(7)	N(4)	N(5)	N(2)	N(3)	N(1)

Multiplication table of the basic positive times the basic negative alternating cycle matrices of dimensions 7 by 7

Figure 16.5.

x	P(1)	P(2)	P(3)	P(4)	P(5)	P(6)	P(7)
N(1)	N(1)	N(2)	N(3)	N(4)	N(5)	N(6)	N(7)
N(2)	N(2)	N(4)	N(1)	N(6)	N(3)	N(7)	N(5)
N(3)	N(3)	N(1)	N(5)	N(2)	N(7)	N(4)	N(6)
N(4)	N(4)	N(6)	N(2)	N(7)	N(1)	N(5)	N(3)
N(5)	N(5)	N(3)	N(7)	N(1)	N(6)	N(2)	N(4)
N(6)	N(6)	N(7)	N(4)	N(5)	N(2)	N(3)	N(1)
N(7)	N(7)	N(5)	N(6)	N(3)	N(4)	N(1)	N(2)

Multiplication table of the basic negative times as basic positive alternating cycle matrices of dimensions 7 by 7

Figure 16.6.

As in the case of matrices with dimensions 3 by 3 analyzed in chapter 14, we see that the products of any basic negative and basic positive alternating cycle matrices of dimensions 7 by 7 is always a basic negative alternating cycle matrix of the same dimensions. By consequence, the products **AC** and **CA** are negative alternating cycle matrices of dimensions 7 by 7. We have already seen in the examples of the previous chapter that **AC** and **CA** are, in general, not equal. Indeed, when we compare the two multiplication tables we see that **N**(2) x **P**(3) and **P**(3) x **N**(2) are different, etc.

We have proved in this manner the following theorem:

If **A** is a positive alternating cycle matrix of dimensions 7 by 7 and **C** is a negative alternating cycle matrix of dimensions 7 by 7, then **AC** and **CA** are negative alternating cycle matrices of the same dimensions.

Let us now look at the multiplication of negative alternating cycle matrices. Let **C** and **D** be two negative alternating cycle matrices of dimensions 7 by 7. Both **C** and **D** may be written as a sum of certain multiples of the seven basic negative alternating cycle matrices. Determining the products **CD** and **DC** corresponds to the elaboration of the products of the two considered sums. Thus, the matrices **CD** and **DC** are sums of multiples of the products of the seven basic negative alternating cycle matrices.

Calculating the product of any two basic negative alternating cycle matrices of dimensions 7 by 7, we observe the interesting phenomenon that such a product is always a **basic positive** alternating cycle matrix, as observed in the case of matrices with dimensions 3 by 3 (Chapter 14). Figure 16.7 presents the multiplication table of the basic negative alternating cycle matrices of dimensions 7 by 7.

x	N(1)	N(2)	N(3)	N(4)	N(5)	N(6)	N(7)
N(1)	P(1)	P(2)	P(3)	P(4)	P(5)	P(6)	P(7)
N(2)	P(3)	P(1)	P(5)	P(2)	P(7)	P(4)	P(6)
N(3)	P(2)	P(4)	P(1)	P(6)	P(3)	P(7)	P(5)
N(4)	P(5)	P(3)	P(7)	P(1)	P(6)	P(2)	P(4)
N(5)	P(4)	P(6)	P(2)	P(7)	P(1)	P(5)	P(3)
N(6)	P(7)	P(5)	P(6)	P(3)	P(4)	P(1)	P(2)
N(7)	P(6)	P(7)	P(4)	P(5)	P(2)	P(3)	P(1)

Multiplication table of the basic negative alternating cycle matrices of dimensions 7 by 7

Figure 16.7.

By consequence, the matrices **CD** and **DC** are sums of multiples of the seven basic positive alternating cycle matrices and hence **CD** and **DC** are positive alternating cycle matrices.

When we compare **CD** and **DC**, we take into account that **CD** is the sum of parts of the type m**N**(r) x **N**(s) whereas **DC** is the sum of parts of the type m**N**(s) x **N**(r). Let us compare first the product **N**(r) x **N**(s) with **N**(s) x **N**(r).

For instance, we have **N**(4) x **N**(2) = **P**(3) whereas **N**(2) x **N**(4) = **P**(2). We see that the two products are different. We already know, however, that the matrices **P**(3) and **P**(2) are symmetrical and we had said that **P**(3) was the transposed matrix of **P**(2): **P**(3) = **P**(2)^T. In this way, we have, in the case of the example under consideration

N(2) x **N**(4) = {**N**(4) x **N**(2)}^{T}.

Following the same reasoning we can present the multiplication table of the basic negative alternating cycle matrices in an alternative way (figure 16.8).

x	**N(1)**	**N(2)**	**N(3)**	**N(4)**	**N(5)**	**N(6)**	**N(7)**
N(1)	P(1)	P(2)	P(3)	P(4)	P(5)	P(6)	P(7)
N(2)	P(2)^T	P(1)	P(5)	P(2)	P(7)	P(4)	P(6)
N(3)	P(3)^T	P(5)^T	P(1)	P(6)	P(3)	P(7)	P(5)
N(4)	P(4)^T	P(2)^T	P(6)^T	P(1)	P(6)	P(2)	P(4)
N(5)	P(5)^T	P(7)^T	P(3)^T	P(6)^T	P(1)	P(5)	P(3)
N(6)	P(6)^T	P(4)^T	P(7)^T	P(2)^T	P(5)^T	P(1)	P(2)
N(7)	P(7)^T	P(6)^T	P(5)^T	P(4)^T	P(3)^T	P(2)^T	P(1)

Alternative presentation of the multiplication table of the basic negative alternating cycle matrices of dimensions 7 by 7

Figure 16.8.

In general, we have

N(r) x **N**(s) = {**N**(s) x **N**(r)}^{T}.

Since **CD** is the sum of certain multiples **N**(r) x **N**(s) and **DC** is the sum of the same multiples of **N**(s) x **N**(r), we can conclude that

CD = (**DC**)^T,

that is, the matrices **CD** and **DC** are mutually symmetrical. The matrix **CD** is composed of alternating cycles. Reflecting matrix **CD** across its principal diagonal, the order of the numbers in its cycles is inverted. Since the result of the same reflection is matrix **DC**, we arrive at the conclusion that the cycles of **CD** and **DC** differ in phase.

We have proved the following beautiful theorem:

If **C** and **D** are any two negative alternating cycle matrices of dimensions 7 by 7, then **CD** and **DC** are positive alternating cycle matrices of the same dimensions and **CD** and **DC** are mutually symmetrical where the cycles of **CD** and **DC** differ in phase.

ACTIVITIES

- Compare the multiplication tables of the basic positive alternating cycle matrices (figure 16.4) with the multiplication tables of the basic negative times the basic positive alternating cycle matrices (figure 16.6). What is the similarity?
- Compare the multiplication tables of the basic negative alternating cycle matrices (figure 16.7) with the multiplication tables of the basic positive times the basic negative alternating cycle matrices (figure 16.5). What is the similarity?
- In chapter 14 we constructed matrices starting with the multiplication tables of basic positive and negative alternating cycle matrices of dimensions 3 by 3. Repeat with the case of dimensions 7 by 7.

ACTIVITY

- We succeeded in proving three theorems in the case of dimensions 3 by 3 and 7 by 7. Will an analogous reasoning be possible in the case of even dimensions? Experiment with the case of dimensions 6 by 6.

In the following chapters we shall return to the questions posed in the last activities.

Chapter 17

CYCLIC STRUCTURE OF MULTIPLICATION TABLES

ABSTRACT

In chapter 17 the structure of the multiplication tables of the basic positive and negative alternating cycle matrices of dimensions 7 by 7 will be analyzed.

In the previous chapter we constructed the multiplication tables of the basic positive and negative alternating cycle matrices of dimensions 7 by 7. Now we shall analyze the structure of these four tables. As we did in chapter 14 for the case of dimensions 3 by 3, we can generate a matrix starting with each multiplication table. Figure 17.1 presents a generated matrix based on the multiplication table of the basic positive alternating cycle matrices of dimensions 7 by 7: from *b* to *c* we isolate the index numbers that appear between parenthesis.

x	**P(1)**	**P(2)**	**P(3)**	**P(4)**	**P(5)**	**P(6)**	**P(7)**
P(1)	P(1)	P(2)	P(3)	P(4)	P(5)	P(6)	P(7)
P(2)	P(2)	P(4)	P(1)	P(6)	P(3)	P(7)	P(5)
P(3)	P(3)	P(1)	P(5)	P(2)	P(7)	P(4)	P(6)
P(4)	P(4)	P(6)	P(2)	P(7)	P(1)	P(5)	P(3)
P(5)	P(5)	P(3)	P(7)	P(1)	P(6)	P(2)	P(4)
P(6)	P(6)	P(7)	P(4)	P(5)	P(2)	P(3)	P(1)
P(7)	P(7)	P(5)	P(6)	P(3)	P(4)	P(1)	P(2)

a

↓

P(1)	P(2)	P(3)	P(4)	P(5)	P(6)	P(7)
P(2)	P(4)	P(1)	P(6)	P(3)	P(7)	P(5)
P(3)	P(1)	P(5)	P(2)	P(7)	P(4)	P(6)
P(4)	P(6)	P(2)	P(7)	P(1)	P(5)	P(3)
P(5)	P(3)	P(7)	P(1)	P(6)	P(2)	P(4)
P(6)	P(7)	P(4)	P(5)	P(2)	P(3)	P(1)
P(7)	P(5)	P(6)	P(3)	P(4)	P(1)	P(2)

B

Figure 17.1 continued on next page.

↓

1	2	3	4	5	6	7
2	4	1	6	3	7	5
3	1	5	2	7	4	6
4	6	2	7	1	5	3
5	3	7	1	6	2	4
6	7	4	5	2	3	1
7	5	6	3	4	1	2

c

Generation of a matrix starting with the multiplication table of the basic positive alternating cycle matrices of dimensions 7 by 7

Figure 17.1.

The matrix in figure 17.1c is a cycle matrix too, and more specifically, it is a negative alternating cycle matrix of dimensions 7 by 7. The multiplication table of the basic negative times the basic positive alternating cycle matrices (figure 16.6) leads us to the same matrix in figure 17.1c.

Surprisingly, the notion of a negative alternating cycle matrix encounters an application in the analysis of two multiplication tables of basic alternating cycle matrices of dimensions 7 by 7.

In the same way the notion of positive alternating cycle matrix has an application in the study of the other two multiplication tables of basic alternating cycle matrices of dimensions 7 by 7. In fact, both multiplication tables of the basic negative alternating cycle matrices of dimensions 7 by 7 (figure 16.7) and of the basic positive times as basic negative alternating cycle matrices (figure 16.5) lead us to the positive alternating cycle matrix presented in figure 17.2.

1	2	3	4	5	6	7
3	1	5	2	7	4	6
2	4	1	6	3	7	5
5	3	7	1	6	2	4
4	6	2	7	1	5	3
7	5	6	3	4	1	2
6	7	4	5	2	3	1

Matrix generated from the multiplication table of the basic negative alternating cycle matrices of dimensions 7 by 7

Figure 17.2.

ACTIVITIES

- Similar to what has been discussed in the last chapters, picture the notion of basic positive and negative alternating cycle matrices of dimensions 6 by 6 and construct the respective multiplication tables of these basic matrices.
- Generate matrices from the multiplication tables of the basic positive and negative alternating cycle matrices of dimensions 6 by 6. Analyze the structures of the obtained matrices and compare them with the ones obtained in the last chapter in the case of dimensions 7 by 7.

QUESTIONS FOR REFLECTION

Let n be any natural number. Will the matrices, generated from the four multiplication tables of the basic positive and negative alternating cycle matrices of dimensions n by n, always have the structure of positive and negative alternating cycle matrices of dimensions n by n?

Chapter 18

MULTIPLICATION TABLES OF BASIC POSITIVE AND NEGATIVE ALTERNATING CYCLE MATRICES OF DIMENSIONS 6 BY 6

ABSTRACT

In chapter 18 the multiplication tables of basic positive and negative alternating cycle matrices of dimensions 6 by 6 will be constructed and analyzed.

In chapter 16 we saw that the particular structure of the multiplication tables of the basic positive and negative cycle matrices with dimensions 7 by 7 was the decisive factor in proving all three theorems involving the multiplication of alternating cycle matrices. Let us now analyze this structure in the case of an example of even dimensions, the case of dimensions 6 by 6.

ACTIVITY

- Similar to the analysis in chapter 16, picture **basic** positive and negative alternating cycle matrices of dimensions 6 by 6 and construct the respective multiplication tables of those basic matrices.

Let us denote the basic positive alternating cycle matrices, of dimensions 6 by 6, by **P**(1), **P**(2), …, **P**(6), where the number between parentheses indicates the place of the number 1 in the first row of the corresponding matrix. Figure 18.1 presents the six basic positive alternating cycle matrices of dimensions 6 by 6.

Matrix **P**(1) is the identity matrix. Observe that **P**(2) and **P**(3) are symmetrical where a reflection across the principal diagonal of matrix **P**(2) transforms it into **P**(3) and vice versa. Hence $\mathbf{P}(3) = \mathbf{P}(2)^T$. In the same way $\mathbf{P}(5) = \mathbf{P}(4)^T$. The only non-zero cycles of **P**(2) and **P**(3) (bold in figure 18.1) differ in phase. The same occurs with the pair **P**(4) and **P**(5). Matrix **P**(6) is symmetrical, and thus $\mathbf{P}(6) = \mathbf{P}(6)^T$.

$$\begin{bmatrix} \mathbf{1} & 0 & 0 & 0 & 0 & 0 \\ 0 & \mathbf{1} & 0 & 0 & 0 & 0 \\ 0 & 0 & \mathbf{1} & 0 & 0 & 0 \\ 0 & 0 & 0 & \mathbf{1} & 0 & 0 \\ 0 & 0 & 0 & 0 & \mathbf{1} & 0 \\ 0 & 0 & 0 & 0 & 0 & \mathbf{1} \end{bmatrix}$$

P(1)

$$\begin{bmatrix} 0 & \mathbf{1} & \mathbf{0} & 0 & 0 & 0 \\ \mathbf{0} & 0 & 0 & \mathbf{1} & 0 & 0 \\ \mathbf{1} & 0 & 0 & 0 & \mathbf{0} & 0 \\ 0 & \mathbf{0} & 0 & 0 & 0 & \mathbf{1} \\ 0 & 0 & \mathbf{1} & 0 & 0 & \mathbf{0} \\ 0 & 0 & 0 & \mathbf{0} & \mathbf{1} & 0 \end{bmatrix}$$

P(2)

$$\begin{bmatrix} 0 & \mathbf{0} & \mathbf{1} & 0 & 0 & 0 \\ \mathbf{1} & 0 & 0 & \mathbf{0} & 0 & 0 \\ \mathbf{0} & 0 & 0 & 0 & \mathbf{1} & 0 \\ 0 & \mathbf{1} & 0 & 0 & 0 & \mathbf{0} \\ 0 & 0 & \mathbf{0} & 0 & 0 & \mathbf{1} \\ 0 & 0 & 0 & \mathbf{1} & \mathbf{0} & 0 \end{bmatrix}$$

P(3)

$$\begin{bmatrix} 0 & 0 & 0 & \mathbf{1} & \mathbf{0} & 0 \\ 0 & 0 & \mathbf{0} & 0 & 0 & \mathbf{1} \\ 0 & \mathbf{1} & 0 & 0 & 0 & \mathbf{0} \\ \mathbf{0} & 0 & 0 & 0 & \mathbf{1} & 0 \\ \mathbf{1} & 0 & 0 & \mathbf{0} & 0 & 0 \\ 0 & \mathbf{0} & \mathbf{1} & 0 & 0 & 0 \end{bmatrix}$$

P(4)

$$\begin{bmatrix} 0 & 0 & 0 & \mathbf{0} & \mathbf{1} & 0 \\ 0 & 0 & \mathbf{1} & 0 & 0 & \mathbf{0} \\ 0 & \mathbf{0} & 0 & 0 & 0 & \mathbf{1} \\ \mathbf{1} & 0 & 0 & 0 & \mathbf{0} & 0 \\ \mathbf{0} & 0 & 0 & \mathbf{1} & 0 & 0 \\ 0 & \mathbf{1} & \mathbf{0} & 0 & 0 & 0 \end{bmatrix}$$

P(5)

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & \mathbf{1} \\ 0 & 0 & 0 & 0 & \mathbf{1} & 0 \\ 0 & 0 & 0 & \mathbf{1} & 0 & 0 \\ 0 & 0 & \mathbf{1} & 0 & 0 & 0 \\ 0 & \mathbf{1} & 0 & 0 & 0 & 0 \\ \mathbf{1} & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

P(6)

The six basic positive alternating cycle matrices of dimensions 6 by 6

Figure 18.1.

Figure 18.2 presents the six basic negative alternating cycle matrices of dimensions 6 by 6.

$$\begin{bmatrix} \mathbf{1} & \mathbf{0} & 0 & 0 & 0 & 0 \\ \mathbf{0} & 0 & \mathbf{1} & 0 & 0 & 0 \\ 0 & \mathbf{1} & 0 & \mathbf{0} & 0 & 0 \\ 0 & 0 & \mathbf{0} & 0 & \mathbf{1} & 0 \\ 0 & 0 & 0 & \mathbf{1} & 0 & \mathbf{0} \\ 0 & 0 & 0 & 0 & \mathbf{0} & \mathbf{1} \end{bmatrix}$$

N(1)

$$\begin{bmatrix} \mathbf{0} & \mathbf{1} & 0 & 0 & 0 & 0 \\ \mathbf{1} & 0 & \mathbf{0} & 0 & 0 & 0 \\ 0 & \mathbf{0} & 0 & \mathbf{1} & 0 & 0 \\ 0 & 0 & \mathbf{1} & 0 & \mathbf{0} & 0 \\ 0 & 0 & 0 & \mathbf{0} & 0 & \mathbf{1} \\ 0 & 0 & 0 & 0 & \mathbf{1} & \mathbf{0} \end{bmatrix}$$

N(2)

Figure 18.2 continued on next page.

0	0	**1**	**0**	0	0
0	**0**	0	0	**1**	0
1	0	0	0	0	**0**
0	0	0	0	0	**1**
0	**1**	0	0	**0**	0
0	0	**0**	**1**	0	0

N(3)

0	0	**0**	**1**	0	0
0	**1**	0	0	**0**	0
0	0	0	0	0	**1**
1	0	0	0	0	**0**
0	**0**	0	0	**1**	0
0	0	**1**	**0**	0	0

N(4)

0	0	0	0	**1**	**0**
0	0	0	**0**	0	**1**
0	0	**1**	0	**0**	0
0	**0**	0	**1**	0	0
1	0	**0**	0	0	0
0	**1**	0	0	0	0

N(5)

0	0	0	0	**0**	**1**
0	0	0	**1**	0	**0**
0	0	**0**	0	**1**	0
0	**1**	0	**0**	0	0
0	0	**1**	0	0	0
1	**0**	0	0	0	0

N(6)

The six negative basic alternating cycle matrices of dimensions 6 by 6

Figure 18.2.

The matrices **N**(1), **N**(2), …, **N**(6) are symmetrical. The only non-zero cycles of **N**(1) and **N**(2) (bold in figure 18.2) differ in phase. The same phenomenon is verified in the case of the pair of matrices **N**(3) and **N**(4) and of the pair **N**(5) and **N**(6), respectively.

Once we picture the basic positive and negative alternating cycle matrices of dimensions 6 by 6, we can construct the four multiplication tables.

Figure 18.3 presents the multiplication table of the basic positive alternating cycle matrices of dimensions 6 by 6.

x	**P(1)**	**P(2)**	**P(3)**	**P(4)**	**P(5)**	**P(6)**
P(1)	P(1)	P(2)	P(3)	P(4)	P(5)	P(6)
P(2)	P(2)	P(4)	P(1)	P(6)	P(3)	P(5)
P(3)	P(3)	P(1)	P(5)	P(2)	P(6)	P(4)
P(4)	P(4)	P(6)	P(2)	P(5)	P(1)	P(3)
P(5)	P(5)	P(3)	P(6)	P(1)	P(4)	P(2)
P(6)	P(6)	P(5)	P(4)	P(3)	P(2)	P(1)

Multiplication table of the basic positive alternating cycle matrices of dimensions 6 by 6

Figure 18.3.

Figures 18.4 and 18.5 present the multiplication tables of the basic positive times the basic negative alternating cycle matrices and of the basic negative times the basic positive alternating cycle matrices.

x	**N(1)**	**N(2)**	**N(3)**	**N(4)**	**N(5)**	**N(6)**
P(1)	N(1)	N(2)	N(3)	N(4)	N(5)	N(6)
P(2)	N(3)	N(1)	N(5)	N(2)	N(6)	N(4)
P(3)	N(2)	N(4)	N(1)	N(6)	N(3)	N(5)
P(4)	N(5)	N(3)	N(6)	N(1)	N(4)	N(2)
P(5)	N(4)	N(6)	N(2)	N(5)	N(1)	N(3)
P(6)	N(6)	N(5)	N(4)	N(3)	N(2)	N(1)

Multiplication table of the basic positive times the basic negative alternating cycle matrices of dimensions 6 by 6

Figure 18.4.

x	P(1)	P(2)	P(3)	P(4)	P(5)	P(6)
N(1)	N(1)	N(2)	N(3)	N(4)	N(5)	N(6)
N(2)	N(2)	N(4)	N(1)	N(6)	N(3)	N(5)
N(3)	N(3)	N(1)	N(5)	N(2)	N(6)	N(4)
N(4)	N(4)	N(6)	N(2)	N(5)	N(1)	N(3)
N(5)	N(5)	N(3)	N(6)	N(1)	N(4)	N(2)
N(6)	N(6)	N(5)	N(4)	N(3)	N(2)	N(1)

Multiplication table of the basic negative times the basic positive alternating cycle matrices of dimensions 6 by 6

Figure 18.5.

Figure 18.6 presents the multiplication table of the basic negative alternating cycle matrices of dimensions 6 by 6.

x	N(1)	N(2)	N(3)	N(4)	N(5)	N(6)
N(1)	P(1)	P(2)	P(3)	P(4)	P(5)	P(6)
N(2)	P(3)	P(1)	P(5)	P(2)	P(6)	P(4)
N(3)	P(2)	P(4)	P(1)	P(6)	P(3)	P(5)
N(4)	P(5)	P(3)	P(6)	P(1)	P(4)	P(2)
N(5)	P(4)	P(6)	P(2)	P(5)	P(1)	P(3)
N(6)	P(6)	P(5)	P(4)	P(3)	P(2)	P(1)

Multiplication table of the basic negative alternating cycle matrices of dimensions 6 by 6

Figure 18.6.

As in the case with dimensions 7 by 7, the multiplication table of the basic negative alternating cycle matrices of dimensions 6 by 6 can be presented in an alternative way that takes into account the symmetrical pairs $P(3) = P(2)^T$, etc. (figure 18.7).

x	N(1)	N(2)	N(3)	N(4)	N(5)	N(6)
N(1)	P(1)	P(2)	P(3)	P(4)	P(5)	P(6)
N(2)	$P(2)^T$	P(1)	P(5)	P(2)	P(6)	P(4)
N(3)	$P(3)^T$	$P(5)^T$	P(1)	P(6)	P(3)	P(5)
N(4)	$P(4)^T$	$P(2)^T$	$P(6)^T$	P(1)	P(4)	P(2)
N(5)	$P(5)^T$	$P(6)^T$	$P(3)^T$	$P(4)^T$	P(1)	P(3)
N(6)	$P(6)^T$	$P(4)^T$	$P(5)^T$	$P(2)^T$	$P(3)^T$	P(1)

Alternative presentation of the multiplication table of the basic negative alternating cycle matrices of dimensions 6 by 6

Figure 18.7.

Summarizing, we can say that the products of the basic positive and negative alternating cycle matrices of dimensions 6 by 6 are always basic matrices of the same dimensions. The results satisfy the rules of ‘signs’: ‘positive’ times ‘positive’ equals ‘positive’; ‘positive’ times ‘negative’ equals ‘negative’; ‘negative’ times ‘positive’ equals ‘negative’ and ‘negative’ times ‘negative’ equals ‘positive’. The matrices associated with the multiplication tables are negative cycle matrices (figure 18.8a) as in the case of the multiplication tables of ‘positive’ times ‘positive’ (figure 18.3) and of ‘negative’ times ‘positive’ (figure 18.5).

Whereas they are positive cycle matrices (figure 18.8b) in the case of the multiplication tables of ‘positive’ times ‘negative’ (figure 18.4) and ‘negative’ times ‘negative’ (figure 18.6).

1	2	3	4	5	6
2	4	1	6	3	5
3	1	5	2	6	4
4	6	2	5	1	3
5	3	6	1	4	2
6	5	4	3	2	1

a

1	2	3	4	5	6
3	1	5	2	6	4
2	4	1	6	3	5
5	3	6	1	4	2
4	6	2	5	1	3
6	5	4	3	2	1

b

Cyclic structures of the multiplication tables

Figure 18.8.

The situation encountered in this chapter concerning the multiplication of basic positive and negative alternating cycle matrices of dimensions 6 by 6 is similar in all respects to the situations previously encountered in the cases of dimensions 3 by 3 (Chapter 14) and of dimensions 7 by 7 (Chapters 15 to 17), only the dimensions differ. By consequence, a reasoning similar to the one used in chapter 16 leads us to a proof of the following theorems:

(1) If **A** and **B** are any two positive alternating cycle matrices of dimensions 6 by 6, then **AB** and **BA** are positive alternating cycle matrices of the same dimensions and **AB** = **BA**.
(2) If **A** is a positive alternating cycle matrix of dimensions 6 by 6 and **C** is a negative alternating cycle matrix of dimensions 6 by 6, then **AC** and **CA** are negative alternating cycle matrices of the same dimensions.
(3) If **C** and **D** are any two negative alternating cycle matrices of dimensions 6 by 6, then **CD** and **DC** are positive alternating cycle matrices of the same dimensions and **CD** and **DC** are mutually symmetrical, whereby the cycles of **CD** and **DC** differ in phase.

ACTIVITIES

- Predict the multiplication tables of the basic positive and negative alternating cycle matrices of dimensions 4 by 4.
- Verify the conjecture. Construct the multiplication tables of the basic positive and negative alternating cycle matrices of dimensions 4 by 4.

QUESTIONS FOR REFLECTION

Will the following be true in general? If the answer is yes, how can we arrive at such a proof?

(1) If **A** and **B** are any two positive alternating cycle matrices of dimensions n by n, then **AB** and **BA** are positive alternating cycle matrices of the same dimensions and **AB** = **BA**.

(2) If **A** is a positive alternating cycle matrix of dimensions n by n and **C** is a negative alternating cycle matrix of dimensions n by n, then **AC** and **CA** are negative alternating cycle matrices of the same dimensions.

(3) If **C** and **D** are any two negative alternating cycle matrices of dimensions n by n, then **CD** and **DC** are positive alternating cycle matrices of the same dimensions and **CD** and **DC** are mutually symmetrical, whereby the cycles of **CD** and **DC** differ in phase.

Chapter 19

OUTLINE OF A PROOF

ABSTRACT

In chapter 19 an outline of a proof of three important theorems concerning the multiplication of positive and negative alternating cycle matrices of any dimensions will be presented. In each case the assumptions will be indicated and they will be analyzed in the following chapters.

The results we obtained in the previous chapters, concerning concrete cases of dimensions 3 by 3, 7 by 7, and 6 by 6, suggest us a general direction in order to prove the following hypotheses formulated at the of the last chapter:

(1) If **A** and **B** are any two positive alternating cycle matrices of dimensions *k* by *k*, then **AB** and **BA** are positive alternating cycle matrices of the same dimensions and **AB** = **BA**.
(2) If **A** is a positive alternating cycle matrix of dimensions *k* by *k* and **C** is a negative alternating cycle matrix of dimensions *k* by *k*, then **AC** and **CA** are negative alternating cycle matrices of the same dimensions.
(3) If **C** and **D** are any two negative alternating cycle matrices of dimensions *k* by *k*, then **CD** and **DC** are positive alternating cycle matrices of the same dimensions and **CD** and **DC** are symmetrical, and the cycles of **CD** and **DC** differ in phase.

As done in the particular cases already studied, we can write each positive alternating cycle matrix of dimensions *k* by *k* as a sum of multiples of basic positive alternating cycle matrices, that is, as a linear combination of the basic positive alternating cycle matrices of dimensions *k* by *k*, denoted by **P**(1), **P**(2), …, **P**(k).

In the same way all negative alternating cycle matrices of dimensions *k* by *k* may be written as sums of multiples of basic negative alternating cycle matrices, that is, as linear combinations of the basic negative alternating cycle matrices of dimensions *k* by *k*, denoted by **N**(1), **N**(2), … , **N**(k).

Let **A** and **B** be any two positive alternating cycle matrices of dimensions *k* by *k*. Both **A** and **B** can be written as a sum of multiples of basic positive alternating cycle matrices of dimensions *k* by *k*. Thus, the multiplication of **A** and **B** corresponds to the multiplication of

two sums of multiples of basic positive alternating cycle matrices of dimensions *k* by *k*, and, by consequence, the product **AB** itself corresponds to a sum of multiples of products of basic positive alternating cycle matrices of dimensions *k* by *k*. **Let us suppose (1a)** that the products of basic positive alternating cycle matrices are themselves basic positive alternating cycle matrices of the same dimensions. In this way, **AB** is equal to the sum of certain multiples of basic positive alternating cycle matrices. Any sum of multiples of basic positive alternating cycle matrices is a positive alternating cycle matrix.

Thus we arrive at the conclusion that **AB** is a positive alternating cycle matrix of dimensions *k* by *k*.

Let us now compare **AB** and **BA**. When we elaborate **AB**, we encounter parts of the type $m\mathbf{P}(r)$ x $\mathbf{P}(s)$, where *m* is any number and *r* and *s* any natural numbers from 1 to *k*. When we elaborate **BA**, however, we encounter a part $m\mathbf{P}(s)$ x $\mathbf{P}(r)$ instead of the part $m\mathbf{P}(r)$ x $\mathbf{P}(s)$. **Supposing (1b)** that the multiplication table of the basic positive alternating cycle matrices is symmetrical, we then always have $\mathbf{P}(r)$ x $\mathbf{P}(s)$ = $\mathbf{P}(s)$ x $\mathbf{P}(r)$. Since corresponding parts are equal, we can conclude that the sums are also equal, such that **AB** = **BA**.

Assuming our suppositions (1a) and (1b) as true, we have proved in this way the following theorem:

If **A** and **B** are any two positive alternating cycle matrices of dimensions *k* by *k*, then **AB** and **BA** are positive alternating cycle matrices of the same dimensions and **AB** = **BA**.

Let us contemplate the multiplication of two matrices **A** and **C**, with **A** being any positive alternating cycle matrix of dimensions *k* by *k* and **C** being any negative alternating cycle matrix of the same dimensions. Matrix **A** can be written as the sum of certain multiples of basic positive alternating cycle matrices, whereas matrix **C** can be written as the sum of certain multiples of basic negative alternating cycle matrices. The calculation of the products **AC** and **CA** corresponds to the multiplication of those two sums of multiples of basic positive and basic negative alternating cycle matrices, respectively, that is, it corresponds to sums of parts of the types $m\mathbf{P}(r)$ x $\mathbf{N}(s)$ and $m\mathbf{N}(s)$ x $\mathbf{P}(r)$, respectively. **Let us suppose (2)** that the products of any basic positive and basic negative alternating cycle matrices of dimensions *k* by *k* are always a basic negative alternating cycle matrix of the same dimensions. By consequence, the products **AC** and **CA** are negative alternating cycle matrices of dimensions *k* by *k*.

Assuming our **supposition (2)** for true, we have proved the following theorem:

If **A** is a positive alternating cycle matrix of dimensions *k* by *k* and **C** is a negative alternating cycle matrix of dimensions *k* by *k*, then **AC** and **CA** are negative alternating cycle matrices of the same dimensions.

Let us now analyze the multiplication of negative alternating cycle matrices. Let **C** and **D** be two negative alternating cycle matrices of dimensions *k* by *k*. Both **C** and **D** can be written as the sum of certain multiples of the *k* basic negative alternating cycle matrices. Determining the products **CD** and **DC** corresponds to the elaboration of the products of the two considered sums. The matrices **CD** and **DC** are thus sums of multiples of the products of the *k* basic negative alternating cycle matrices.

Considering the experiences of the previous chapters, we are led to **suppose (3a)** that when calculating the product of any two basic negative alternating cycle matrices of dimensions k by k, we always obtain a basic positive alternating cycle matrix of the same dimensions.

By consequence, matrices **CD** and **DC** are sums of multiples of the k basic positive alternating cycle matrices and hence **CD** and **DC** are positive alternating cycle matrices.

When we compare **CD** and **DC**, we take into account that **CD** is sum of parts of the type $m\mathbf{N}(r)$ x $\mathbf{N}(s)$ whereas **DC** is the sum of the parts of the type $m\mathbf{N}(s)$ x $\mathbf{N}(r)$. **Supposing (3b)** that the following is valid

$$\mathbf{N}(r) \text{ x } \mathbf{N}(s) = \{\mathbf{N}(s) \text{ x } \mathbf{N}(r)\}^{T},$$

and being **CD** the sum of certain multiples of $\mathbf{N}(r)$ x $\mathbf{N}(s)$ and **DC** the sum of the same multiples of $\mathbf{N}(s)$ x $\mathbf{N}(r)$, we can conclude that

$$\mathbf{CD} = (\mathbf{DC})^{T},$$

that is, the matrices **CD** and **DC** are mutually symmetrical. Matrix **CD** is composed of alternating cycles. When we reflect matrix **CD** across its principal diagonal, the order of the numbers on its cycles is inverted. Since the result of the same reflection is matrix **DC**, we may conclude that the cycles of **CD** and **DC** differ in phase.

Taking into account the **suppositions (3a)** and **(3b)**, we have proved in this way the following theorem:

If **C** and **D** are any two negative alternating cycle matrices of dimensions k by k, then **C**D and **DC** are positive alternating cycle matrices of the same dimensions and **CD** and **DC** are symmetrical, and the cycles of **CD** and **DC** differ in phase.

The proofs are complete if our suppositions (1a), (1b), (2), (3a) and (3b) are true.

In the following chapters we shall analyze our suppositions.

Chapter 20

MULTIPLICATION OF BASIC POSITIVE ALTERNATING CYCLE MATRICES OF EVEN DIMENSIONS: FORMULATION OF HYPOTHESES

ABSTRACT

In chapter 20 hypotheses concerning the multiplication of basic positive alternating cycle matrices of even dimensions will be discovered and formulated. Eight distinct situations will be distinguished.

Let us consider alternating cycle matrices of even dimensions. Let us say that the dimensions are *2m* by *2m*, where *m* represents any natural number greater than 1. Let *r* be a natural number smaller than or equal to 2*m*. We will try to describe the matrix **P**(*r*) in such a way that it is indicated exactly where one can find the 1's of this matrix. Figure 20.1 presents a sketch of its only cycle that contains the alternating 1's and 0's. There are four parts of segments indicated by (a), (b), (c) and (d), respectively,

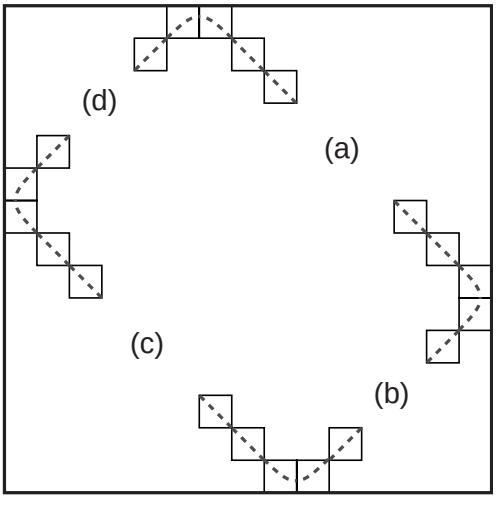


Figure 20.1.

We have two possibilities for the position of the 1 in the first row: if r is an even number, the 1 of the first row is located in the left unit square; if r is an odd number, the 1 of the first row is located in the right unit square.

Figure 20.2 presents an example, where one can observe the positions of the 1's in the cases of **P**(2) and **P**(3) if m=3. The first number in each square indicates the number of the row, whereas the second number indicates the number of the column. The 1's of the matrix **P**(2) are encountered in the positions that have as indices (1, 2) (start of the cycle and end of the part d), (2, 4) and (4, 6) (part a), (6, 5) (part b), and (5, 3) and (3, 1) (part c). The 1's of the matrix **P**(3) are located in the positions that have as indices (1, 3) and (3, 5) (part a), (5, 6) (part b), (6, 4) and (4, 2) (part c) and (2, 1) (part d), respectively.

	12	13			
21			24		
31				35	
	42				46
		53			56
			64	65	

The indices of the 1's of the matrices **P**(2) and **P**(3)

Figure 20.2.

ACTIVITIES

- Compare the indices of the 1's of matrix **P**(2) in the case m=3.
- Compare the indices of the 1's of matrix **P**(3) in the case m=3.
- Compare the indices of the 1's of matrix **P**(4) in the case m=5.
- Compare the indices of the 1's of matrix **P**(5) in the case m=5.

For the indices of the 1's of part (a) of matrix **P**(2) (figure 20.2) we see that they are even numbers and that the differences are equal: 2 – 4 = 4 – 6 = -2. For the indices of the 1's of part (a) of matrix **P**(3) we see that they are odd numbers whereas the differences are equal to -2: 1 – 3 = 3 – 5 = -2. Etc.

In general, it will be necessary to distinguish between two situations: r even and r odd. If r is even, a natural number exists, let us say n, such that $r = 2n$. If r is odd, a natural number exists, let us say n, such that $r = 2n + 1$.

part	indices (f, g)	conditions
(a)	$f - g = -2n$	f even, g even
(b)	$f + g = (4m+1) - 2n$	f even, g odd
(c)	$f - g = 2n$	f odd, g odd
(d)	$f + g = 2n + 1$	f odd, g even

Case $r = 2n$

Figure 20.3.

The central column of figure 20.3 presents the relationships between the indices f and g and of the places where one finds the 1's of matrix $\mathbf{P}(r) = \mathbf{P}(2n)$, whereas the last column indicates the parity (even or odd) of f and g.

ACTIVITY

- Construct a table similar to the table in figure 20 for the case $r = 2n+1$.

In the case $\mathbf{P}(r) = \mathbf{P}(2n+1)$ the relationships between the indices are the same as in the previous case, whereas the conditions of parity are inverted (see figure 20.4).

Part	Indices (f, g)	Conditions
(a)	$f - g = -2n$	f odd, g odd
(b)	$f + g = (4m+1) - 2n$	f odd, g even
(c)	$f - g = 2n$	f even, g even
(d)	$f + g = 2n + 1$	f even, g odd

Case $r = 2n + 1$

Figure 20.4.

We have already completed the construction of our instruments to enable analysis of the structure of the multiplication table containing the basic positive alternating cycle matrices.

Let r and s be two natural numbers smaller than or equal to $2m$. What can be said about the product $\mathbf{P}(r)$ times $\mathbf{P}(s)$?

Let $\mathbf{A}$ be the matrix resulting from the multiplication. The elements of matrix $\mathbf{A}$ can only be 0 or 1, and in each row and in each column the number 1 appears only once. Let us begin by analyzing the placement of the 1 in the first row of matrix $\mathbf{A}$.

Imagining that the 1 may be found in the j-th column, what can we say about the value of j? In other words, $\mathbf{A}_{(1,j)} = 1$. This element $\mathbf{A}_{(1,j)}$ results from the multiplication of the first row of the matrix $\mathbf{P}(r)$ by the j-th column of the matrix $\mathbf{P}(s)$. The only non-zero element of the first row of matrix $\mathbf{P}(r)$ is found in its r-th column: $\mathbf{P}(r)_{(1,r)} = 1$.

By consequence, $1 = \mathbf{A}_{(1,j)} = \mathbf{P}(r)_{(1,r)} \times \mathbf{P}(s)_{(r,j)} = \mathbf{P}(s)_{(r,j)}$. With the knowledge that $\mathbf{P}(s)_{(r,j)} = 1$, what can be deduced about the value of j?

We need to distinguish eight cases, as the numbers r, s and j can be either even or odd (figure 20.5). If r is even, let us write $r = 2n$; if r is odd, $r = 2n+1$. If s is even, let us write $s = 2t$ for a certain natural number t; if s is odd, then $s = 2t+1$.

case	*r*	*s*	*j*
1	even	even	even
2	even	even	odd
3	even	odd	even
4	even	odd	odd
5	odd	even	even
6	odd	even	odd
7	odd	odd	even
8	odd	odd	odd

Figure 20.5.

Let us analyze the situation case by case.

Case 1: Part (a) in figure 20.3: $r - j = -2t = -s$, that is, $j = r\ s$, supposing $r+s \leq 2m$.

Case 2: Part (b) in figure 20.3: $r+j = (4m+1) - 2t = (4m+1) - s$, that is, $j = (4m+1) - (r+s)$. As $j \leq 2m$, we have $r+s > 2m$.

Case 3: Part (c) in figure 20.4: $r - j = 2t = s - 1$, that is, $j = r - s + 1$. As j has to be greater than 0, we have $r > s$.

Case 4: Part (d) in figure 20.4: $r + j = 2t + 1 = s$, that is, $j = s - r$, supposing $s > r$.

Case 5: Part (d) in figure 20.3: $r + j = 2t + 1 = s + 1$, that is, $j = s - r + 1$. As j has to be greater than 0, $s > r$ holds.

Case 6: Part (c) in figure 20.3: $r - j = 2t = s$, that is, $j = r - s$, assuming $r > s$.

Case 7: Part (b) in figure 20.4: $r + j = (4m+1) - 2t = (4m+1) - s + 1$, that is, $j = 4m+2 - (r+s)$. As j has to be smaller than or equal to $2m$, the following holds: $r + s > 2m$.

Case 8: Part (a) in figure 20.4: $r - j = -2t = -s + 1$, that is, $j = r + s - 1$. As j has to be smaller than or equal to $2m$, the following has to be valid $r + s \leq 2m$.

We wanted to know the placement of the 1 in the first row of matrix **A**, where **A** = **P**(r) x **P**(s). Now we know that there are eight different situations. We want to show that **A** is always a basic positive alternating cycle matrix, that is, we want to show that one can always find a natural number y such that

P(r) x **P**(s) = **P**(y).

The only element in the first row of the matrix **P**(y) different from 0 is the y-th. We know already that the j-th element of matrix **A** is not 0. In order for the matrices **A** and **P**(y) to be equal y has to equal to j, and j has to satisfy all equalities found in the eight distinct cases.

In this manner, we can formulate the hypothesis

P(r) x **P**(s) = **P**(j),

composed of eight sub-hypotheses presented in figure 20.6.

case	r	s	conditions	j
1	even	even	$r+s \leq 2m$	$r+s$
2	even	even	$r+s > 2m$	$(4m+1) - (r+s)$
3	even	odd	$r > s$	$r - s + 1$
4	even	odd	$s > r$	$s - r$
5	odd	even	$s > r$	$s - r + 1$
6	odd	even	$r > s$	$r - s$
7	odd	odd	$r+s > 2m$	$(4m+2) - (r+s)$
8	odd	odd	$r+s \leq 2m$	$r+s - 1$

Figure 20.6.

ACTIVITIES

- Test the hypotheses in the case of dimensions 6 by 6 where $m = 3$.
- Test the hypotheses in the case of dimensions 10 by 10 where $m = 5$.
- Test the hypotheses in the case of dimensions 12 by 12.
- Evaluate the hypotheses in general.

Let us conclude the chapter with an analysis of the case of dimensions 10 by 10.

Calculating the multiplication table of the basic positive alternating cycle matrices of dimensions 10 by 10 we obtain the associated matrix presented in figure 20.7.

	1	2	3	4	5	6	7	8	9	10
1	1	2	3	4	5	6	7	8	9	10
2	2	4	1	6	3	8	5	10	7	9
3	3	1	5	2	7	4	9	6	10	8
4	4	6	2	8	1	10	3	9	5	7
5	5	3	7	1	9	2	10	4	8	6
6	6	8	4	10	2	9	1	7	3	5
7	7	5	9	3	10	1	8	2	6	4
8	8	10	6	9	4	7	2	5	1	3
9	9	7	10	5	8	3	6	1	4	2
10	10	9	8	7	6	5	4	3	2	1

Figure 20.7.

Let us compare this matrix derived from the multiplication table with what the hypotheses predict.

In agreement with the first hypothesis we have: **P**(2) x **P**(2) = **P**(2+2) = **P**(4), **P**(2) x **P**(4) = **P**(2+4) = **P**(6), **P**(2) x **P**(6) = **P**(8), **P**(2) x **P**(8) = **P**(10), etc. The predicted results correspond to the values of the calculated matrix (figure 20.7).

In agreement with the second hypothesis we have: **P**(2) x **P**(10) = **P**(20+1-2-10) = **P**(9), **P**(4) x **P**(8) = **P**(20+1-4-8) = **P**(9), etc. These results correspond to the values in the associated matrix. Figure 20.8 indicates the values foreseen in cases 1 and 2.

	1	2	3	4	5	6	7	8	9	10
1										
2		4		6		8		10		9
3										
4		6		8		10		9		7
5										
6		8		10		9		7		5
7										
8		10		9		7		5		3
9										
10		9		7		5		3		1

Cases 1 and 2

Figure 20.8.

In the same way figures 20.9, 20.10 and 20.11 indicate the values foreseen in the cases 3 and 4, 5 and 6, and 7 and 8, respectively.

	1	2	3	4	5	6	7	8	9	10
1										
2	2		1		3		5		7	
3										
4	4		2		1		3		5	
5										
6	6		4		2		1		3	
7										
8	8		6		4		2		1	
9										
10	10		8		6		4		2	

Cases 3 and 4

Figure 20.9.

	1	2	3	4	5	6	7	8	9	10
1		2		4		6		8		10
2										
3		1		2		4		6		8
4										
5		3		1		2		4		6
6										
7		5		3		1		2		4
8										
9		7		5		3		1		2
10										

Cases 5 and 6

Figure 20.10.

	1	2	3	4	5	6	7	8	9	10
1	1		3		5		7		9	
2										
3	3		5		7		9		10	
4										
5	5		7		9		10		8	
6										
7	7		9		10		8		6	
8										
9	9		10		8		6		4	
10										

Cases 7 and 8

Figure 20.11.

Chapter 21

MULTIPLICATION OF BASIC POSITIVE ALTERNATING CYCLE MATRICES OF EVEN DIMENSIONS: SOME PROOFS

ABSTRACT

In chapter 21 some results concerning the multiplication of basic positive alternating cycle matrices of dimensions $2m$ by $2m$ will be proven.

Let us consider basic positive alternating cycle matrices of dimensions $2m$ by $2m$, where m represents a natural number greater than 1. The first of the eight hypotheses formulated at the end of chapter 20 asserts the following. If r and s are two even numbers ($r=2n$, $s=2t$) that satisfy the condition $r+s \leq 2m$, then

$\mathbf{P}(r) \times \mathbf{P}(s) = \mathbf{P}(r+s)$.

ACTIVITY

- Attempt to prove $\mathbf{P}(r) \times \mathbf{P}(s) = \mathbf{P}(r+s)$ for any even numbers r and s that satisfy the condition $r+s \leq 2m$.

Knowing that $\mathbf{P}(2m)$ is an exceptional basic positive alternating cycle matrix in the sense that all its non-zero elements are lying on the secondary diagonal of the matrix, we can distinguish two subcases: $r+s = 2m$ and $r+s < 2m$. Let us start with the first subcase.

Subcase $r+s = 2m$

If $\mathbf{A} = \mathbf{P}(r) \times \mathbf{P}(s)$ and supposing that $r+s = 2m$, we have to show that $\mathbf{A} = \mathbf{P}(2m)$, that is, to show that all elements of the secondary diagonal of matrix $\mathbf{A}$ are equal to 1.

The elements of the secondary diagonal of matrix **A** are the ones that belong to the i-th row and $(2m-i-1)$-th column, where i is any natural number between 1 and $2m$. In other words, we have to show that the following always holds true:

$$\mathbf{A}_{(i,\, 2m-i-1)} = 1.$$

In the case where i is an even number, we can consider two types of situation: $i \leq s$ and $i > s$.

1) Supposing $i \leq s$ that is $i+r \leq 2m$, we have $\mathbf{P}(r)_{(i,\, i+r)} = 1$ [see part (a), figure 20.3] and $\mathbf{P}(s)_{(i+r,\, 2m+1-i)} = 1$ [see part (b), figure 20.3], what implies immediately:

$$\mathbf{A}_{(i,\, 2m-i-1)} = \mathbf{P}(r)_{(i,\, i+r)} \text{ x } \mathbf{P}(s)_{(i+r,\, 2m+1-i)} = 1.$$

2) Supposing $i > s$, we have $\mathbf{P}(r)_{(i,\, 2m+1-i+s)} = 1$ [see part (b), figure 20.3] and $\mathbf{P}(\mathrm{s})_{(2m+1-i+s,\, 2m+1-i)} = 1$ [see part (c), figure 20.3], what implies immediately:

$$\mathbf{A}_{(i,\, 2m-i-1)} = \mathbf{P}(r)_{(i,\, 2m+1-i+s)} \text{ x } \mathbf{P}(s)_{(2m+1-i+s,\, 2m+1-i)} = 1.$$

In the case where i is an odd number, we may consider two situations: $i < r$ and $i > r$.

1) Supposing $i < r$, we have $\mathbf{P}(r)_{(i,\, r+1-i)} = 1$ [see part (d), figure 20.3] and $\mathbf{P}(s)_{(r+1-i,\, 2m+1-i)} = 1$ [see part (a), figure 20.3], what implies immediately:

$$\mathbf{A}_{(i,\, 2m-i-1)} = \mathbf{P}(r)_{(i,\, r+1-i)} \text{ x } \mathbf{P}(s)_{(r+1-i,\, 2m+1-i)} = 1.$$

2) Supposing $i > r$, we have $\mathbf{P}(r)_{(i,\, i-r)} = 1$ [see part (c), figure 20.3] and $\mathbf{P}(s)_{(i-r,\, 2m+1-i)} = 1$ [see part (d), figure 20.3], what implies immediately:

$$\mathbf{A}_{(i,\, 2m-i-1)} = \mathbf{P}(r)_{(i,\, i-r)} \text{ x } \mathbf{P}(s)_{(i-r,\, 2m+1-i)} = 1.$$

In this manner it follows that all elements of the secondary diagonal are equal to 1, and $\mathbf{P}(r)$ x $\mathbf{P}(s) = \mathbf{P}(2m)$ when $r+s = 2m$. In other words, the following is always valid

$$\mathbf{P}(r) \text{ x } \mathbf{P}(2m-r) = \mathbf{P}(2m)$$

Subcase $r+s < 2m$

In the case where i is an even number, we may consider two types of situation: $i+r+s \leq 2m$ and $i+r+s > 2m$.

1) Supposing $i+r+s \leq 2m$, we have $\mathbf{P}(r)_{(i,\, i+r)} = 1$ [see part (a), figure 20.3] and $\mathbf{P}(s)_{(i+r,\, i+r+s)} = 1$ [see part (a), figure 20.3], what implies immediately:

$$\mathbf{A}_{(i,\, i+r+s)} = \mathbf{P}(r)_{(i,\, i+r)} \text{ x } \mathbf{P}(s)_{(i+r,\, i+r+s)} = 1 = \mathbf{P}(r+s)_{(i,\, i+r+s)}$$

[part (a) of the non-zero cycle of the matrix $\mathbf{P}(r+s)$].

2) Supposing $i+r+s > 2m$, we have $\mathbf{P}(r)_{(i,\ 4m+1-i-r)} = 1$ [see part (b), figure 20.3] and $\mathbf{P}(s)_{(4m+1-i-r,\ 4m+1-r-s-i)} = 1$ [see part (c), figure 20.3], what implies immediately:

$$\mathbf{A}_{(i,\ 4m+1-r-s-i)} = \mathbf{P}(r)_{(i,\ 4m+1-i-r)} \text{ x } \mathbf{P}(s)_{(4m+1-i-r,\ 4m+1-r-s-i)} =$$
$$= 1 = \mathbf{P}(r+s)_{(i,\ 4m+1-r-s-i)}.$$

[part (b) of the non-zero cycle of the matrix $\mathbf{P}(r+s)$].

In the case where i is an odd number, we may consider two types of situation: $i-r-s > 0$ and $i-r-s \leq 0$.

1) Supposing $i-r-s > 0$, we have $\mathbf{P}(r)_{(i,\ i-r)} = 1$ [see part (c), figure 20.3] and $\mathbf{P}(s)_{(i-r,\ i-r-s)} = 1$ [see part (c), figure 20.3], what implies immediately:

$$\mathbf{A}_{(i,\ i-r-s)} = \mathbf{P}(r)_{(i,\ i-r)} \text{ x } \mathbf{P}(s)_{(i-r,\ i-r-s)} = 1 = \mathbf{P}(r+s)_{(i,\ i-r-s)}$$

[part (c) of the non-zero cycle of matrix $\mathbf{P}(r+s)$].

2) Supposing $i-r-s \leq 0$, we have $\mathbf{P}(r)_{(i,\ i-r)} = 1$ [see part (c), figure 20.3] and $\mathbf{P}(s)_{(i-r,\ r+s+1-i)} = 1$ [see part (d), figure 20.3], what implies immediately:

$$\mathbf{A}_{(i,\ r+s+1-i)} = \mathbf{P}(r)_{(i,\ i-r)} \text{ x } \mathbf{P}(s)_{(i-r,\ r+s+1-i)} = 1 = \mathbf{P}(r+s)_{(i,\ r+s+1-i)}$$

[part (d) of the non-zero cycle of matrix $\mathbf{P}(r+s)$].

In this way, we have successfully proved that $\mathbf{P}(r)$ x $\mathbf{P}(s) = \mathbf{P}(r+s)$ when $r+s < 2m$.

Combining the results of the two subcases, we have just proved that:

If r and s are two even numbers ($r=2n$, $s=2t$) satisfying the condition of $r+s \leq 2m$, then we have

$$\mathbf{P}(r) \text{ x } \mathbf{P}(s) = \mathbf{P}(r+s).$$

Let us analyze the second hypothesis next. The second of the eight hypotheses formulated at the end of the chapter 20 asserts the following. If r and s are two even numbers ($r=2n$, $s=2t$) satisfying the condition $r+s > 2m$, then

$$\mathbf{P}(r) \text{ x } \mathbf{P}(s) = \mathbf{P}(4m+1-r-s).$$

ACTIVITY

- Try to show that $\mathbf{P}(r)$ x $\mathbf{P}(s) = \mathbf{P}(4m+1-r-s)$ for $r+s > 2m$, where r and s are two even numbers.

The number $4m+1-r-s = 2(2m-n-t) +1$ is odd. By consequence, the four parts of the non-zero cycle of matrix $\mathbf{P}(4m+1-r-s)$ are governed by the conditions explained in figure 21.1 (compare with figure 20.4).

part	indices (f, g)	conditions
(a)	$f - g = -4m+r+s$	f odd, g odd
(b)	$f + g = r+s+1$	f odd, g even
(c)	$f - g = 4m-r-s$	f even, g even
(d)	$f + g = 4m+1-r-s$	f even, g odd

Figure 21.1.

Let matrix $\mathbf{A}$ be defined by $\mathbf{A} = \mathbf{P}(r) \times \mathbf{P}(s)$. If $\mathbf{A}_{(f,g)} = 1$, then a number natural exists j such that $\mathbf{P}(r)_{(f,j)} \times \mathbf{P}(s)_{(j,g)} = 1$, that is, $\mathbf{P}(r)_{(f,j)} = \mathbf{P}(s)_{(j,g)} = 1$. Let us analyze the eight possibilities for the parities of f, j and g.

Case 1: The three numbers are even. Since $\mathbf{P}(r)_{(f,j)} = 1$, we have $f - j = -r$ [part (a) in figure 20.3]. Since $\mathbf{P}(s)_{(j,g)} = 1$, we have $j - g = -s$ [part (a) in figure 20.3]. By consequence, $f - g = - r - s$. As $r+s > 2m$, the difference of g and f would be greater than $2m$, which is not possible. In other words, the three numbers cannot be even simultaneously.

Case 2: The numbers f and g are even, whereas j is odd. Since $\mathbf{P}(r)_{(f,j)} = 1$, we have $f + j = (4m+1)-r$ [part (b) in figure 20.3]. Since $\mathbf{P}(s)_{(j,g)} = 1$, we have $j + g = s+1$ [part (b) in figure 20.3]. It follows that $f - g = 4m - r - s$, as in part (c) of the non-zero cycle of $\mathbf{P}(4m+1-r-s)$.

Case 3: The numbers f and j are even, whereas g is odd. Since $\mathbf{P}(r)_{(f,j)} = 1$, we have $f - j = -r$ [part (a) in figure 20.3]. Since $\mathbf{P}(s)_{(j,g)} = 1$, we have $j + g = 4m+1-s$ [part (b) in figure 20.3]. By consequence, $f + g = 4m+1 - r - s$, as in part (d) of the non-zero cycle of $\mathbf{P}(4m+1-r-s)$.

Case 4: The number f is even, whereas j and g are odd. Since $\mathbf{P}(r)_{(f,j)} = 1$, we have $f + j = (4m+1)-r$ [part (b) in figure 20.3]. Since $\mathbf{P}(s)_{(j,g)} = 1$, we have $j - g = s$ [part (c) in figure 20.3]. By consequence, $f + g = 4m+1 - r - s$, as in part (d) of the non-zero cycle of $\mathbf{P}(4m+1-r-s)$.

Case 5: The number f is odd, whereas j and g are even. Since $\mathbf{P}(r)_{(f,j)} = 1$, we have $f + j = r+1$ [part (d) in figure 20.3]. Since $\mathbf{P}(s)_{(j,g)} = 1$, we have $j - g = -s$ [part (a) in figure 20.3]. By consequence, $f + g = r+s+1$, as in part (b) of the non-zero cycle of $\mathbf{P}(4m+1-r-s)$.

Case 6: The number g *is* even, whereas f and j are odd. Since $\mathbf{P}(r)_{(f,j)} = 1$, we have $f - j = r$ [part (c) in figure 20.3]. Since $\mathbf{P}(s)_{(j,g)} = 1$, we have $j + g = s+1$ [part (a) in figure 20.3]. By consequence, $f + g = r+s+1$, as in part (b) of the non-zero cycle of $\mathbf{P}(4m+1-r-s)$.

Case 7: The number j *is* even, whereas f and g are odd. Since $\mathbf{P}(r)_{(f,j)} = 1$, we have $f + j = r+1$ [part (d) in figure 20.3]. Since $\mathbf{P}(s)_{(j,g)} = 1$, we have $j + g = 4m+1 - s$ [part (b) in figure 20.3]. By consequence, $f - g = r+s - 4m$, as in part (a) of the non-zero cycle of $\mathbf{P}(4m+1-r-s)$.

Case 8: The three numbers are odd. Since $\mathbf{P}(r)_{(f,j)} = 1$, we have $f - j = r$ [part (c) in figure 20.3]. Since $\mathbf{P}(s)_{(j,g)} = 1$, we have $j - g = s$ [part (c) in figure 20.3]. By consequence, $f - g = r + s$. As $r+s > 2m$, the difference of f and g could not be equal to $r+s$. In other words, the three numbers cannot be odd simultaneously.

In this manner, we have proven:
If r and s are two even numbers (r=2n, s=2t) satisfying the condition of $r+s > 2m$, then $\mathbf{P}(r)$ x $\mathbf{P}(s) = \mathbf{P}(4m+1-r-s)$.
There are six more situations to analyze.

ACTIVITIES

Try to prove:

- If r is an even number and s an odd number, satisfying the condition $r > s$, then $\mathbf{P}(r)$ x $\mathbf{P}(s) = \mathbf{P}(r - s + 1)$.
- If r is an even number and s an odd number, satisfying the condition $s > r$, then $\mathbf{P}(r)$ x $\mathbf{P}(s) = \mathbf{P}(s - r)$.
- If r is an odd number and s an even number, satisfying the condition $s > r$, then $\mathbf{P}(r)$ x $\mathbf{P}(s) = \mathbf{P}(s - r + 1)$.
- If r is an odd number and s an even number, satisfying a condition $r > s$, then $\mathbf{P}(r)$ x $\mathbf{P}(s) = \mathbf{P}(r - s)$.
- If r and s are odd numbers, satisfying the condition $r+s > 2m$, then $\mathbf{P}(r)$ x $\mathbf{P}(s) = \mathbf{P}((4m+2) - (r+s))$.
- If r and s are odd numbers, satisfying the condition $r+s \leq 2m$, then $\mathbf{P}(r)$ x $\mathbf{P}(s) = \mathbf{P}(r+s-1)$.

With the proof of these last six results, one completes the proof of the theorem.

Theorem 21.1.

If $\mathbf{P}(r)$ and $\mathbf{P}(s)$ are any two basic positive alternating cycle matrices of dimensions 2m by 2m, then the product, $\mathbf{P}(r)$ x $\mathbf{P}(s)$, is also a basic positive alternating cycle matrix, $\mathbf{P}(j)$, where the value of j is indicated in the last column of figure 21.2, according to the eight distinct cases in agreement with the parity and conditions for r and s.

case	***r***	***s***	**conditions**	***j***
1	even	even	$r+s \leq 2m$	$r+s$
2	even	even	$r+s > 2m$	$(4m+1) - (r+s)$
3	even	odd	$r > s$	$r - s + 1$
4	even	odd	$s > r$	$s - r$
5	odd	even	$s > r$	$s - r + 1$
6	odd	even	$r > s$	$r - s$
7	odd	odd	$r+s > 2m$	$(4m+2) - (r+s)$
8	odd	odd	$r+s \leq 2m$	$r+s - 1$

Figure 21.2.

With this theorem it will be easy to prove that the multiplication of basic positive alternating cycle matrices is commutative, that is, for any numbers r and s, $\mathbf{P}(r)$ x $\mathbf{P}(s)$ = $\mathbf{P}(s)$ x $\mathbf{P}(r)$ is always valid.

Let us compare $\mathbf{P}(r)$ x $\mathbf{P}(s)$ with $\mathbf{P}(s)$ x $\mathbf{P}(r)$, case by case.

Case 1:	$\mathbf{P}(r)$ x $\mathbf{P}(s) = \mathbf{P}(r+s) = \mathbf{P}(s+r) = \mathbf{P}(s)$ x $\mathbf{P}(r)$.
Case 2:	$\mathbf{P}(r)$ x $\mathbf{P}(s) = \mathbf{P}(4m+1-r-s) = \mathbf{P}(4m+1-s-r) = \mathbf{P}(s)$ x $\mathbf{P}(r)$.
Cases 3 and 5:	We have $\mathbf{P}$(r) x $\mathbf{P}$(s) = $\mathbf{P}(r-s+1)$ (case 3), whereas $\mathbf{P}(s)$ x $\mathbf{P}(r)$ = $\mathbf{P}(r-s+1)$ (case 5), and thus $\mathbf{P}(r)$ x $\mathbf{P}(s) = \mathbf{P}(s)$ x $\mathbf{P}(r)$.
Cases 4 and 6:	We have $\mathbf{P}(r)$ x $\mathbf{P}(s) = \mathbf{P}(s-r)$ (case 4), whereas $\mathbf{P}(s)$ x $\mathbf{P}(r) = \mathbf{P}(s-r)$ (case 6), and thus $\mathbf{P}(r)$ x $\mathbf{P}(s) = \mathbf{P}(s)$ x $\mathbf{P}(r)$.
Case 7:	$\mathbf{P}(r)$ x $\mathbf{P}(s) = \mathbf{P}(4m+2-r-s) = \mathbf{P}(4m+2-s-r) = \mathbf{P}(s)$ x $\mathbf{P}(r)$.
Case 8:	$\mathbf{P}$(r) x $\mathbf{P}$(s) = $\mathbf{P}(r+s-1) = \mathbf{P}(s+r-1) = \mathbf{P}(s)$ x $\mathbf{P}(r)$.

Hence we have succeeded in proving that the multiplication of basic positive alternating cycle matrices of dimensions $2m$ by $2m$ is commutative.

The commutativity of basic positive alternating cycle matrices implies the commutativity of positive alternating cycle matrices in general.

Theorem 21.2.

The multiplication of basic positive alternating cycle matrices of dimensions $2m$ by $2m$ is commutative. For any numbers r and s, the following is always true:

$$\mathbf{P}(r) \text{ x } \mathbf{P}(s) = \mathbf{P}(s) \text{ x } \mathbf{P}(r).$$

From theorem 21.1 we can derive, as seen in Chapter 19, that the product of two positive alternating cycle matrices is also a positive alternating cycle matrix.

Theorem 21.3.

The product of two positive alternating cycle matrices **A** and **B** of dimensions $2m$ by $2m$ is a positive alternating cycle matrix of the same dimensions. Furthermore, **A** and **B** commute: **AB** = **BA**.

From theorem 21.1 we can also deduce that a matrix associated with the multiplication table of the basic positive alternating cycle matrices of dimensions $2m$ by $2m$ is a negative alternating cycle matrix.

Theorem 21.4.

The matrix associated with the multiplication table of the basic positive alternating cycle matrices of dimensions $2m$ by $2m$ is a negative alternating cycle matrix.

Chapter 22

ACTIVITIES OF PROOF

ABSTRACT

In chapter 22 a series of activities aiming to prove several theorems concerning the multiplication of positive and negative alternating cycle matrices of even and odd dimensions will be presented.

We can analyze the multiplication of negative alternating cycle matrices by negative or positive alternating cycle matrices of even dimensions in the same way a proof of four theorems involving the multiplication of positive alternating cycle matrices of even dimensions was analyzed in the previous chapter.

Let us start by analyzing the products of basic negative alternating cycle matrices, distinguishing eight cases.

ACTIVITY

- Indicate each distinct case of the multiplication of basic negative alternating cycle matrices and try to formulate a specific hypothesis for each case.

ACTIVITIES

Try to prove:

- If r and s are even numbers, satisfying the condition $r \geq s$, then $\mathbf{N}(r)$ x $\mathbf{N}(s)$ = $\mathbf{P}$ $(r\text{-}s+1)$.
- If r and s are even numbers, satisfying the condition $r < s$, then $\mathbf{N}(r)$ x $\mathbf{N}(s) = \mathbf{P}(s\text{-}r)$.
- If r is an even number and s an odd number, satisfying the condition $r+s < 2m$, then $\mathbf{N}(r)$ x $\mathbf{N}(s) = \mathbf{P}(r+s)$.
- If r is an even number and s n odd number, satisfying the condition $r+s > 2m$, then $\mathbf{N}(r)$ x $\mathbf{N}(s) = \mathbf{P}((4m+1) - (r+s))$.

- If r is an odd number and s an even number, satisfying the condition $r+s \leq 2m+1$, then $\mathbf{N}(r)$ x $\mathbf{N}(s) = \mathbf{P}(r+s-1)$.
- If r is an odd number and s a number par, satisfying the condition $r+s > 2m+1$, then $\mathbf{N}(r)$ x $\mathbf{N}(s) = \mathbf{P}((4m+2) - (r+s))$.
- If r and s are odd numbers, satisfying the condition $r > s$, then $\mathbf{N}(r)$ x $\mathbf{N}(s) = \mathbf{P}(r-s)$.
- If r and s are odd numbers, satisfying the condition $r \leq s$, then $\mathbf{N}(r)$ x $\mathbf{N}(s) = \mathbf{P}(s-r+1)$.

With the set of proofs of these eight results, one completes a proof of the following theorem.

Theorem 22.1.

If $\mathbf{N}$(r) and $\mathbf{N}$(s) are any two basic negative alternating cycle matrices of dimensions $2m$ by $2m$, the product $\mathbf{N}(r)$ x $\mathbf{N}(s)$ is a basic positive alternating cycle matrix, $\mathbf{P}(y)$, where the value of y is indicated in the last column in figure 22.1, in agreement with the eight distinct cases depending on the parity and the conditions for r and s.

case	***r***	***s***	**conditions**	***y***
1	even	even	$r \geq s$	$r - s + 1$
2	even	even	$r < s$	$s - r$
3	even	odd	$r+s < 2m$	$r+s$
4	even	odd	$r+s > 2m$	$(4m+1) - (r+s)$
5	odd	even	$r+s \leq 2m+1$	$r+s - 1$
6	odd	even	$r+s > 2m+1$	$(4m+2) - (r+s)$
7	odd	odd	$r > s$	$r - s$
8	odd	odd	$r \leq s$	$s - r + 1$

Figure 22.1.

This result leads, as we saw in Chapter 19, immediately to the following theorem.

Theorem 22.2.

If $\mathbf{A}$ and $\mathbf{B}$ are any two negative alternating cycle matrices of dimensions $2m$ by $2m$, then the product $\mathbf{AB}$ is a positive alternating cycle matrix.

ACTIVITY

- Taking into account the results of the theorem 22.1, compare $\mathbf{N}(r)$ x $\mathbf{N}(s)$ with $\mathbf{N}(s)$ x $\mathbf{N}(r)$. On the basis of the experience with dimensions 4 by 4 and 6 by 6, what relationship between $\mathbf{N}(r)$ x $\mathbf{N}(s)$ and $\mathbf{N}(s)$ x $\mathbf{N}(r)$ may be expected, in general?

Taking into account the concrete experience, for example, in the case of dimensions 6 by 6 (see figure 18.7), we may conjecture that $\mathbf{N}(s)$ x $\mathbf{N}(r)$ is the transposed matrix of $\mathbf{N}(r)$ x $\mathbf{N}(s)$. Using theorem 22.1, we may try to compare $\mathbf{N}(r)$ x $\mathbf{N}(s)$ with $\mathbf{N}(s)$ x $\mathbf{N}(r)$. Let us do this case-by-case, presupposing r and s are different numbers.

Cases 1 and 2: We have $\mathbf{N}(r)$ x $\mathbf{N}(s) = \mathbf{P}(r\text{-}s+1)$ (case 1), since $r\text{-}s+1$ is an odd number. $\mathbf{N}(s)$ x $\mathbf{N}(r) = \mathbf{P}(r\text{-}s)$ (case 2, where the roles of r and s are inverted), since $r\text{-}s$ is the even number before the odd number $r\text{-}s+1$. The non-zero cycle of matrix $\mathbf{P}(r\text{-}s)$ coincides with the non-zero cycle of $\mathbf{P}(r\text{-}s+1)$, having its 0's and 1's inverted. Reflecting the non-zero cycle of matrix $\mathbf{P}(r\text{-}s)$ across its principal diagonal, we obtain the non-zero cycle of matrix $\mathbf{P}(r\text{-}s+1)$, that is, $\mathbf{P}(r\text{-}s) = \mathbf{P}(r\text{-}s+1)^{\mathrm{T}}$. In this way, $\mathbf{N}(r)$ x $\mathbf{N}(s) = [\mathbf{N}(s)$ x $\mathbf{N}(r)]^{\mathrm{T}}$.
Cases 3 and 5: We have $\mathbf{N}(r)$ x $\mathbf{N}(s) = \mathbf{P}(r+s)$ (case 3), since $r+s$ is an odd number. $\mathbf{N}(s)$ x $\mathbf{N}(r) = \mathbf{P}(s+r\text{-}1) = \mathbf{P}(r+s\text{-}1)$ (case 5), since $r+s\text{-}1$ is the even number that antecedes the odd number $r+s$. Hence $\mathbf{P}(r+s) = \mathbf{P}(r+s\text{-}1)^{\mathrm{T}}$.
Cases 4 and 6: We have $\mathbf{N}(r)$ x $\mathbf{N}(s) = \mathbf{P}((4m+1) - (r+s))$ (case 4), since $(4m+1) - (r+s)$ is an even number. $\mathbf{N}(s)$ x $\mathbf{N}(r) = \mathbf{P}((4m+2) - (r+s))$ (case 6), since $(4m+1) - (r+s)$ is the even number that antecedes the odd number $(4m+2) - (r+s)$. Thus it follows that $\mathbf{P}((4m+1) - (r+s)) = \mathbf{P}((4m+2) - (r+s))^{\mathrm{T}}$.
Cases 7 and 8: We have $\mathbf{N}(r)$ x $\mathbf{N}(s) = \mathbf{P}(r\text{-}s)$ (case 7), since $r\text{-}s$ is an even number. $\mathbf{N}(s)$ x $\mathbf{N}(r) = \mathbf{P}(r–s+1)$ (case 8), since $r–s$ is the even number that antecedes the odd number $r–s+1$. Thus, it follows that $\mathbf{P}(r\text{-}s) = \mathbf{P}(r–s+1)^{\mathrm{T}}$.

If $r = s$, then, in agreement with case 1, we have $\mathbf{N}(r)$ x $\mathbf{N}(r) = \mathbf{P}(r\text{-}r+1) = \mathbf{P}(1)$. Matrix $\mathbf{P}(1)$ is the identity matrix that is equal to its own transposed matrix.

By consequence, we have finished proving the following:

Theorem 22.3.

If $\mathbf{N}(r)$ and $\mathbf{N}(s)$ are any two basic negative alternating cycle matrices of dimensions $2m$ by $2m$, then the product $\mathbf{N}(r)$ x $\mathbf{N}(s)$ is equal to the transposed matrix of the product $\mathbf{N}(s)$ x $\mathbf{N}(r)$:

$$\mathbf{N}(r) \text{ x } \mathbf{N}(s) = [\mathbf{N}(s) \text{ x } \mathbf{N}(r)]^{\mathrm{T}}.$$

This result leads, as in Chapter 19, immediately to the following theorem.

Theorem 22.4.

If **A** and **B** are two negative alternating cycle arbitrary matrices of dimensions $2m$ by $2m$, then $\mathbf{AB} = [\mathbf{BA}]^{\mathrm{T}}$, that is, the matrices **AB** and **BA** are mutually symmetrical. The corresponding cycles of **AB** and **BA** differ in phase.

From theorem 22.1 one may also deduce the structure of the matrix associated with the multiplication table of the basic negative alternating cycle matrices of dimensions *2m* by *2m*: it is a positive alternating cycle matrix.

Theorem 22.5.

The matrix associated with the multiplication table of the basic negative alternating cycle matrices of dimensions *2m* by *2m* is a positive alternating cycle matrix.

We already analyzed multiplication tables of the basic positive and negative alternating cycle matrices of dimensions *2m* by *2m*. We have also seen some implications. We verified that 'positive' times 'positive' equals 'positive' and 'negative' times 'negative' equals 'positive' holds true.

The reader is invited to try to prove the theorems that imply 'positive' times 'negative' equals 'negative' and 'negative' times 'positive' equals 'negative'.

Theorem 22.6.

If **P**(r) is any basic positive alternating cycle matrix of dimensions *2m* by *2m* and **N**(s) is any basic negative alternating cycle matrix with the same dimensions, then the product **P**(r) x **N**(s) is a basic negative alternating cycle matrix, **N**(y), where the value of y is indicated in the last column in figure 22.2, in agreement with the eight distinct cases depending on the parity and the conditions for r and s.

case	***r***	***s***	**conditions**	***y***
1	even	even	$r \geq s$	$r - s + 1$
2	even	even	$r < s$	$s - r$
3	even	odd	$r+s < 2m$	$r+s$
4	even	odd	$r+s > 2m$	$(4m+1) - (r+s)$
5	odd	even	$r+s \leq 2m+1$	$r+s - 1$
6	odd	even	$r+s > 2m+1$	$(4m+2) - (r+s)$
7	odd	odd	$r > s$	$r - s$
8	odd	odd	$r \leq s$	$s - r + 1$

Figure 22.2.

This result leads, as we saw in chapter 19, to the following theorem.

Theorem 22.7.

If **C** is any positive alternating cycle matrix of dimensions *2m* by *2m* and **D** is any negative alternating cycle matrix of the same dimensions, then the product **CD** is a negative alternating cycle matrix.

From theorem 22.6 we may also deduce the structure of the matrix associated with the multiplication table of basic positive alternating cycle matrices of dimensions *2m* by *2m* times basic negative alternating cycle matrices of the same dimensions: it is a positive alternating cycle matrix.

Theorem 22.8.

The matrix associated with the multiplication table of basic positive alternating cycle matrices of dimensions *2m* by *2m* times basic negative alternating cycle matrices of the same dimensions is a positive alternating cycle matrix.

And now, in the inverse order, we have analogously.

Theorem 22.9.

If **N**(r) is any basic negative alternating cycle matrix of dimensions *2m* by *2m* and **P**(s) is any basic positive alternating cycle matrix of the same dimensions, then the product **N**(r) x **P**(s) is a basic negative alternating cycle matrix, **N**(j), where the value of j is indicated in the last column in figure 22.3, in agreement with the eight distinct cases depending on the parity and conditions for r and s.

case	***r***	***s***	**conditions**	***j***
1	even	even	$r+s \leq 2m$	$r+s$
2	even	even	$r+s > 2m$	$(4m+1) - (r+s)$
3	even	odd	$r > s$	$r - s + 1$
4	even	odd	$s > r$	$s - r$
5	odd	even	$s > r$	$s - r + 1$
6	odd	even	$r > s$	$r - s$
7	odd	odd	$r+s > 2m$	$(4m+2) - (r+s)$
8	odd	odd	$r+s \leq 2m$	$r+s - 1$

Figure 22.3.

This result leads, as we saw in Chapter 19, immediately to the following theorem.

Theorem 22.10.

If **E** is any negative alternating cycle matrix of dimensions *2m* by *2m* and **F** is any positive alternating cycle matrix of the same dimensions, then the product **EF** is a negative alternating cycle matrix.

Theorem 22.11.

The matrix associated with the multiplication table of basic negative alternating cycle matrices of dimensions *2m* by *2m* times basic positive alternating cycle matrices of the same dimensions is a negative alternating cycle matrix.

ACTIVITIES

In chapters 21 and 22 we formulated and proved a series of theorems, which concern the properties of the multiplication of alternating cycle matrices of even dimensions.

- Formulate the corresponding theorems for the multiplication of alternating cycle matrices of odd dimensions.
- Try to prove these theorems distinguishing, if necessary, the various possible subcases.

In the next chapter we shall introduce cycle matrices that do not alternate.
Let us end chapter 22 with two more application theorems.

Theorem 22.12.

If one knows any one row or column of an alternating cycle matrix and one knows its structure – positive or negative –, then one may reconstruct the whole matrix.

Figure 22.4 presents an example. Only the third column of a negative alternating cycle matrix of dimensions 8 by 8 is given.

		3					
		2					
		1					
		5					
		6					
		-2					
		4					
		9					

Figure 22.4.

Let us mark in the matrix the first alternating cycle. We see that * corresponds to a 2, and # to a 5 (figure 22.5).

*	#	3					
#		2					
	*	1	#				
		5		*			
		6	*		#		
		-2		#		*	
		4			*		#
		9				#	*

Figure 22.5.

In this way we can fill in the elements of the first alternating cycle (figure 22.6).

2	5	3					
5		2					
	2	1	5				
		5		2			
		6	2		5		
		-2		5		2	
		4			2		5
		9				5	2

Figure 22.6.

In the same way we obtain the elements of the second cycle (figure 22.7), etc.

2	5	**3**	**-2**				
5	**-2**	2		**3**			
3	2	1	5		**-2**		
-2		5		2		**3**	
	3	6	2		5		**-2**
		-2		5		2	**3**
		4	**3**		2	**-2**	5
		9		**-2**	**3**	5	2

Figure 22.7.

An immediate and practical consequence of the last theorem is the following:

Theorem 22.13.

To calculate the product of two alternating cycle matrices it is sufficient to determine the elements of one row or of one column of the matrix product.

Knowing that row or column, one may easily complete the whole matrix, as the nature of the alternating cycle matrix – positive or negative – is determined by the nature of the factors themselves.

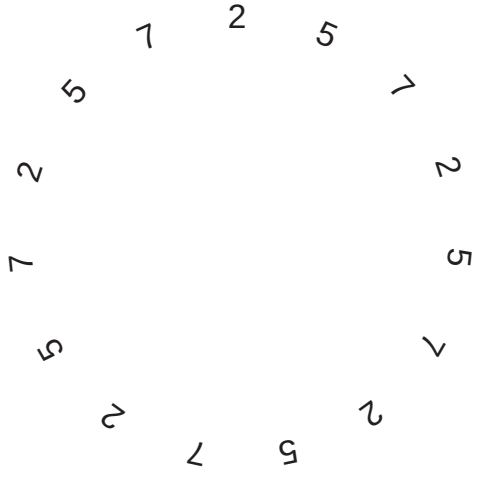
Chapter 23

CYCLE MATRICES OF DIMENSIONS 6 BY 6 AND OF PERIOD 3

ABSTRACT

In chapter 23 cycle matrices of dimensions 6 by 6 and of period 3 will be constructed and some of their properties will be analyzed.

All cycle matrices considered so far had alternating cycles. In an alternating cycle two numbers alternate. One may, however, construct other types of cycles where more than two numbers are repeated. Figure 23.1 illustrates a cycle of period 3, where the three numbers 2, 5, and 7 alternate.



Cycle of period 3

Figure 23.1.

ACTIVITIES

- Try to picture matrices of dimensions 6 by 6 that have a negative cyclic structure of period 3.
- Try to construct matrices of dimensions 6 by 6 that have a positive cyclic structure of period 3.

2	3	-1	5	-4	0
4	-3	4	-2	-3	-2
5	3	0	2	-4	-1
-1	-4	2	0	3	5
-2	-3	-2	4	-3	4
0	-4	5	-1	3	2

A

Figure 23.2.

Figure 23.2 presents a matrix that has a negative cyclic structure. In the first cycle (figure 23.3a), the numbers 2, 3 and 4 alternate; in the second cycle (figure 23.3b), the numbers -1, 5 and -3 alternate and in the third cycle the numbers -4, 0 and -2 are repeated (figure 23.3c).

2	3				
4		4			
	3		2		
		2		3	
			4		4
				3	2

a

		-1	5		
	-3			-3	
5					-1
-1					5
	-3			-3	
		5	-1		

b

				-4	0
			-2		-2
		0		-4	
	-4		0		
-2		-2			
0	-4				

c

Figure 23.3.

Matrix **B** in figure 23.4 has the same negative cyclic structure of period 3.

3	1	2	-2	-1	5
0	4	0	-4	4	-4
-2	1	5	3	-1	2
2	-1	3	5	1	-2
-4	4	-4	0	4	0
5	-1	-2	2	1	3

B

Figure 23.4.

ACTIVITIES

- Construct some more matrices that present a negative cyclic structure of period 3.
- Multiply some negative cycle matrices of period 3 by a number and observe the structure of the results. Formulate a general hypothesis.
- Add some pairs of negative cycle matrices of period 3 and observe the structure of the results. Formulate a general hypothesis.

8	12	-4	20	-16	0
16	-12	16	-8	-12	-8
20	12	0	8	-16	-4
-4	-16	8	0	12	20
-8	-12	-8	16	-12	16
0	-16	20	-4	12	8

4**A**

5	4	1	3	-5	5
4	1	4	-6	1	-6
3	4	5	5	-5	1
1	-5	5	5	4	3
-6	1	-6	4	1	4
5	-5	3	1	4	5

A+B

Figure 23.5.

Figure 23.5 presents the matrices 4**A** and **A**+**B**. Both are indeed negative cycle matrices of period 3. In general, a multiple of a negative cycle matrix of period 3 is also a negative cycle matrix of period 3. The other conjecture that may be easily proven is the following: The sum of two negative cycle matrices of period 3, having the same dimensions, is also a negative cycle matrix of period 3.

ACTIVITIES

- Multiply some pairs of negative cycle matrices of dimensions 6 by 6 and of period 3. Analyze the structure of the results of the multiplication. What is a particularity of the diagonals of the products?
- Visualize the notion of a positive cycle matrix of period 3.

34	-8	30	6	0	-14
2	-12	38	2	-36	38
30	0	34	-14	-8	6
6	-8	-14	34	0	30
38	-36	2	38	-12	2
-14	0	6	30	-8	34

AB

24	3	24	8	-11	-8
12	8	-20	12	-48	-20
24	-11	24	-8	3	8
8	3	-8	24	-11	24
-20	-48	12	-20	8	12
-8	-11	8	24	3	24

BA

Figure 23.6.

Figure 23.6 presents the products **AB** and **BA** that are very different one from another. As could be expected, the products are not negative cycle matrices of period 3. It might be expected that negative' times 'negative' would equal 'positive.' Let us observe the two positive cycles of matrix **AB** (figure 23.7): Both have period 3.

	-8	30			
2			2		
30				-8	
	-8				30
		2			2
			30	-8	

1st cycle

			6	0	
		38			38
	0				6
6				0	
38			38		
	0	6			

2nd cycle

Figure 23.7.

In the case of alternating cycle matrices of period 2 the diagonals are constant, but now this does not occur (see figure 23.8).

34					
	-12				
		34			
			34		
				-12	
					34

Principal diagonal of **AB**

					-14
				-36	
			-14		
		-14			
	-36				
-14					

Secondary diagonal of **AB**

Figure 23.8.

In each one of the diagonals two numbers appear, in the sequence: 1st number, 2nd number, 1st ,1st, 2nd and, finally, once more the 1st number. Will this be a particularity only of matrix **AB**? Let us observe also the diagonals of matrix **BA** (figure 23.9).

24					
	8				
		24			
			24		
				8	
					24

Principal diagonal of **BA**

					-8
				-48	
			-8		
		-8			
	-48				
-8					

Secondary diagonal of **BA**

Figure 23.9.

The same phenomenon occurs. We can suppose that to define positive cycle matrices of dimensions 6 by 6 and of period 3, the diagonals have to present the following structure:

$$a, b, a, a, b, a$$

The question emerges why this structure and not another.

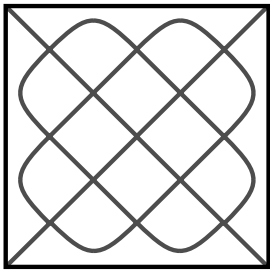
The triplet (*a*, *b*, *a*) is repeated on a diagonal, and reading the triplet from the left to the right or from the right to the left, we observe the same sequence. We can read the diagonal from the top to the bottom and then continue to read from the bottom to the top and, in this way, complete the 'cycle'. This special 'cycle' has period 3.

By consequence, we can present the general form and structure of a positive cycle matrix of dimensions 6 by 6 and of period 3 (figure 23.10).

a	c	d	f	g	i
e	b	h	e	j	h
d	g	a	i	c	f
f	c	i	a	g	d
h	j	e	h	b	e
i	g	f	d	c	a

General form

a



General structure

b

Positive cycle matrices of dimensions 6 by 6 and of period 3

Figure 23.10.

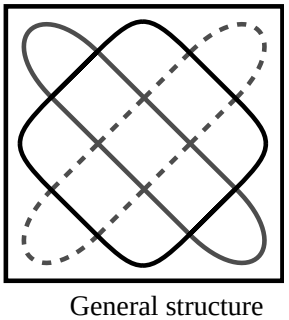
ACTIVITIES

- Construct some negative cycle matrices of dimensions 6 by 6 and of period 3 (figure 23.11) and calculate some products of two of these matrices. Will all results be positive cycle matrices of period 3?
- Multiply a negative cycle matrix of dimensions 6 by 6 and period 3 by a positive cycle matrix of the same dimensions and period, and vice versa. What may be asserted about the products? Formulate a general hypothesis.

a	b	d	e	g	h
c	f	c	i	f	i
e	b	h	a	g	d
d	g	a	h	b	e
i	f	i	c	f	c
h	g	e	d	b	a

General form

a



General structure

b

Negative cycle matrices of dimensions 6 by 6 and of period 3

Figure 23.11.

Matrices **C** and **D** constitute examples of positive cycle matrices of dimensions 6 by 6 and of period 3 (figure 23.12).

1	-4	5	6	-2	3
-3	2	0	-3	4	0
5	-2	1	3	-4	6
6	-4	3	1	-2	5
0	4	-3	0	2	-3
3	-2	6	5	-4	1

C

3	-5	1	-3	-1	4
0	2	-2	0	1	-2
1	-1	3	4	-5	-3
-3	-5	4	3	-1	1
-2	1	0	-2	2	0
4	-1	-3	1	-5	3

D

Positive cycle matrices e of dimensions 6 by 6 and of period 3

Figure 23.12.

The products **CD** and **DC** are also positive cycle matrices of dimensions 6 by 6 and period 3 (figure 23.13). In other words, in the example ‘positive’ times ‘positive’ equals ‘positive’ is true.

6	-53	39	42	-55	12
-8	38	-19	-8	16	-19
39	-55	6	12	-53	42
42	-53	12	6	-55	39
-19	16	-8	-19	38	-8
12	-55	42	39	-53	6

CD

17	-24	34	53	-42	7
-22	16	-17	-22	26	-17
34	-42	17	7	-24	53
53	-24	7	17	-42	34
-17	26	-22	-17	16	-22
7	-42	53	34	-24	17

DC

Figure 23.13.

18	-36	36	5	-6	37
15	-30	15	33	-30	33
5	-36	37	18	-6	36
36	-6	18	37	-36	5
33	-30	33	15	-30	15
37	-6	5	36	-36	18

AC

9	0	14	12	21	31
-3	-15	-3	-3	-15	-3
12	0	31	9	21	14
14	21	9	31	0	12
-3	-15	-3	-3	-15	-3
31	21	12	14	0	9

CA

Figure 23.14.

When we calculate the products **AC** and **CA**, we observe that the results are negative cycle matrices of dimensions 6 by 6 and period 3 (figure 23.14). In other words, in these particular examples one verifies that ‘negative’ times ‘positive’ equals ‘negative’ and that ‘positive’ times ‘negative’ equals ‘negative.’ The reader is invited to experiment with other matrices to determine if the same occurs.

ACTIVITIES

- Formulate general hypotheses concerning the multiplication of cycle matrices, both positive and negative, of dimensions 6 by 6 and period 3. Try to prove them.
- Picture cycle matrices of dimensions 9 by 9 and period 3 and analyze them. What will be the general form of the diagonals of positive cycle matrices of dimensions 9 by 9 and period 3?
- Construct cycle matrices of dimensions 6 by 6 and of period 4. Analyze them. What will be the general form of the diagonals of the respective positive cycle matrices?
- Why are the diagonals of alternating cycle matrices constant?
- Will it be possible to conceive of cycle matrices of period 1?
- Will it be possible to conceive of cycle matrices of period 5? If yes, give examples and analyze them.
- What values are possible for the periods of cycle matrices of dimensions $2m$ by $2m$?

Some of the questions posed in these activities will be analyzed in the next chapters.

Chapter 24

OTHER PERIODIC CYCLE MATRICES OF DIMENSIONS 6 BY 6

ABSTRACT

In chapter 24 it will be determined which periods are possible for cycle matrices of dimensions 6 by 6, where some properties of periodic cycle matrices of dimensions 6 by 6 will be encountered.

We met already two types of cycle matrices of dimensions 6 by 6, on one hand, the alternating ones of period 2 (Chapters 8 to 10), and on the other hand, the ones of period 3 (Chapter 23).

Will we be able to construct cycle matrices of dimensions 6 by 6 that present other periods?

ACTIVITIES

- Consider a negative alternating cycle matrix of dimensions 6 by 6. Multiply the matrix by itself. What will be the periodicity of the product? Why?
- Determine all possible periods for cycle matrices of dimensions 6 by 6. Characterize in each case the corresponding positive and negative cycle matrices.
- When one multiplies two cycle matrices of dimensions 6 by 6 and the same period, what can be asserted about the periodicity of the resulting matrix?
- When one multiplies an alternating cycle matrix of dimensions 6 by 6 and a cycle matrix of the same dimensions and period 3, what can be said about the periodicity of the result? Experiment with various matrices.

Let us analyze the questions posed in the activities.

-1	5	-2	1	3	4
5	1	-1	4	-2	3
-2	-1	3	5	4	1
1	4	5	3	-1	-2
3	-2	4	-1	1	5
4	3	1	-2	5	-1

A

56	12	12	1	1	18
12	56	1	12	18	1
12	1	56	18	12	1
1	12	18	56	1	12
1	18	12	1	56	12
18	1	1	12	12	56

$\mathbf{A}^2$

Figure 24.1.

Let **A** be a negative alternating cycle matrix (see the example in figure 24.1). Since **B** is a matrix of the same type, we have, in agreement with theorem 22.4,

$$\mathbf{AB} = (\mathbf{BA})^T.$$

Replacing matrix **B** with matrix **A** we have

$$\mathbf{A}^2 = \mathbf{AA} = (\mathbf{AA})^T = (\mathbf{A}^2)^T.$$

This equality means that the positive alternating cycle matrix $\mathbf{A}^2$ is a symmetrical matrix, and thus all cycles of $\mathbf{A}^2$ are constant. In other words, matrix $\mathbf{A}^2$ is a positive cycle matrix of period 1. Figure 24.1 presents an example of dimensions 6 by 6.

In this way we proved the following theorem.

Theorem 24.1.

If **A** is a negative alternating cycle matrix of any dimensions, then $\mathbf{A}^2$ is a positive cycle matrix of period 1.

Any cycle of a matrix of dimensions 6 by 6 has 12 elements. By consequence, cycle matrices of dimensions 6 by 6 may only have the factors of 12 as a period, that is, the periods may be 1 (see the example of matrix $\mathbf{A}^2$ in figure 24.1), 2 (See the chapters 8 to 10), 3 (chapter 23), 4, 6 and 12. The case of period 12 is not interesting, as any matrix of dimensions 6 by 6 is a negative cycle matrix of period 12. We shall now see the cases of periods 4 and 6.

PERIOD 4

1	3	0	-2	-4	-3
4	5	2	6	-1	-5
-1	2	-5	4	6	5
-2	-3	3	-4	1	0
-4	0	-3	1	-2	3
6	-5	5	-1	4	2

A

2	1	-2	3	0	-1
-3	6	3	-4	4	-6
4	3	-6	-3	-4	6
3	-1	1	0	2	-2
0	-2	-1	2	3	1
-4	-6	6	4	-3	3

B

Figure 24.2.

Figure 24.2 presents two negative cycle matrices of period 4. In each cycle the numbers appear three times: in the first cycle of matrix **A** we have 1, 3, 2, 4, **1**, **3**, **2**, **4**, 1, 3, 2 and 4 (figure 24.3), etc.

1	3				
4		2			
	2		4		
		3		1	
			1		3
				4	2

First cycle of matrix **A**

Figure 24.3.

Let us calculate the products **AB** and **BA**.

-1	47	-9	-29	5	-28
39	66	-28	-36	36	-50
-36	-50	66	36	39	-28
5	-9	-28	-1	-29	47
-29	-28	47	5	-1	-9
36	-28	-50	39	-36	66

AB

-4	3	16	-17	-22	-23
-26	69	-57	80	-12	-6
80	-6	69	-12	-26	-57
-22	16	-23	-4	-17	3
-17	-23	3	-22	-4	16
-12	-57	-6	-26	80	69

BA

Figure 24.4.

As might be expected – transpiring the whole beauty of the theory of cycle matrices constructed so far! – both products present (figure 24.4) a **positive** cyclic structure of period 4. Figure 24.5 illustrates the first positive cycle of matrix **AB**.

	47	-9			
39			-36		
-36				39	
	-9				47
		47			-9
			39	-36	

Figure 24.5.

Let us observe the form of the diagonals of the matrices **AB** and **BA**: (-1, 66, 66, -1, -1. 66) and (36, -28, -28, 36, 36, -28) of **AB** and (-4, 69, 69, -4, -4, 69) and (-12, -23, -23, -12, -12, -23) of **BA**. The four diagonals have the same particular form (*a*, *b*, *b*, *a*, *a*, *b*) that we can understand as having period 4: reading in figure 24.6 in the top row from the left to the right, we start with quadruple ***abba***, followed by ***ab***. When we continue to read as a cycle in the bottom row from right to left, we start with ***ba***, thus completing the second quadruple, followed by ***abba***, closing the cycle.

a	b	b	a	a	b
a	b	b	a	a	b

Figure 24.6.

We can interpret in this way the diagonal as a **degenerated** cycle: the two halves were superposed. If the quadruple would have the form ***abcd***, the cycle would present the form in figure 24.7. But when the two rows in figure 24.7 are superposed, then the elements that remain in the same place have to be equal. Hence *c=b* and *d=a*.

a	b	c	d	a	b
d	c	b	a	d	c

Figure 24.7.

In this manner we obtain the cycle in figure 24.6, leading us to the general form of the particular diagonals of the matrices **AB** and **BA**.

Now we are in the condition to define positive cycle matrices of period 4 and dimensions 6 by 6. The normal cycles have period 4 and the diagonals have as general form (*a*, *b*, *b*, *a*, *a*, *b*).

Satisfying this definition, we may construct some positive cycle matrices of period 4 and dimensions 6 by 6. Figure 24.8 presents two examples.

3	-1	5	-4	6	4
0	-2	-5	2	1	-3
2	-3	-2	1	0	-5
6	5	4	3	-4	-1
-4	4	-1	6	3	5
1	-5	-3	0	2	-2

C

5	2	-1	-3	-4	2
1	3	6	-2	0	-5
-2	-5	3	0	1	6
-4	-1	2	5	-3	2
-3	2	2	-4	5	-1
0	6	-5	1	-2	3

D

Figure 24.8.

Calculating the products **CD** and **DC** (figure 24.9), we observe that both are also positive cycle matrices of dimensions 6 by 6 and period 4, that is, in this example one verifies ‘positive’ times ‘positive’ equals ‘positive.’

2	18	-10	-47	27	39
-3	1	-6	7	0	-26
7	-26	1	0	-3	-6
27	-10	39	2	-47	18
-47	39	18	27	2	-10
0	-6	-26	-3	7	1

CD

13	-47	3	-50	36	-2
-2	-10	-15	2	7	-23
2	-23	-10	7	-2	-15
36	3	-2	13	-50	-47
-50	-2	-47	36	13	3
7	-15	-23	-2	2	-10

DC

Figure 24.9.

Figure 24.10 presents the products **AC** (‘negative’ times ‘positive’) and **CA** (‘positive’ times ‘negative’), and in both the cases the result is a negative cycle matrix of the same dimensions and period 4.

4	-18	-5	-28	-1	-17
51	31	31	8	-8	-10
-8	31	-10	51	8	31
-28	-17	-18	-1	4	-5
-1	-5	-17	4	-28	-18
8	-10	31	-8	51	31

AC

2	6	-37	26	19	47
-29	-11	9	-36	-40	-18
-40	9	-18	-29	-36	-11
26	47	6	19	2	-37
19	-37	47	2	26	6
-36	-18	-11	-40	-29	9

CA

Figure 24.10.

PERIOD 6

Figure 24.11 presents two negative cycle matrices of dimensions 6 by 6 and period 6. These matrices have a rotational symmetry of 180°.

3	4	2	5	8	-3
-4	6	1	-8	-1	-5
7	0	-7	-2	-9	-6
-6	-9	-2	-7	0	7
-5	-1	-8	1	6	-4
-3	8	5	2	4	3

E

2	0	3	4	1	0
3	-4	6	5	-1	-6
-8	-3	7	-2	8	-5
-5	8	-2	7	-3	-8
-6	-1	5	6	-4	3
0	1	4	3	0	2

F

Figure 24.11.

Figure 24.12 presents the products **EF** and **FE**. Both display a rotational symmetry of 180° and a negative cyclic structure. Will this time ‘negative’ times ‘negative’ be equal to ‘negative’?

-71	7	65	102	-32	-56
48	-95	22	-65	26	10
134	8	-93	-44	-7	12
12	-7	-44	-93	8	134
10	26	-65	22	-95	48
-56	-32	102	65	7	-71

EF

-2	-29	-33	-23	-5	0
60	-104	-72	-13	-56	-4
24	-80	-153	-18	-96	-64
-64	-96	-18	-153	-80	24
-4	-56	-13	-72	-104	60
0	-5	-23	-33	-29	-2

FE

Figure 24.12.

The two matrices have special diagonals. Their general form is (a, b, c, c, b, a). In other words, the diagonals are once more degenerated cycles. In order for the reading from left to right of the top sextuple in figure 24.13 to coincide with the reading from right to left in the

bottom sextuple, *a=f*, *b=e* and *c=d* has to be valid, which leads us to discover the general form (*a*, *b*, *c*, *c*, *b*, *a*).

a	b	c	d	e	f
f	e	d	c	b	a

Figure 24.13.

In this manner the products **EF** and **FE** are positive cycle matrices of period 6, and 'negative' times 'negative' equals 'positive' remains valid. Nevertheless, a positive matrix of this type has rotational symmetry of 180° and, by consequence, is a negative cycle matrix too.

The last question posed was about what happens if we multiply two cycle matrices of the same dimensions but of different periods. Let us observe an example. Figure 24.14 presents the negative cycle matrices **G** and **H**, the first having 2 as a period and the second having 3 as period.

-1	5	-2	1	3	4
5	1	-1	4	-2	3
-2	-1	3	5	4	1
1	4	5	3	-1	-2
3	-2	4	-1	1	5
4	3	1	-2	5	-1

G

2	3	-1	5	-4	0
4	-3	4	-2	-3	-2
5	3	0	2	-4	-1
-1	-4	2	0	3	5
-2	-3	-2	4	-3	4
0	-4	5	-1	3	2

H

Figure 24.14.

The products **GH** and **HG** have neither period 2 nor 3. Observing the form of their diagonals we see that they have period 6. The number 6 is the lowest common multiple of 2 and 3.

1	-53	37	-11	3	17
9	-13	26	10	8	17
-6	-30	5	13	5	42
42	5	13	5	-30	-6
17	8	10	26	-13	9
17	3	-11	37	-53	1

GH

8	42	-1	28	-13	-14
-46	5	-17	13	23	2
-4	41	-20	29	-2	6
6	-2	29	-20	41	-4
2	23	13	-17	5	-46
-14	-13	28	-1	42	8

HG

Figure 24.15.

ACTIVITY

- Based on the observations made in this chapter, formulate some hypotheses and determine how to prove them.

Chapter 25

PERIODIC CYCLE MATRICES OF ODD DIMENSIONS

ABSTRACT

In chapter 25 some periodic cycle matrices of odd dimensions will be presented and several properties of these matrices will be analyzed.

ACTIVITIES

- Determine the possible periods for cycle matrices of dimensions 5 by 5. Characterize in each case the corresponding positive and negative matrices.
- Determine the possible periods for cycle matrices of dimensions 7 by 7.
- Determine the possible periods for cycle matrices of dimensions 9 by 9.
- When two cycle matrices of dimensions 9 by 9 and the same period are multiplied, what can be asserted about the periodicity of the resulting matrix?

Let us analyze the questions posed in the activities.

Each cycle of a cycle matrix of dimensions 5 by 5 has 10 elements. By consequence, the possible periods are 1, 2, 5 and 10. The case 10 does not restrict the matrix in anything, as all matrices of dimensions 5 by 5 have cycles of period 10. Thus, only three periods really remain: 1 (all cycles are constant; see the general forms in figure 25.1), 2 (already analyzed in chapter 13) and 5.

$$\begin{array}{|ccccc|}\hline a & b & b & c & c \\ b & a & c & b & c \\ b & c & a & c & b \\ c & b & c & a & b \\ c & c & b & b & a \\ \hline \end{array} \qquad \begin{array}{|ccccc|}\hline d & d & e & e & f \\ d & e & d & f & e \\ e & d & f & d & e \\ e & f & d & e & d \\ f & e & e & d & d \\ \hline \end{array}$$

Positive Negative

Period 1

Figure 25.1.

In the case of period 5 we first analyze the form of a diagonal as a degenerated cycle (figure 25.2). Immediately we see that $a = e$ and $b = d$ has to hold, and hence the general form of a diagonal is (a, b, c, b, a).

a	b	c	d	e
e	d	c	b	a

Figure 25.2.

In this manner the general forms of positive and negative cycle matrices of dimensions 5 by 5 and period 5 are the ones presented in figure 25.3.

a	c	d	h	i
g	b	l	e	j
f	k	c	k	f
j	e	l	b	g
i	h	d	c	a

Positive

c	d	h	i	a
g	l	e	b	j
k	f	c	f	k
j	b	e	l	g
a	i	h	d	c

Negative

General forms of positive and negative cycle matrices of dimensions 5 by 5 and period 5

Figure 25.3.

Cycle matrices of dimensions 7 by 7 can only have the numbers 1, 2 (Chapters 15 and 16) and 7 as a period, except for the trivial case of 14.

Cycle matrices of dimensions 9 by 9 may have a period of the factors of 18, that is 1, 2, 3, 6, and 9, excluding the trivial case of 18.

Let us analyze the general form of a diagonal when the period is 6.

a	b	c	d	e	f	a	b	c
f	e	d	c	b	a	f	e	d

Figure 25.4.

From figure 25.4 we draw the conclusion that $a = f$, $b = e$ and $c = d$. Thus the general form of the diagonal of a cycle matrix of dimensions 9 by 9 and period 6 has to be (a, b, c, c, b, a, a, b, c).

2	3	-3	-2	8	5	0	1	4
-6	-4	-1	6	1	3	4	5	3
-1	5	5	0	0	6	7	-3	2
6	-3	0	2	5	7	-1	0	5
4	1	3	-1	5	-6	3	-4	6
5	1	-2	4	3	0	2	8	-3
0	8	4	-3	1	2	5	3	-2
3	5	6	3	-4	4	-6	1	-1
7	0	2	5	-3	-1	6	5	0

P

1	4	5	-3	4	-2	5	-4	2
-1	-2	-2	3	2	2	0	-3	-2
1	4	6	2	1	-5	5	2	3
-5	2	2	3	4	5	1	1	6
0	2	-2	-2	-3	-1	2	-2	3
-2	-4	-3	2	4	5	1	4	5
5	4	2	5	-4	1	-2	4	-3
2	-3	3	-2	-2	0	-1	2	-2
5	1	3	6	2	1	-5	4	2

Q

Figure 25.5.

Figure 25.5 presents two matrices **P** and **Q** that are negative cycle matrices of dimensions 9 by 9 and period 6. Both products, **PQ** and **QP** (see figure 25.6), are – is it not miraculous?! – positive cycle matrices of the same dimensions and of the same period 6.

18	-17	-34	7	5	28	-7	-3	32
6	-14	5	54	-20	60	-52	88	31
26	21	8	93	17	26	5	53	22
5	17	22	8	53	26	26	21	93
60	88	54	31	-14	-52	6	-20	5
-7	5	32	-34	-3	18	28	-17	7
28	-3	7	32	-17	-7	18	5	-34
-52	-20	31	5	88	6	60	-14	54
26	53	93	22	21	5	26	17	8

PQ

-37	43	34	-13	26	-6	123	-5	35
25	-25	-25	-17	39	-24	-9	-10	10
-10	57	74	11	0	76	80	-9	47
80	0	47	74	-9	-10	76	57	11
-24	-10	-17	10	-25	-9	25	39	-25
123	26	35	34	-5	-37	-6	43	-13
-6	-5	-13	35	43	123	-37	26	34
-9	39	10	-25	-10	25	-24	-25	-17
76	-9	11	47	57	80	-10	0	74

QP

Figure 25.6.

Figure 25.7 presents the matrices **R** and **S** that are positive cycle matrices of dimensions 9 by 9 and period 6.

5	-3	-1	4	7	-4	2	4	3
-3	5	4	2	-5	5	0	-3	-6
2	-3	1	6	0	2	4	5	-5
4	0	-5	1	5	2	2	-3	6
5	-3	2	-6	5	0	-3	-5	4
2	7	3	-1	4	5	-4	-3	4
-4	4	4	3	-3	2	5	7	-1
0	-5	-6	4	-3	-3	5	5	2
2	5	6	-5	-3	4	2	0	1

R

-3	-5	1	4	5	4	6	4	2
4	2	6	-2	0	2	-3	-3	-1
-2	6	8	-5	-1	2	5	4	0
5	-1	0	8	4	-2	2	6	-5
2	-3	-2	-1	2	-3	4	0	6
6	5	2	1	4	-3	4	-5	4
4	4	4	2	-5	6	-3	5	1
-3	0	-1	6	-3	4	2	2	-2
2	4	-5	0	6	5	-2	-1	8

S

Figure 25.7.

The products **RS** and **SR** are presented in figure 25.8. Once more, both are positive cycle matrices of dimensions 9 by 9 and period 6.

-13	-62	-54	80	36	38	50	84	37
48	63	112	-34	-28	-40	-3	-24	-73
13	-10	32	97	-24	5	54	82	-61
54	-24	-61	32	82	13	5	-10	97
-40	-24	-34	-73	63	-3	48	-28	112
50	36	37	-54	84	-13	38	-62	80
38	84	80	37	-62	50	-13	36	-54
-3	-28	-73	112	-24	48	-40	63	-34
5	82	97	-61	-10	54	13	-24	32

RS

31	8	-2	-22	29	25	29	21	80
32	-8	26	36	37	11	-12	4	-38
-53	29	59	82	-93	52	58	83	-105
58	-93	-105	59	83	-53	52	29	82
11	4	36	-38	-8	-12	32	37	26
29	29	80	-2	21	31	25	8	-22
25	21	-22	80	8	29	31	29	-2
-12	37	-38	26	4	32	11	-8	36
52	83	82	-105	29	58	-53	-93	59

SR

Figure 25.8.

Figure 25.9 presents the products **PR** and **RP**. Remember that **P** is a negative cycle matrix whereas **R** is a positive cycle matrix. Will it also be verified in these cases that ‘negative’ times ‘positive’ equals ‘negative’ (**PR**) and ‘positive’ times ‘negative’ equals ‘negative’ (**RP**)?

45	44	66	-75	34	35	-43	-60	49
5	25	-26	-24	-11	34	32	4	72
-22	108	102	29	-26	100	18	22	-45
100	22	29	-45	108	18	-22	-26	102
32	-11	72	-26	4	5	34	25	-24
35	-60	-75	49	44	-43	45	34	66
-43	34	49	66	-60	45	35	44	-75
34	4	-24	72	25	32	5	-11	-26
18	-26	-45	102	22	-22	100	108	29

PR

94	49	50	-22	57	13	-6	-42	75
-74	-30	-31	26	11	62	23	37	-21
47	64	34	-24	48	82	-47	-8	12
82	-8	-24	12	64	-47	47	48	34
23	11	-21	-31	37	-74	62	-30	26
13	-42	-22	75	49	-6	94	57	50
-6	57	75	50	-42	94	13	49	-22
62	37	26	-21	-30	23	-74	11	-31
-47	48	12	34	-8	47	82	64	-24

RP

Figure 25.9.

Indeed, both matrices **PR** and **RP** are negative cycle matrices of period 6. For example, in the first cycle of matrix **RP** the numbers 94, 49, -31, -24, 64, -74 repeat.

This no longer constitutes a surprise: for all periodic cycle matrices studied, we observed the same wonderful phenomenon, and in the case of alternating cycle matrices some proofs were presented. Let us attempt the formulation of some general hypotheses concerning the multiplication of periodic cycle matrices.

ACTIVITY

- Formulate some general hypotheses concerning the multiplication of periodic cycle matrices.

Chapter 26

THE WORLD OF THE PERIODIC CYCLE MATRICES

ABSTRACT

In chapter 26 some general theorems concerning the properties of positive and negative cycle matrices of any dimensions and any period will be presented. The theorems will summarize and generalize the experiences obtained throughout the book.

After all experimentation and all reflection with cycle matrices of various dimensions and of several periods, realized throughout the present book, the moment has come to formulate some general regularities concerning the properties of cycle matrices.

Let us consider matrices of dimensions n by n, where n represents any natural number. The number n can be even or odd.

The number of elements in any cycle of matrices of dimensions n by n is always equal to $2n$. The period p of a cycle matrix of dimensions n by n is a divisor of $2n$, except for $2n$ itself.

Two classes of cycle matrices of dimensions n by n and period p may be distinguished. Matrices of the first class are called **positive**. Matrices of the second class are called **negative**.

When n is even ($n = 2m$), the negative matrices are composed of m cycles, while the positive matrices are composed of m-1 cycles and two periodic diagonals.

When n is odd ($n = 2s+1$), the negative matrices are composed of s cycles and a periodic secondary diagonal, whereas the positive matrices are constituted by s cycles and a periodic principal diagonal.

Easy to prove are the following theorems concerning the multiplication of a cycle matrix by an arbitrary number:

Theorem 26.1a.

When one multiplies a **positive** cycle matrix of dimensions n by n and period p by any number, the matrix resulting from this operation is also a positive cycle matrix of the same dimensions and equal period.

Theorem 26.1b.

When one multiplies a **negative** cycle matrix of dimensions *n* by *n* and period *p* by any number, the matrix resulting from this operation is also a positive cycle matrix of the same dimensions and equal period.

Equally easy to prove are the following theorems concerning the addition of two cycle matrices:

Theorem 26.2a.

If **A** and **B** are two **positive** cycle matrices of dimensions *n* by *n* and period *p*, then **A**+**B** is also a positive cycle matrix of the same dimensions and equal period.

Theorem 26.2a.

If **A** and **B** are two **negative** cycle matrices of dimensions *n* by *n* and period *p*, then **A**+**B** is also a negative cycle matrix of the same dimensions and equal period.

ACTIVITIES

- Prove theorems 26.1a and 26.1b.
- Prove theorems 26.2a and 26.2b.

Truly spectacular and wonderful is the situation arising from the multiplication of positive and negative cycle matrices of dimensions *n* by *n* and period *p*.

Theorem 26.3a.

If **A** and **B** are two positive cycle matrices of dimensions *n* by *n* and period *p*, then **AB** is also a positive cycle matrix of the same dimensions and equal period.

Theorem 26.3b.

If **A** is a positive cycle matrix of dimensions *n* by *n* and period *p* and **B** is a negative cycle matrix of the same dimensions and equal period, then **AB** and **BA** are negative cycle matrices of the same dimensions and the same period.

Theorem 26.3c.

If **A** and **B** are two negative cycle matrices of dimensions *n* by *n* and period *p*, then **AB** is a positive cycle matrix of the same dimensions and equal period.

Relative to the multiplication of two cycle matrices of the dimensions *n* by *n* and distinct periods *p* and *q*, we have the following theorem:

Theorem 26.4.

If **A** is a cycle matrix of dimensions *n* by *n* and period *p* and **B** is a cycle matrix of the same dimensions and of period *q*, then **AB** and **BA** are cycle matrices of the same dimensions that have a period equal to **the lowest common multiple of *p* and *q***. The matrices **AB** and **BA** are negative if one of the matrices is negative and the other positive; the matrices **AB** and **BA** are positive if both matrices **A** and **B** are positive or if both are negative.

Besides these surprising rules of 'signs', we observed more interesting particularities in specific cases, for instance, p = 2:

Theorem 26.5.

If **A** and **B** are two positive alternating cycle matrices of dimensions *n* by *n*, then **A** and **B** commute, that is, **AB** = **BA**.

Theorem 26.6.

If **A** and **B** are two negative alternating cycle matrices of dimensions *n* by *n*, then **AB** and **BA** are mutually symmetrical in the sense that a reflection across the principal diagonal of **AB** transforms matrix **AB** into matrix **BA**, and the cycles of **AB** and **BA** differ in phase:

$$\mathbf{AB} = (\mathbf{BA})^{T}.$$

Here we conclude our adventurous trip through the new world of cycle matrices. The next chapter explains how I had the good luck to discover or invent this world.

The reader is invited to join me on further adventurous trips to other matrix worlds like those containing **cylinder** and **helix** matrices. In the bibliography the reader will find some references to papers published on-line about those types of matrices.

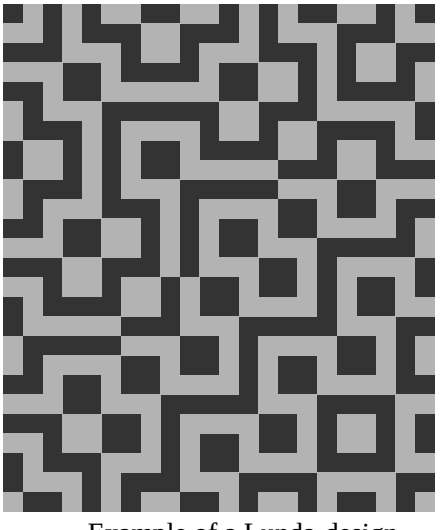
Chapter 27

DISCOVER THE WORLD OF THE CYCLE MATRICES

ABSTRACT

In chapter 27 I shall describe the context that led me to enter, unexpectedly, into the world of cycle matrices.

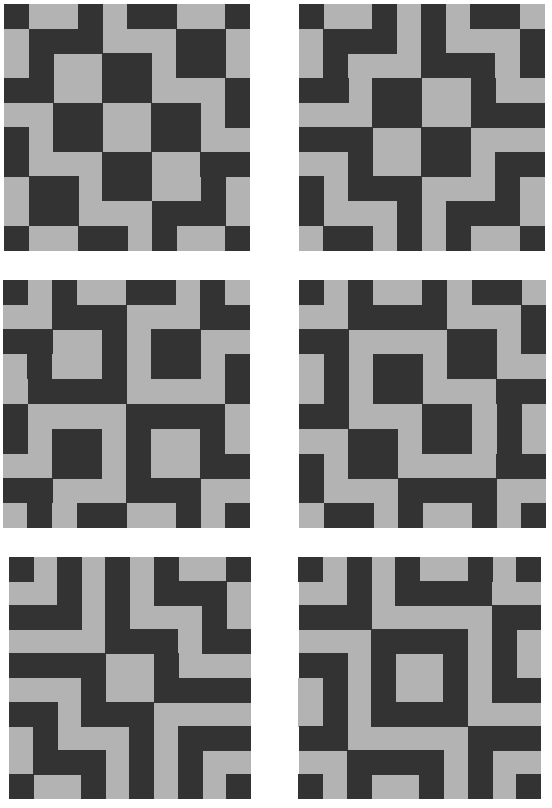
It happened on May 13, 2001. My daughter Likilisa celebrated her birthday and I was analyzing some aspects of a type of design, that I had called Lunda-designs. Figure 27.1 presents an example of a Lunda-design.



Example of a Lunda-design

Figure 27.1.

I had formulated the concept of **Lunda-design** for the first time in 1989. It had been a beautiful and unexpected surprise too. It emerged in the context of my mathematical analysis of traditional drawings from a region of Angola, called Lunda. The story of the discovery of Lunda-designs is already another story, told in other publications (see, for instance, my book *Sona Geometry from Angola. Mathematics of an African Tradition* [2006]).

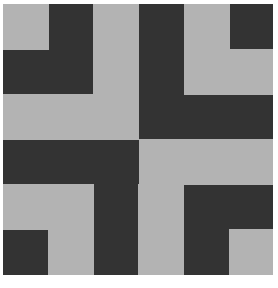


Examples of Liki-designs of dimensions 10 by 10

Figure 27.2.

Let us return to May 13, 2001. I was considering a special class of Lunda-designs, noting that the designs belonging to this class had specific attractive properties that other Lunda-designs do not display. As it was the birthday of my daughter Likilisa, I decided to call this type design **Liki-design**. Figure 27.2 illustrates some examples of Liki-designs of dimensions 10 by 10. The diagonals are symmetry axes.

Figure 27.3 presents a Liki-design of dimensions 6 by 6 together with the matrix associated to the design. One obtains the matrix from the design by the substitution of the darker unit squares by the number 1 and of the lighter colored unit squares by the number 0.



$$\begin{matrix} 0 & 1 & 0 & 1 & 0 & 1 \\ 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & 1 \\ 1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 1 \\ 1 & 0 & 1 & 0 & 1 & 0 \end{matrix}$$

A Liki-design and its associated matrix

Figure 27.3.

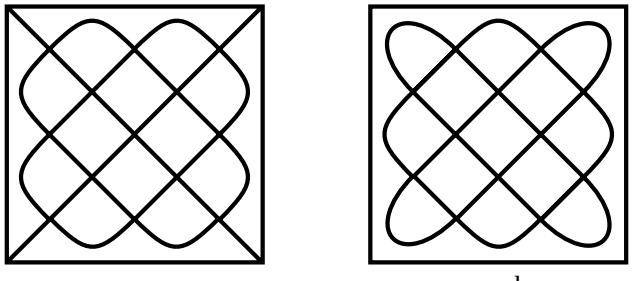
Then I started to experiment, to ‘play’, with associated matrices. For instance, since **B** is the associated matrix in figure 27.3, I calculated the matrices $\mathbf{B}^2$, $\mathbf{B}^3$, $\mathbf{B}^4$, etc. (figure 27.4).

$$\begin{array}{|cccccc|}\hline 3 & 2 & 2 & 1 & 1 & 0 \\ 2 & 3 & 1 & 2 & 0 & 1 \\ 2 & 1 & 3 & 0 & 2 & 1 \\ 1 & 2 & 0 & 3 & 1 & 2 \\ 1 & 0 & 2 & 1 & 3 & 2 \\ 0 & 1 & 1 & 2 & 2 & 3 \\ \hline \end{array} \qquad \begin{array}{|cccccc|}\hline 3 & 6 & 2 & 7 & 3 & 6 \\ 6 & 7 & 3 & 6 & 2 & 3 \\ 2 & 3 & 3 & 6 & 6 & 7 \\ 7 & 6 & 6 & 3 & 3 & 2 \\ 3 & 2 & 6 & 3 & 7 & 6 \\ 6 & 3 & 7 & 2 & 6 & 3 \\ \hline \end{array} \qquad \begin{array}{|cccccc|}\hline 19 & 16 & 16 & 11 & 11 & 8 \\ 16 & 19 & 11 & 16 & 8 & 11 \\ 16 & 11 & 19 & 8 & 16 & 11 \\ 11 & 16 & 8 & 19 & 11 & 16 \\ 11 & 8 & 16 & 11 & 19 & 16 \\ 8 & 11 & 11 & 16 & 16 & 19 \\ \hline \end{array}$$

$$\mathbf{B}^2 \qquad\qquad \mathbf{B}^3 \qquad\qquad \mathbf{B}^4$$

Figure 27.4.

I noted that the diagonals of $\mathbf{B}^2$ and $\mathbf{B}^4$ were constant, and that there were other numbers repeated in the matrices. The repeated numbers formed certain cycles. I found that the even powers of **B** had a certain cyclic structure (figure 27.5a) and that the odd powers of **B** had another cyclic structure (figure 27.5b). In this way the door to the world of cycle matrices was discovered.



a b

Cyclic structures

Figure 27.5.

Will it be that someone had entered this new world before me? Entering through some other door? I do not know the answer to this question, both historical and philosophical. Nevertheless, this book constitutes an invitation to the readers to enter into the beautiful new world of cycle matrices and to encounter interesting algebraic-geometric structures.

The reader may ask if the concept of cycle matrix has already had applications.

In fact, in this book some applications were presented in the context itself of the analysis of cycle matrices. For instance, we verified that the matrix associated to the multiplication table of basic negative alternating cycle matrices of given dimensions constitutes a positive cycle matrix (Chapters 17 and 18).

But have there been applications outside the direct context of the cycle matrices themselves?

The matrix associated to the table of multiplication modulo 5 of the natural numbers 1 to 4, constitutes an example of a negative alternating cycle matrix, as figure 27.6 shows.

X	**1**	**2**	**3**	**4**
1	1	2	3	4
2	2	4	1	3
3	3	1	4	2
4	4	3	2	1

1	2	3	4
2	4	1	3
3	1	4	2
4	3	2	1

Multiplication table modulo 5 and associated matrix

Figure 27.6.

The reader is invited to find other examples.

In the following chapter I shall give an example of an application in an unexpected context…

Chapter 28

CYCLE MATRICES, GENETIC MATRICES AND THE GOLDEN SECTION

ABSTRACT

Chapter 28 will present an application of cycle matrices in biology.

In September 2004 I was reading some recent papers about genetic matrices by the Russian mathematician-biologist Sergei Petoukhov and of the Chinese mathematician Matthew He, residing in the United States of America. The concept of a genetic matrix had been introduced by Petoukhov to facilitate the study of genetic codes.

The matrices in figure 28.1 constitute two examples of genetic matrices presented in the works of Petoukhov and He.

$$\begin{array}{|cccc|}\hline 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ \hline \end{array} \qquad \begin{array}{|cccc|}\hline 6 & 5 & 5 & 4 \\ 5 & 6 & 4 & 5 \\ 5 & 4 & 6 & 5 \\ 4 & 5 & 5 & 6 \\ \hline \end{array}$$

a b

Figure 28.1.

The first genetic matrix is also a Lunda-matrix and both a negative alternating cycle matrix ( ) as a positive cycle matrix of period 1 ( ) of dimensions 4 by 4. The second genetic matrix constitutes another example of a positive cycle matrix of period 1 and of dimensions 4 by 4. The genetic matrices of dimensions 8 by 8, 16 by 16, 32 by 32, etc., presented by Petoukhov and He, are composed of blocs of these positive cycle matrices of period 1. By consequence, various interesting properties of the genetic matrices result immediately from the general properties of positive cycle matrices.

In our book only whole numbers in the cycle matrices we analyzed appear. This was just to facilitate the calculations and the representation of the matrices. Thus any rational, real or complex numbers can be used.

Next we shall see matrices in which a famous irrational number, abbreviated by the Greek letter φ (phi), appears. The number φ, called the **golden section**, is the positive solution of the quadratic equation $x^2 - x - 1 = 0$, that is, $\varphi = (1+\sqrt{5})/2 = 1.61803398\ldots$ The golden section has been applied in architecture since antiquity and frequently appears in studies of Nature, particularly in biological studies.

Petoukhov analyzes the relationship between the golden section and genetic matrices and introduces the concept of **golden genetic matrices**, that is, genetic matrices of which all elements are powers of the number φ. With the theory of cycle matrices we can construct golden genetic matrices that have not yet appeared in the studies of Petoukhov.

We know that the square of a negative cycle matrix of period 1 or 2 (Theorem 24.1) is a positive cycle matrix of period 1. The genetic matrix in figure 28.1b is a positive cycle matrix of period 1. By consequence, we can try to see if it is possible to construct negative cycle matrices that have as elements powers of the golden ratio *φ*, such that the squares of the same matrices are equal to the genetic matrix under consideration.

ACTIVITY

- Construct negative cycle matrices that have as elements powers of the golden section, φ, such that the squares of the same matrices are equal to the genetic matrix in figure 28.1b.

$$\begin{array}{|cccc|} \varphi & \varphi & \varphi^{-1} & \varphi^{-1} \\ \varphi & \varphi^{-1} & \varphi & \varphi^{-1} \\ \varphi^{-1} & \varphi & \varphi^{-1} & \varphi \\ \varphi^{-1} & \varphi^{-1} & \varphi & \varphi \end{array} \qquad \begin{array}{|cccc|} \varphi & \varphi^{-1} & \varphi & \varphi^{-1} \\ \varphi^{-1} & \varphi^{-1} & \varphi & \varphi \\ \varphi & \varphi & \varphi^{-1} & \varphi^{-1} \\ \varphi^{-1} & \varphi & \varphi^{-1} & \varphi \end{array}$$

a b

Figure 28.2.

It is possible to construct the matrices in figure 28.2. The first is a negative cycle matrix of period 1, whereas the second is a negative cycle matrix of period 2. The squares of these two matrices are positive cycle matrices of period 1. Let us calculate the first element of the first row of the square of the first matrix:

The first element of the first row is equal to

$$\varphi\varphi + \varphi\varphi + \varphi^{-1}\varphi^{-1} + \varphi^{-1}\varphi^{-1} = 2\varphi^2 + 2(\varphi^{-1})^2.$$

According to the definition of φ we have $\varphi^2 = \varphi + 1$ and $\varphi^{-1} = \varphi - 1$. Thus, the first element of the first row is equal to

$$2\varphi^2 + 2(\varphi - 1)^2 = 2\varphi^2 + 2\varphi^2 - 4\varphi + 2 = 4\varphi + 4 - 4\varphi + 2 = 6,$$

that is, equal to first element of the genetic matrix in figure 28.1b.

In a similar way the other elements of the two matrices may be calculated, arriving at the conclusion that the squares of both matrices in figure 28.2 are equal to the genetic matrix in figure 28.1b. In other words, the negative cycle matrices of period 1 and 2, respectively, in figure 28.2 are golden genetic matrices.

The design associated with the second matrix in figure 28.2 is a Liki-design, just like the one associated with the first matrix in figure 28.1. I sent these and other considerations about the relationships between cycle matrices and genetic matrices to Petoukhov, who revealed that he was quite interested in studying cycle matrices aiming for further applications in the field of genetic codes of life.

As happens frequently in the development of mathematics, it is not possible to foretell in which areas of knowledge mathematical concepts may find applications, but the rich structures and forms analyzed by mathematicians constitute useful instruments for scientists in other fields and often reflect profound relationships in Nature.

Chapter 29

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Chapter 30

NOTE ON THE USE OF A COMPUTER

We can calculate the product of two matrices using paper and pencil. However, the reader who has access to a computer can multiply quickly matrices using, for example, the Microsoft Excel program.

To multiply two matrices of dimensions 5 by 5, one marks a field of 5 by 5 and one writes

{=MMULT(A3:E7,G3:K7)}

supposing that the first matrix to multiply may be encountered in the cells A3 (first element of the first row) to E7 (fifth and last element of the fifth row) (see figure 30.1), and the second matrix in the cells from G3 to K7.

	A	**B**	**C**	**D**	**E**	**F**
2						
3	2	3	0	-1	4	
4	3	-1	2	4	0	
5	0	2	4	3	-1	
6	-1	4	3	0	2	
7	4	0	-1	2	3	
8						

Figure 30.1.

In order to vary easily the elements of a cycle matrix of certain dimensions, one can write the initial matrix in such a way that by changing some elements the other elements change automatically. For example, in order to introduce a negative alternating cycle matrix of dimensions 5 by 5 in figure 30.1 in the cells from A3 to E7, one writes only in the first row the five numbers. In the other rows one does not have to indicate concrete values, but, taking into account the cyclic structure of the matrix, one indicates only the coordinates of the element of the first row that has the same value. Figure 30.2 presents what has to be typewritten to conclude the first cycle.

	A	B	C	D	E	F
2						
3	2	3	0	-1	4	
4	=B3		=A3			
5		=A3		=B3		
6			=B3		=A3	
7				=A3	=B3	
8						

Figure 30.2.

Next, one advances with the other cycles (figure 30.3).

	A	B	C	D	E	F
2						
3	2	3	0	-1	4	
4	=B3	=D3	=A3	=E3	=C3	
5	=C3	=A3	=E3	=B3	=D3	
6	=D3	=E3	=B3	=C3	=A3	
7	=E3	=C3	=D3	=A3	=B3	
8						

Figure 30.3.

When one completes this process, what one sees on the screen of the computer is the matrix in the form of figure 30.1. When changing the values of the first row of the matrix, the computer will automatically change the other elements.

In this manner one can use the computer in the search of properties of cycle and other matrices. The computer facilitates much the experimentation.

APPENDICES

Chapter 31

INVERSE MATRICES

When the reader has access to a computer and to a spreadsheet program like Microsoft Excel, the reader may execute various operations on square matrices. The function MINVERSE calculates the inverse matrix of a square matrix.

Remember that the **identity matrix** of dimensions *k* by *k* is the matrix that has only 1's on its principal diagonal and only 0's as other elements. Figure 31.1 presents the identity matrix in the case *k*=4.

1	0	0	0
0	1	0	0
0	0	1	0
0	0	0	1

The identity matrix in the case *k*=4

Figure 31.1.

A square matrix **B** is called the inverse matrix of square matrix **A** of dimensions *k* x *k*, if **AB** = **BA** = **I**, where I represents the identity matrix of dimensions *k* x *k*. A square matrix does not always have an inverse matrix. When it does, it is usually called a non-singular matrix and has a unique inverse matrix.

We may ask ourselves whether or not non-singular cycle matrices have interesting inverse matrices in the sense that they display some particular property.

ACTIVITY

- Choose a negative alternating cycle matrix of dimensions 6 x 6, and, using the function MINVERSE, determine if it has an inverse matrix. In the case it has an inverse matrix, discover some particularity it displays.
- Choose some positive and negative cycle matrices of various dimensions with distinct periods, and by utilizing the function MINVERSE, determine if they have inverse matrices. When the chosen matrix has an inverse matrix, analyze if it possesses an interesting property.

Figure 31.2 shows a negative alternating cycle matrix of dimensions 6 x 6.

-1	5	-2	1	3	4
5	1	-1	4	-2	3
-2	-1	3	5	4	1
1	4	5	3	-1	-2
3	-2	4	-1	1	5
4	3	1	-2	5	-1

Negative alternating cycle matrix **R** of dimensions 6 x 6

Figure 31.2.

Matrix **R** has matrix **S** as its inverse matrix (figure 31.3).

-0.065	0.1034	-0.021	-0.016	0.0064	0.093
0.1034	-0.016	-0.065	0.093	-0.021	0.0064
-0.021	-0.065	0.0064	0.1034	0.093	-0.016
-0.016	0.093	0.1034	0.0064	-0.065	-0.021
0.0064	-0.021	0.093	-0.065	-0.016	0.1034
0.093	0.0064	-0.016	-0.021	0.1034	-0.065

S, the inverse matrix of **R**

Figure 31.3.

Observing the values of the elements in matrix **S** we can see that there are three alternating cycles: the inverse matrix of **R** is also a negative alternating cycle matrix.

Is the case of matrix **R** exceptional? Or is the following affirmation true: All inverse matrices of negative cycle matrices dimensions 6 x 6 and any period are negative cycle matrices too?

Let us experiment further. Let us consider the negative cycle matrix of period 3, presented in figure 31.4.

0.2	-2.1	-0.3	1.2	-1.3	0
0.5	-2.1	0.5	2	-2.1	-2
1.2	-2.1	0	0.2	-1.3	-0.3
-0.3	-1.3	0.2	0	-2.1	1.2
-2	-2.1	-2	0.5	-2.1	0.5
0	-1.3	1.2	-0.3	-2.1	0.2

Negative cycle matrix **T** (dimensions 6 x 6, period 3)

Figure 31.4.

We can calculate the inverse matrix de **T**. It is matrix **U** in figure 31.5.

-0.618	0.25	0.6181	0.8104	-0.25	-0.81
-1.057	0.5313	0.0297	1.0858	-0.118	-0.863
0.8164	-0.25	-0.533	-0.612	-0.209	0.8956
0.4808	0.25	-0.481	0.4808	-0.25	-0.481
1.0985	-0.531	-0.274	-1.044	-0.03	0.6192
0.4867	-0.25	-0.203	0.4867	-0.209	-0.203

Matrix **U**, the inverse matrix of **T**

Figure 31.5.

In matrix **U** some numbers appear various times: 0.25, -0.25, -0.209, …, but there does not appear to be any cycles. By consequence, the inverse matrix of **T** is not a cycle matrix.

Let us return, therefore, to the alternating cycle matrices. Consider a positive alternating cycle matrix.

-1	1.2	2.2	0.3	-2	1.3
2.2	-1	-2	1.2	1.3	0.3
1.2	0.3	-1	1.3	2.2	-2
-2	2.2	1.3	-1	0.3	1.2
0.3	1.3	1.2	-2	-1	2.2
1.3	-2	0.3	2.2	1.2	-1

A positive alternating cycle matrix **V** of dimensions 6 x 6

Figure 31.6.

The positive alternating cycle matrix **V** (figure 31.6) has as inverse matrix: matrix **W** illustrated in figure 31.7.

0.1018	0.001	0.295	-0.301	0.3821	0.0216
0.295	0.1018	0.3821	0.001	0.0216	-0.301
0.001	-0.301	0.1018	0.0216	0.295	0.3821
0.3821	0.295	0.0216	0.1018	-0.301	0.001
-0.301	0.0216	0.001	0.3821	0.1018	0.295
0.0216	0.3821	-0.301	0.295	0.001	0.1018

Matrix **W**, the inverse matrix of matrix **V**

Figure 31.7.

The diagonals of matrix **W** are constant. Also two alternating cycles may be observed. In other words, matrix **W** is a positive alternating cycle matrix too.

ACTIVITY

- Experiment with some more alternating cycle matrices. When done, formulate a general conjecture and try to prove it.

Let **A** be a non-singular positive alternating cycle matrix of dimensions *k* x *k*. Let **B** be the inverse matrix of **A** and *f* be the first row of matrix **B**.

We know already (Theorem 22.12) that there exists a unique positive alternating cycle matrix **C** that has the row *f* as its first row. If **B** is the inverse matrix of **A**, then the first row of the product **BA** is the first row of the identity matrix of dimensions *k* x *k*, that is, it is composed of one 1 followed by (k-1) 0's. As the first row of matrix **C** is equal to the first row of matrix **B**, the first row of the product **CA** is equal to the first row of the product **BA**, that is, equal to the first row of the identity matrix.

The product of two positive alternating cycle matrices is a positive alternating cycle matrix (Theorem 22.3). Thus, **CA** is a positive alternating cycle matrix.

As we know already that the first row of matrix **CA** is equal to the first row of the identity matrix, and that the identity matrix is a positive alternating cycle matrix, we may conclude that **CA** is equal to the identity matrix. In this way it follows that **C** is the inverse matrix of **A**.

We have proved the following theorem:

Theorem 31.1.

The inverse matrix of a non-singular positive alternating cycle matrix is a positive alternating cycle matrix too.

In a similar manner one may prove:

Theorem 31.2.

The inverse matrix of a non-singular negative alternating cycle matrix is a negative alternating cycle matrix too.

Chapter 32

DETERMINANTS

In this chapter we shall analyze the values of the determinants of alternating cycle matrices. The readers who have not yet studied the concept of the determinant of a square matrix may advance directly to Chapter 33.

Figure 32.1 presents two alternating cycle matrices, the first positive and the second negative, such that they have the same first row.

3	-2	-3	7
-3	3	7	-2
-2	7	3	-3
7	-3	-2	3

3	-2	-3	7
-2	7	3	-3
-3	3	7	-2
7	-3	-2	3

Two positive and negative alternating cycle matrices of dimensions
4 x 4 with the same first row

Figure 32.1.

Utilizing the function MDETERM in the Microsoft Excel program we can calculate the determinants of these two matrices. The determinant of the first is equal to 1275, whereas the determinant of the second is equal to -1275.

ACTIVITIES

- Construct some pairs of alternating cycle matrices of various dimensions with the first of each pair positive and the second negative, such that they have the same first row. Compare the values of the respective determinants.
- Try to formulate a general hypothesis and try to find a proof.

2	3	0	-1	4
0	2	4	3	-1
3	-1	2	4	0
4	0	-1	2	3
-1	4	3	0	2

2	3	0	-1	4
3	-1	2	4	0
0	2	4	3	-1
-1	4	3	0	2
4	0	-1	2	3

Positive and negative alternating cycle matrices of dimensions 5 x 5

Figure 32.2.

In the example of the pair of positive and negative alternating cycle matrices of dimensions 5x5, presented in figure 32.2, we find that both determinants are equal to 88.

-1	3	-2	4	-3	2
-2	-1	-3	3	2	4
3	4	-1	2	-2	-3
-3	-2	2	-1	4	3
4	2	3	-3	-1	-2
2	-3	4	-2	3	-1

-1	3	-2	4	-3	2
3	4	-1	2	-2	-3
-2	-1	-3	3	2	4
4	2	3	-3	-1	-2
-3	-2	2	-1	4	3
2	-3	4	-2	3	-1

Positive and negative alternating cycle matrices of dimensions 6 x 6

Figure 32.3.

In the case of the pair of positive and negative alternating cycle matrices of dimensions 6x6, presented in figure 32.3, we see that both determinants are equal to -3159.

1	3	0	-1	4	2	-2
0	1	4	3	-2	-1	2
3	-1	1	2	0	-2	4
4	0	-2	1	2	3	-1
-1	2	3	-2	1	4	0
-2	4	2	0	-1	1	3
2	-2	-1	4	3	0	1

1	3	0	-1	4	2	-2
3	-1	1	2	0	-2	4
0	1	4	3	-2	-1	2
-1	2	3	-2	1	4	0
4	0	-2	1	2	3	-1
2	-2	-1	4	3	0	1
-2	4	2	0	-1	1	3

Positive and negative alternating cycle matrices of dimensions 7 x 7

Figure 32.4.

In the case of figure 32.4 with the pair of positive and negative alternating cycle matrices of dimensions 7x7, we have the determinant of the positive matrix equal to 30233, whereas the determinant of the negative matrix is equal to -30233.

In each case we observe that the absolute values of the two determinants are equal. In the case where the dimensions are 4x4, the determinants have opposite signs. When the dimensions are 5x5 or 6x6, we note that the respective determinants are equal. In the case that the dimensions are 7x7, the determinants have opposite signs. What will the general rule be? And how do we prove it?

An important theorem of determinants says that, if one interchanges two rows of a matrix, keeping the other rows in the same place, then the sign of the determinant is inverted. Thus, knowing the number of times (let us say q times) that we have to interchange two rows to advance from a positive alternating cycle matrix to a negative alternating cycle matrix, maintaining the first row intact, the sign of the determinant changes q times. If q is an odd number, the two determinants have to have opposite signs; if q is an even number, the two determinants will be equal.

Figure 32.5 presents any positive alternating cycle matrix of dimensions 5x5. Interchanging the 2nd row with the 3rd, and the 4th with the 5th, we obtain a negative alternating cycle matrix with the same first row. The sign was changed twice and therefore we may conclude – without calculating the two determinants – that these determinants are equal.

$$\begin{matrix} a & b & c & d & e \\ c & a & e & b & d \\ b & d & a & e & c \\ e & c & d & a & b \\ d & e & b & c & a \end{matrix} \quad \longrightarrow \quad \begin{matrix} a & b & c & d & e \\ b & d & a & e & c \\ c & a & e & b & d \\ d & e & b & c & a \\ e & c & d & a & b \end{matrix}$$

k=5, q = 2

Figure 32.5.

In the case that the dimensions are 6x6 two interchanges are also sufficient, as illustrated by figure 32.6.

$$\begin{matrix} a & b & c & d & e & f \\ c & a & e & b & f & d \\ b & d & a & f & c & e \\ e & c & f & a & d & b \\ d & f & b & e & a & c \\ f & e & d & c & b & a \end{matrix} \quad \longrightarrow \quad \begin{matrix} a & b & c & d & e & f \\ b & d & a & f & c & e \\ c & a & e & b & f & d \\ d & f & b & e & a & c \\ e & c & f & a & d & b \\ f & e & d & c & b & a \end{matrix}$$

k=6, q = 2

Figure 32.6.

The reasoning we used can be easily generalized. When the dimensions are k by k, one interchanges the 1st and the 2nd row, the 3rd and the 4th row, the 5th and the 6th row, etc. If k is

an odd number, let us say $k=2q+1$, we need q interchanges of rows; if k is an even number, let us say $k=2q+2$, one needs q interchanges too. Letting **P** be the positive alternating cycle matrix under consideration and **N** the negative alternating cycle matrix with the same first row as **P**, we have the following general result

$$\text{Det } \mathbf{N} = (-1)^q \text{ Det } \mathbf{P},$$

where Det means determinant.

Chapter 33

TRANSFORMATIONS OF ALTERNATING CYCLE MATRICES. PERMUTATIONS

Let us consider any positive alternating cycle matrix **P** of dimensions 5x5 (figure 33.1).

a	b	c	d	e
c	a	e	b	d
b	d	a	e	c
e	c	d	a	b
d	e	b	c	a

Matrix **P**

Figure 33.1.

Let us construct a new matrix (**Q**) of the same nature, wherein the first row of matrix **P** appears as the fourth row (figure 33.2).

?	?	?	?	?
?	?	?	?	?
?	?	?	?	?
a	*b*	*c*	*d*	*e*
?	?	?	?	?

Beginning the construction of matrix **Q**

Figure 33.2.

What will the other rows of matrix **Q** be? The fourth element of the fourth row of matrix **Q** is *d*. Thus, the principal diagonal of matrix **Q** has *d*'s (figure 33.3a).

d	?	?	?	?
?	d	?	?	?
?	?	d	?	?
a	b	c	d	e
?	?	?	?	d

a

d	e	b	?	?
b	d	?	e	?
e	?	d	?	b
a	b	c	d	e
?	?	e	b	d

b

d	e	b	c	a
b	d	a	e	c
e	c	d	a	b
a	b	c	d	e
c	a	e	b	d

c

Completing the construction of matrix **Q**

Figure 33.3.

Now we can construct the two cycles of matrix **Q** and complete the entire matrix (figure 33.3b and c).

ACTIVITY

- Observe the rows and the columns of matrix **Q**. Do you note some particularity?

Observing matrix **Q** we can see that, surprisingly (!), the rows of **Q** are equal to the rows of **P** with only their sequence changed. The columns of matrix **Q** are the same as the columns of matrix **P** with only the order of the columns altered. Matrices **P** and **Q** have the same rows and the same columns. In this sense, the transformation of **P** into **Q** is a **permutation**: the order of the rows is changed; the rows are permuted.

Let us look at how the order of the rows was changed:

- the first row (1) of matrix **P** became the fourth row of matrix **Q**;
- the fourth row (4) of matrix **P** became the third row of matrix **Q**;
- the third row (3) of matrix **P** became the second row of matrix **Q**;
- the second row (2) of matrix **P** became the fifth row of matrix Q;
- the fifth row (5) of matrix **P** became the first row of matrix **Q**.

This permutation may be abbreviated by (1 4 3 2 5). This notation is usually called the **cycle notation** of a permutation.

ACTIVITIES

- Transform matrix **P** into a positive alternating cycle matrix **R**, such that the first row of **P** becomes the third row of **R**. If the transformation is a permutation represent it in cycle notation.
- Transform matrix **P** into a positive alternating cycle matrix **S**, such that its first row becomes the second row of **S**. If the transformation is a permutation represent it in cycle notation.

Let us analyze the case of the transformation of matrix **P** onto matrix **T**, where **T** is the positive alternating cycle matrix that has the first row of **P** as fifth row (figure 33.4).

e	c	d	a	b
d	e	b	c	a
c	a	e	b	d
b	d	a	e	c
a	b	c	d	e

Matrix **T**

Figure 33.4.

Once more, we observe that all rows (and all columns) of matrix **P** appear as rows (and columns) in matrix **T**. We are dealing with a permutation. The transformation of the rows of matrix **P** to arrive at matrix **T** may be represented by (1 5 2 3 4). In the same way, the permutations of the last activities may be represented in the cyclic notation by (1 3 5 4 2) and (1 2 4 5 3), respectively.

ACTIVITIES

- Consider any positive alternating cycle matrix **U** of dimensions 6x6 (figure 32.5).

a	b	c	d	e	f
c	a	e	b	f	d
b	d	a	f	c	e
e	c	f	a	d	b
d	f	b	e	a	c
f	e	d	c	b	a

Matrix **U**

Figure 33.5.

  Construct a positive alternating cycle matrix **V**, such that the fourth row of **V** is equal to the first row of **U**. Will matrix **V** have the same rows as matrix **U**? And will it have the same columns?
  If the answer is yes, how can one denote the permutation of the rows?
- Analyze the situation when the first row of **T** becomes the fifth row of a matrix **W**.
- Analyze the two situations when **U** has dimensions 7x7.
- Analyze the two situations when **U** has dimensions 10x10.

Let us construct matrix **V**. Its diagonals are constant, and thus we have only *d*'s on the principal diagonal and on the secondary diagonal only *c*'s (figure 33.6b). Now we only need to complete the two cycles of alternating numbers to conclude the construction of matrix **V** (figure 33.6c and d).

?	?	?	?	?	?
?	?	?	?	?	?
?	?	?	?	?	?
a	*b*	*c*	*d*	*e*	*f*
?	?	?	?	?	?
?	?	?	?	?	?

a

d	?	?	?	?	c
?	*d*	?	?	c	?
?	?	*d*	c	?	?
a	*b*	*c*	*d*	*e*	*f*
?	c	?	?	*d*	?
c	?	?	?	?	*d*

b

d	f	b	?	?	c
b	d	?	f	c	?
f	?	d	c	b	?
a	b	c	d	e	f
?	c	f	?	d	b
c	?	?	b	f	d

c

d	f	b	e	a	c
b	d	a	f	c	e
f	e	d	c	b	a
a	b	c	d	e	f
e	c	f	a	d	b
c	a	e	b	f	d

d

Construction of matrix V

Figure 33.6.

When we observe matrix **V**, we see that all rows of matrix **U** appear in it, but in a different order. The rows have been interchanged.

The first row of matrix **U** became the fourth row of matrix **V**; the fourth row of matrix **U** became the fifth row of matrix **V**; the fifth row of matrix **U** became the first row of matrix **V**. To this part of the transformation corresponds the cycle notation (1 4 5).

The second row of matrix **U** became the sixth row of matrix **V**; the sixth row of matrix **U** became the third row of matrix **V** and the third row of matrix **U** became the second row of matrix **V**. The second part of the transformation corresponds to the abbreviation (2 6 3).

Thus the entire permutation of the rows may be represented in cycle notation as (1 4 5) (2 6 3).

Figures 33.7 and 33.8 present the matrices **U** and **V** with dimensions 7x7 and 10x10.

a	b	c	d	e	f	g
c	a	e	b	g	d	f
b	d	a	f	c	g	e
e	c	g	a	f	b	d
d	f	b	g	a	e	c
g	e	f	c	d	a	b
f	g	d	e	b	c	a

d	f	b	g	a	e	c
b	d	a	f	c	g	e
f	g	d	e	b	c	a
a	b	c	d	e	f	g
g	e	f	c	d	a	b
c	a	e	b	g	d	f
e	c	g	a	f	b	d

Matrices **U** and **V** (dimensions 7x7)

Figure 33.7.

a	b	c	d	e	f	g	h	i	j
c	a	e	b	g	d	i	f	j	h
b	d	a	f	c	h	e	j	g	i
e	c	g	a	i	b	j	d	h	f
d	f	b	h	a	j	c	i	e	g
g	e	i	c	j	a	h	b	f	d
f	h	d	j	b	i	a	g	c	e
i	g	j	e	h	c	f	a	d	b
h	j	f	i	d	g	b	e	a	c
j	i	h	g	f	e	d	c	b	a

d	f	b	h	a	j	c	i	e	g
b	d	a	f	c	h	e	j	g	i
f	h	d	j	b	i	a	g	c	e
a	b	c	d	e	f	g	h	i	j
h	j	f	i	d	g	b	e	a	c
c	a	e	b	g	d	i	f	j	h
j	i	h	g	f	e	d	c	b	a
e	c	g	a	i	b	j	d	h	f
i	g	j	e	h	c	f	a	d	b
g	e	i	c	j	a	h	b	f	d

Matrices **U** and **V** (dimensions 10x10)

Figure 33.8.

In both of these cases we are dealing with permutations once again. They may be denoted by (1 4 7 3 2 6 5) and (1 4 8 9 5) (2 6 10 7 3), respectively.

ACTIVITIES

- Does there exist an *easier* or a more *beautiful* (!) notation to represent the permutations that correspond to the transformation of a positive alternating cycle matrix into another positive alternating cycle matrix that has a row of the first matrix as one of its rows? Experiment!
- Suggestion: place the numbers that appear in the cycle notation around a regular polygon (figure 32.9 illustrates the case of dimensions 5x5: pentagon) and join the numbers that appear in the cycle notation by segments.

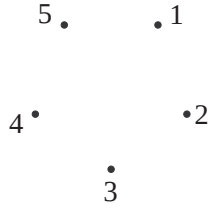


Figure 33.9.

- In this chapter we dealt with positive alternating cycle matrices. Analyze the situations created by negative alternating cycle matrices. Will the transformations of these negative matrices be permutations too?

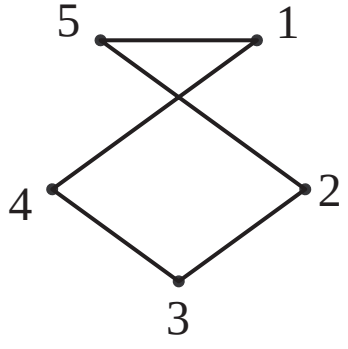
Chapter 34

POLYGONAL AND CIRCULAR REPRESENTATIONS

We concluded the last chapter with the following question

- Does there exist an *easier* or a more *beautiful* (!) notation, to represent the permutations that correspond to the transformation of a positive alternating cycle matrix into another positive alternating cycle matrix that has a row of the first matrix as one of its rows?

Following the presented suggestion, let us consider the case of the transformation of matrix **P** onto matrix **Q** (figure 33.3). The permutation may be denoted by (1 4 3 2 5). Connecting the five points placed at the vertices of a regular pentagon, numbered 1 to 5, by segments in agreement with the cycle notation, point 1 with point 4, point 4 with point 3, point 3 with point 2, point 2 with point 5, and, closing the cycle, point 5 with point 1, we obtain the image illustrated in figure 34.1.



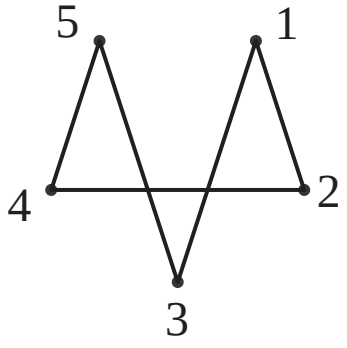
Polygonal representation of permutation (1 4 3 2 5)

Figure 34.1.

The drawing presents an axial symmetry: the vertical line constitutes the axis of symmetry.

The transformation of matrix **P** into matrix **T** (figure 33.4), denoted by (1 5 2 3 4), may be represented by the same graph, but passing through in the opposite direction. In the same

way, the other two transformations in the case of dimensions 5x5, (1 3 5 4 2) and (1 2 4 5 3), respectively, may be represented by the symmetrical design illustrated in figure 34.2.



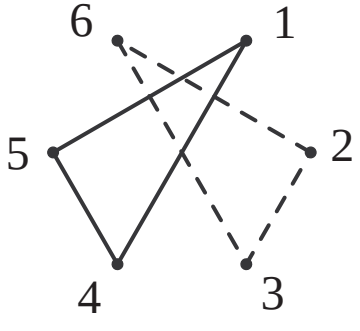
Polygonal representation of the permutations (1 3 5 4 2) and (1 2 4 5 3)

Figure 34.2.

ACTIVITIES

- In a similar manner find polygonal representations for the other permutations analyzed in the previous chapter.

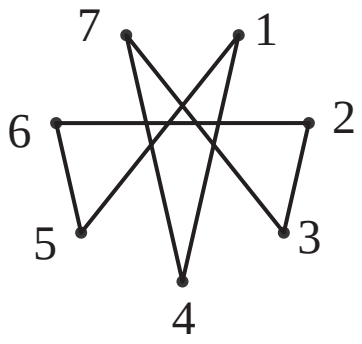
When the dimensions are 6x6, the transformation of the positive alternating cycle matrix **U** into matrix **V** (figure 33.6), has the cycle notation (1 4 5) (2 6 3). The polygonal representation is composed of two triangles that are symmetrical to one another (figure 34.3).



Polygonal representation of the permutation (1 4 5) (2 6 3)

Figure 34.3.

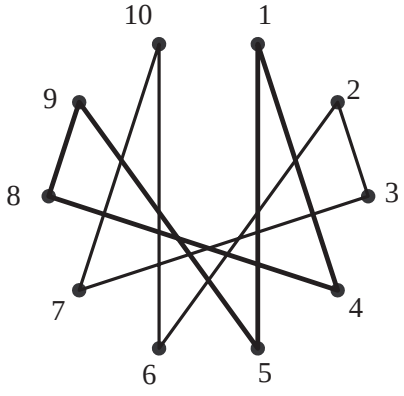
When the dimensions are 7x7, the transformation of the positive alternating cycle matrix **U** into matrix **V** (figure 33.7) has the cycle notation (1 4 7 3 2 6 5). The polygonal representation is the symmetrical design presented in figure 34.4.



Polygonal representation of the permutation (1 4 7 3 2 6 5)

Figure 34.4.

When the dimensions are 10x10, the transformation of the positive alternating cycle matrix **U** into matrix **V** (figure 33.8) has the cycle notation (1 4 8 9 5) (2 6 10 7 3). The polygonal representation is the design presented in figure 34.5, composed of two closed polygonal lines that are mutually symmetrical.



Polygonal representation of permutation (1 4 8 9 5) (2 6 10 7 3)

Figure 34.5.

All polygonal representations in figures 33.1 to 33.5 have a vertical axis of symmetry. Although the representations are interesting, we remain with one doubt: why do the numbers in the cycle notation appear in their particular sequence? Does there exist a more interesting, more logical or more evident structure behind it?

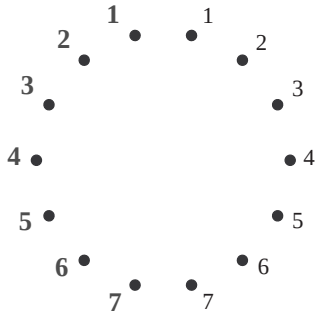
Let us look at one cycle of a positive alternating cycle matrix. Figure 34.6 presents one cycle of matrix **U** (figure 33.7). The number of each row is indicated on both the left and right side of the matrix **U**.

1	.	.	.	d	e	.	.	1
2	.	.	e	.	.	d	.	2
3	.	d	.	.	.	.	e	3
4	e	.	.	.	.	.	d	4
5	d	.	.	.	.	e	.	5
6	.	e	.	.	d	.	.	6
7	.	.	d	e	.	.	.	7

A cycle of matrix **U**

Figure 34.6.

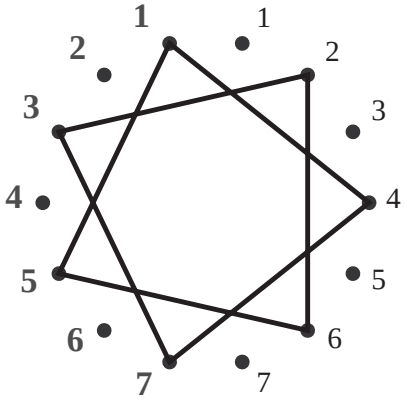
We know the cycle passes exactly twice through each of the rows of the matrix. Inspired by this fact we may group the seven numbers of the rows *twice* around a circle (figure 34.7).



Basic structure for the circular representation when the dimensions are 7x7

Figure 34.7.

What will the representation be of the permutation (1 4 7 3 2 6 5) that had let us to the polygonal representation of figure 34.4? Starting with point (1) at the left, and advancing clockwise to point (4) at the right side, etc., we obtain the beautiful representation illustrated in figure 34.8. We are dealing with a regular heptagonal star.



Circular representation of permutation (1 4 7 3 2 6 5)

Figure 34.8.

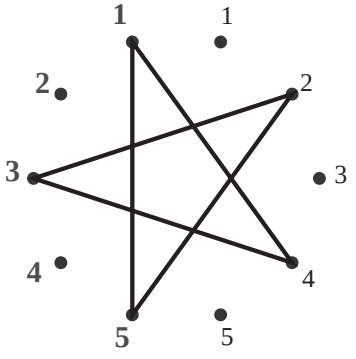
Behind the sequence of the numbers in the cycle notation of the permutation, (1 4 7 3 2 6 5) – which appears to lack a certain order – , exists the perfect order of the regular heptagonal star!

We may call this type of representation a **circular representation** to be able to distinguish it from the previous polygonal representation.

ACTIVITIES

- Analyze if there exist circular representations that correspond to the polygonal representations (figures 34.1, 34.2, 34.3 and 34.5) of the permutations of positive alternating cycle matrices.

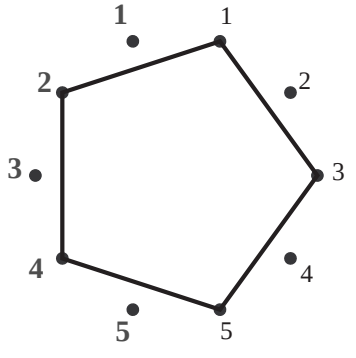
The permutation (1 4 3 2 5) (figure 34.1) may be represented by a pentagonal star (figure 34.9).



Circular representation of permutation (1 4 3 2 5)

Figure 34.9.

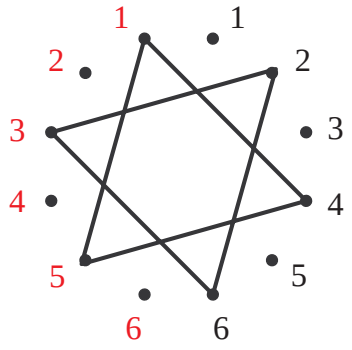
Permutation (1 3 5 4 2) (see figure 34.2), however, may be represented by a regular pentagon (figure 34.10).



Circular representation of permutation (1 3 5 4 2)

Figure 34.10.

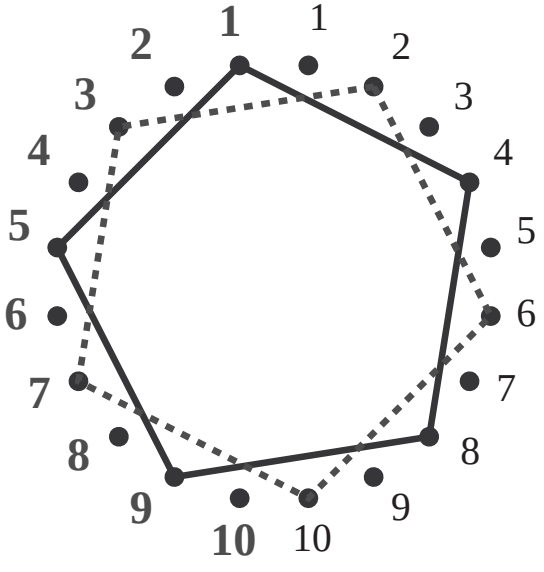
Considering the case of dimensions 6x6, the permutation (1 4 5) (2 6 3) (figure 34.3), may be represented by two intersecting equilateral triangles that together form a hexagonal star (figure 34.11).



Circular representation of permutation (1 4 5) (2 6 3)

Figure 34.11.

Finally, in the case of dimensions 10x10, the permutation (1 4 8 9 5) (2 6 10 7 3) (figure 34.5) may be represented by two intersecting regular pentagons, forming a decagonal star (figure 34.12).



Circular representation of the permutation (1 4 8 9 5) (2 6 10 7 3)

Figure 34.12.

ACTIVITIES

- Find the circular representations for all permutations of positive alternating cycle

matrices of dimensions 6x6.

- Find the circular representations for all permutations of positive alternating cycle matrices of dimensions 7x7.
- Find the circular representations for all permutations of positive alternating cycle matrices of dimensions 10x10.
- Try to formulate a general theorem about the circular representations of permutations of positive alternating cycle matrices.
- Experiment with negative alternating cycle matrices.

We discovered a beautiful world behind the permutations of positive alternating cycle matrices. For the positive alternating cycle matrices we may formulate the following theorem:

Theorem 34.1.

Let **A** be a positive alternating cycle matrix of dimensions m by m. Let **B** be the positive alternating cycle matrix that has the first row of matrix **A** as its j-th row. Consider $2m$ points placed at equal distances around a circle, numerated clockwise 1, 2, …, m-1, m, m, m-1, …, 2, 1, keeping the first m points on the right side and the other m points on the left side. The circular representation of the permutation that transforms matrix **A** into matrix **B** is characterized by the following construction steps:

(1) If j is an even number, start at 1-point on the left side and advance j points clockwise each time, until returning to the initial point and closing thus a polygonal line;
(2) If j is an odd number, start at the 1-point on the right side and advance j-1 points clockwise until returning to the initial point and closing a polygonal line;
(3) If the polygonal line has m vertices, stop. If the polygonal line has s vertices, where s is a divisor of m (let us say $m = st$), then turn the polygonal about an angle of (360° / t) around the centre of the circle and copy it. Repeat this process t-1 times.

The resulting representation is a regular m-gon, a regular star with m vertices, or a singular star with m vertices.

We saw already examples of circular representations in the form of a regular polygon and a regular polygonal star. Figure 34.13 presents an example of a singular star with 10 vertices, that corresponds to the permutation (1 10) (2 9) (3 8) (4 7) (5 6).

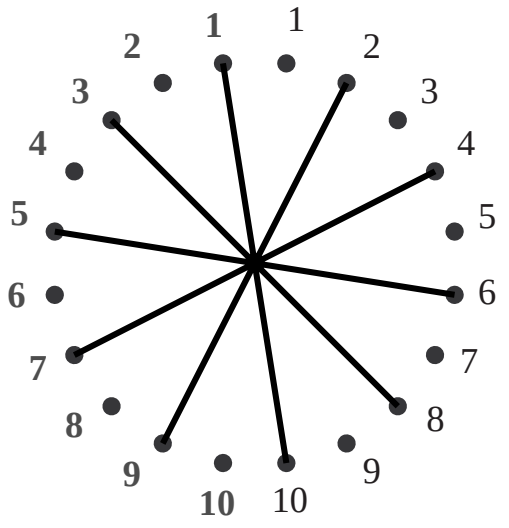


Figure 34.13.

In this way we found a surprising and beautiful relationship between the permutations of positive alternating cycle matrices and regular polygons and polygonal stars.

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